Roadmap 4: theorem-boundary packet program

Roadmap target R4.3b: Wilson-loop dictionary lemma.

Roadmap target R4.3a should now be read as factoring through R4.3b.

We build the Wilson-loop / Schwinger dictionary by isolating a common-realization witness package, canonical Euclidean representatives for the packet generators, and the OS word-expectation formula as abstract packet-side ingredients. This file is a roadmap note, not a claim of a complete proof.

Hamiltonian identification theorem boundary

This appendix records the canonical audited boundary for the Yang–Mills-specific identification step. The repository status is *theorem boundary*, not a completed proof of the four-dimensional $SU(3)$ mass-gap conjecture.

Three identification clauses already isolated

The remaining bridge is organized around three abstract clauses that are already separated cleanly in the audit:

- the *vacuum clause*, identifying the convergence projector with the vacuum-complement sector of the OS reconstruction;
- the *algebra clause*, identifying the convergence observable algebra with the Wilson-loop / GNS algebra and its separating family;
- the *generator clause*, identifying the limiting semigroup generator and the operative common core with the OS Hamiltonian domain.

In this reduced formulation the vacuum clause, algebra clause, and generator clause are not the open problem by themselves; the open problem is the Yang–Mills-specific realization needed to verify them in one common model. The packet reductions below are the mechanism that tries to make that realization finite and checkable. They use the dictionary lemma, the packet generators, a common finite domain, and the OS word-expectation formula.

Finite packet reductions that are already abstractly available

The current audit treats the following reductions as abstractly available and therefore not the final Yang–Mills obstruction:

- finite Wilson-loop packet truncation is abstractly available;
- finite rational-time packet truncation is abstractly available;
- the finite Wilson-loop moment packet is the state-side finite datum;
- a common-realization witness package is the bookkeeping device used to compare the convergence and OS sides on finite packets.

Wilson-loop packet branch on W_N

The state-side branch is the finite Wilson-loop packet entrywise equality theorem boundary. Its internal reduction is the following theorem-sized ladder:

1. common-Schwinger-family criterion for Wilson-loop packet equality;
2. common-realization witness package for Wilson-loop packet equality;
3. tuple-fiber criterion for packet Schwinger witness existence;
4. common packet Schwinger witness from two descents;
5. the packet comparison factors through the tuple fibers;
6. common continuum Schwinger family theorem boundary on W_N;
7. convergence-side tuple-fiber constancy theorem boundary on W_N;
8. OS-side tuple-fiber constancy theorem boundary on W_N;
9. OS word formula implies tuple-fiber constancy;
10. tuple-image witness identification theorem boundary on W_N;
11. convergence-side Schwinger realization theorem boundary on W_N;
12. OS-side Schwinger realization theorem boundary on W_N.

The shortcut architecture is also explicit: injective packet dictionary implies tuple-fiber constancy; letter-level injectivity implies packet-dictionary injectivity; and the letter-level injectivity theorem boundary on the packet alphabet is reduced to the statement that generator injectivity plus reflection-extension injectivity imply letter-level injectivity. That leaves the generator-level injectivity theorem boundary on the finite packet generators, the reflection-extension injectivity theorem boundary on the packet alphabet, and therefore a shared shortcut obstruction. The intended shortcut is an injective single-letter dictionary, and the scheduler now records this injectivity shortcut explicitly together with the unresolved shortcut subgoals. This is why the appendix keeps both the exact live state-side blocker and the blocked-action view visible at once: the dictionary shortcut is shared even when the two packet branches are written separately.

Rational-time packet branch on F_N

The action-side branch is the finite rational-time packet entrywise equality theorem boundary. Its finite reduction is:

1. common Euclidean-action criterion for rational-time packet equality;
2. common-realization witness package for rational-time packet equality;

3. representative-fiber criterion for packet action witness existence;
4. common packet action witness from two descents;
5. packet representative fibers;
6. common Euclidean action theorem boundary on F_N;
7. convergence-side representative-fiber constancy theorem boundary on F_N;
8. OS-side representative-fiber constancy theorem boundary on F_N;
9. injective representative map implies representative-fiber constancy;
10. labeled representative encoding implies representative-map injectivity;
11. representative-image witness identification theorem boundary on F_N;
12. convergence-side action realization theorem boundary on F_N;
13. OS-side action realization theorem boundary on F_N.

The same generator-level injectivity theorem boundary on the finite packet generators enters here because the representative map must remain faithful on the same packet alphabet that feeds the state-side dictionary.

Dictionary infrastructure shared by both branches

The packet reductions only make sense after the following common dictionary statements are isolated:

- single-letter dictionary theorem boundary;
- adjoint/reflection dictionary theorem boundary;
- word-to-tuple assembly theorem boundary;
- tuple-domain admissibility theorem boundary.

These shared preliminaries explain why both packet branches are localized but not independent: one common-realization witness package still depends on a single Yang–Mills-specific identification mechanism.

Exact live theorem boundary

The exact live state-side blocker is the convergence-side Schwinger realization theorem boundary on W_N: one must prove that the convergence-side continuum state is realized by the same continuum Schwinger family used on the OS side. The parallel action-side blocker is the convergence-side action realization theorem boundary on F_N: one must prove that the convergence-side packet action is realized by the same Euclidean time-translation action that enters the OS Hamiltonian construction. The current verifier now also requires the Schwinger,

OS reconstruction, operator-convergence, and semigroup artifacts to carry the same action-specification hash before this action-side support is accepted. The downstream countable localization layer uses explicit continuity estimates: for contraction letters,

$$\|g_1 \cdots g_n - d_1 \cdots d_n\| \le n\varepsilon,$$

so state differences are bounded by $2n\varepsilon$, and rational-time action agreement is upgraded to real-time agreement by strong continuity. Until the packet realization statements are proved, packetwise equalities cannot be extended to the full continuum Wilson-loop state or the full Euclidean semigroup.

Any manuscript claim that the mass gap is already proved must therefore first discharge the vacuum clause, algebra clause, and generator clause in the same Yang–Mills setting by proving the packet dictionary, the two packet realization theorems, and the downstream countable extension steps.

1 rigorous HST program

This appendix records a conjectural research direction. It is not part of the repository's audited proof spine and it does not resolve the theorem boundary identified in Appendix .

The earlier version of this file claimed to "rigorously discharge" several Yang–Mills blockers by introducing Holographic Stochastic Transport (HST) and a "Quantum Weak KAM" mechanism. Those claims were unsupported: no precise Yang–Mills construction, no verified curvature-dimension estimate, no proved continuum OS tightness theorem, no justified Trotter–Kato identification, and no proved clustering-to-gap implication were supplied in the manuscript. Accordingly, those claims are retracted here.

1.1 What an HST program would need to prove

For HST to become part of a rigorous mass-gap proof, one would need theorem-level results of the following kind:

1. a well-posed gauge-covariant stochastic flow on a Yang–Mills configuration space together with a proved invariant or asymptotically invariant continuum measure;

2. a non-circular coercive inequality, such as an LSI or spectral-gap bound, with constants uniform in the relevant approximating volumes and cutoffs;

3. a proof that reflection positivity and the other required Osterwalder–Schrader axioms survive the continuum limit produced by that flow;

4. an identification theorem showing that the limiting semigroup generator and its domain coincide with the OS-reconstructed Hamiltonian in the same Yang–Mills model;

5. a separate argument transferring any such verified semigroup or clustering estimate into a positive lower bound for the physical continuum Hamiltonian.

1.2 Why the current file does not close those steps

At the current revision, none of the required results above is proved here. In particular, this appendix does not provide:

- a mathematically defined HST dynamics on the non-abelian continuum theory;
- a proved Bakry–Émery curvature lower bound in the Yang–Mills setting;
- a non-circular derivation of a volume-uniform LSI constant;
- a rigorous construction of the continuum OS measure by the proposed transport map;
- a proved resolvent-convergence theorem identifying the continuum generator with the OS Hamiltonian;
- a valid deduction of vacuum uniqueness or a positive mass gap from the proposed framework.

1.3 Correct status

HST should therefore be cited only as a rigorous side program. It may suggest possible intermediate lemmas or model problems, but it does not currently prove any of the four open obligations listed in Appendix , and it does not convert the repository into a proof of the four-dimensional $SU(3)$ Yang–Mills mass-gap conjecture.

2 rigorous QGW foliation

This appendix is a second rigorous side program. It is not part of the repository's audited proof spine and it does not discharge the Hamiltonian identification theorem boundary.

The intended role of a Quantum Gromov–Wasserstein foliation theory would be to supply a rigidity mechanism for the finite packet realization problems described in Appendix . Concretely, one would need theorem-level results of the following kind:

1. packetwise reconstruction estimates on the Wilson-loop packets W_N and rational-time packets F_N;
2. a rigidity theorem showing that packetwise agreement under the common OS constraints forces equality of the continuum state and equality of the Euclidean action;

3. an upgrade theorem from packet equality to the full Wilson-loop state and the full Euclidean semigroup.

No such theorems are proved here. In particular, the present appendix does *not* prove tuple-fiber constancy, does *not* prove generator-level injectivity for the Yang–Mills packet dictionary, and does *not* prove the blocked action/state extension steps. Those remain open obligations in the audit artifacts and in the roadmap.

Accordingly, this appendix should be cited only as a conjectural research framework. Any statement that it "resolves" Blocker 3 or closes the mass-gap proof is false at the current revision.

3 Final synthesis: audited status of the mass-gap program

The current repository does not prove the four-dimensional $SU(3)$ Yang–Mills mass-gap conjecture. What it does provide is an audited theorem-boundary organization of the existing evidence and of the remaining Yang–Mills-specific obstructions.

3.1 What is established in the audited spine

The verification artifacts record the following items as present at the current revision:

- approximant gap / decay input for the semigroup bridge;
- semigroup-shaped operator convergence evidence;
- continuum Schwinger-limit and OS-reconstruction evidence artifacts;
- abstract projector/algebra/generator identification scaffolding.

These ingredients support only a theorem-boundary audit, not a completed proof.

3.2 What remains open

Appendix isolates the missing Yang–Mills-specific bridge into four live theorem obligations:

1. convergence-side realization by the same continuum Schwinger family on finite Wilson-loop packets;
2. convergence-side realization by the same Euclidean time-translation action on finite rational-time packets;
3. extension of packetwise state agreement to equality of the full continuum Wilson-loop state;

4. extension of packetwise action agreement to equality of the full Euclidean semigroup.

Until those four items are proved in the same Yang–Mills setting, no argument in this manuscript yields a rigorous positive spectral gap for the OS-reconstructed continuum Hamiltonian.

3.3 Status conclusion

The honest conclusion of the present manuscript is therefore:

> The repository organizes constructive evidence, abstract bridge lemmas, and explicit open theorem obligations, but it does not currently establish a Clay-level proof of the $4D$ $SU(3)$ Yang–Mills existence and mass-gap conjecture.

4 Conditional route: dual lattice current realization

This appendix records a new candidate mechanism for the two live realization theorems. It is not an established theorem package in the present repository. It should be read as a precise research target: if the lemmas below are proved with the stated compatibility, then the two packet realization blockers in Appendix would be discharged.

4.1 Finite current charts

For each finite Wilson-loop packet W_N and rational-time packet F_N, the proposed construction introduces a finite dual-current chart J_N. The intended role of J_N is to carry both Wilson-loop packet observables and rational-time time-translation representatives in one finite-dimensional gauge-invariant observable space. The first required lemma is:

> For every N, each packet observable factors through J_N up to a gauge-exact cylindrical coboundary.

This would turn both realization problems into finite current-law comparison problems.

A finite-coordinate version of this step is discharged first. Since W_N and F_N are finite, enumerate all packet observables as $O_1, \dots, O_m$ and set

$$J_N(A) = (O_1(A), \dots, O_m(A)).$$

Every packet observable is exactly recovered by a coordinate projection. This proves finite-dimensional packet packaging.

The finite gauge-theoretic upgrade is also discharged on each packet graph. Let Γ_N be the finite graph spanned by the packet paths and choose a maximal

tree T. Tree gauge sets every tree-edge holonomy to the identity. The remaining non-tree edge variables form a finite cycle-current chart with

$$|E(\Gamma_N)| - |V(\Gamma_N)| + c(\Gamma_N)$$

group-valued coordinates. Every packet path holonomy is a word in these non-tree variables and their inverses. A closed Wilson loop changes only by residual root conjugation, and its normalized trace is therefore unchanged. The factorization error is exactly zero. Thus the finite DLCR chart lemma is proved packetwise; the remaining DLCR work is analytic, not finite graph packaging.

4.2 Exact shadow coupling

The second required lemma is an exact shadow coupling. Its finite probability core is discharged. If the convergence-side chart law and the OS chart law on J_N are equal, the diagonal coupling gives equality almost surely and hence zero error for all packet tests. More generally, if their total variation distance is $\delta_N(a)$, then every bounded packet test Φ obeys

$$|\mathbf{E}_{\mathrm{conv}}\Phi - \mathbf{E}_{\mathrm{OS}}\Phi| \le 2\|\Phi\|_\infty \delta_N(a).$$

Thus the remaining shadow-coupling problem is a chart-law theorem for Yang–Mills. Equivalently, it is the finite chart-law convergence problem. This problem has a standard compact-group reduction. By Peter–Weyl, the gauge-invariant matrix-coefficient polynomials are uniformly dense in the continuous class functions on the finite chart $SU(3)^m/\mathrm{Ad}$. If a packet test Φ is approximated by a finite Peter–Weyl polynomial P with $\|\Phi - P\|_\infty \le \varepsilon$, then

$$|\mathbf{E}_\mu\Phi - \mathbf{E}_\nu\Phi| \le |\mathbf{E}_\mu P - \mathbf{E}_\nu P| + 2\varepsilon.$$

Thus finite chart-law convergence reduces to convergence of finitely truncated spin-network moments plus a uniform approximation tail.

The finite chart-law tests can be made more concrete on the same packet graph. After tree gauge fixing, let $x_1, \dots, x_m$ denote the non-tree $SU(3)$ variables and, for a word w in the free group on these variables, set

$$\tau_w(x) = \frac{1}{3}\,\mathrm{Tr}(w(x_1, \dots, x_m)).$$

The unital $*$-algebra generated by the τ_w is dense in $C(SU(3)^m/\mathrm{Ad})$. Indeed it is closed under conjugation since $\overline{\tau_w} = \tau_{w^{-1}}$, and it separates quotient points: equality of all trace words is equality of the characters of the corresponding three-dimensional unitary representations of the free group. Finite-dimensional unitary representations are semisimple and irreducible characters are linearly independent, hence the representations are unitarily equivalent, and the conjugating unitary can be rescaled to determinant one. Stone–Weierstrass applies. Therefore, if $Q = \sum_\ell c_\ell M_\ell$ is a finite polynomial in trace words and $\|\Phi - Q\|_\infty \le \varepsilon$, then

$$|\mathbf{E}_\mu\Phi - \mathbf{E}_\nu\Phi| \le \sum_\ell |c_\ell|\,|\mathbf{E}_\mu M_\ell - \mathbf{E}_\nu M_\ell| + 2\varepsilon.$$

Thus the chart-law problem is now reduced to finite mixed Wilson trace-word moment convergence plus the explicit approximation tail.

The resulting finite certificate has explicit constants. Allow the approximant to depend on the lattice spacing, $Q_a = \sum_\ell c_\ell(a) M_\ell(a)$; a fixed polynomial is the special case. If the selected monomials satisfy

$$|\mathbf{E}_a M_\ell(a) - \mathbf{E}_{\mathrm{OS}} M_\ell(a)| \le B_\ell(a) a^{\beta_\ell(a)}, \qquad \beta_\ell(a) \ge \beta > 0,$$

with uniform coefficient-weighted constant $\sum_\ell |c_\ell(a)| B_\ell(a) \le B$, and the approximation tail is power-controlled, $\|\Phi - Q_a\|_\infty \le E a^\gamma$, then

$$|\mathbf{E}_a \Phi - \mathbf{E}_{\mathrm{OS}} \Phi| \le (B + 2E)\, a^\alpha, \qquad \alpha = \min(\beta, \gamma).$$

If only $\|\Phi - Q_a\|_\infty \le \varepsilon$ is available, the same argument gives the weaker bound

$$|\mathbf{E}_a \Phi - \mathbf{E}_{\mathrm{OS}} \Phi| \le B a^\beta + 2\varepsilon.$$

Thus a $C_N a^{\alpha_N}$ chart-law estimate now follows from finite trace-word moment estimates and a power-controlled trace-polynomial approximation scheme. Those two Yang–Mills analytic inputs remain proved in this repository.

There is one further finite reduction for the moment estimates. Put

$$d_\infty(x, y) = \max_j \|x_j - y_j\|_{\mathrm{op}}$$

on the finite chart representatives. A normalized trace word $\tau_w = 3^{-1} \operatorname{Tr}(w)$ of word length $|w|$ is $|w|$-Lipschitz: the product telescoping identity changes one letter at a time, inverses preserve operator-norm distance on unitary matrices, and $|3^{-1} \operatorname{Tr} A| \le \|A\|_{\mathrm{op}}$. Hence a monomial $M = \prod_r \tau_{w_r}$ is Lipschitz with constant at most $\sum_r |w_r|$. Consequently any finite-chart coupling with

$$\mathbf{E}\, d_\infty(X_a, Y) \le D a^\eta$$

implies

$$|\mathbf{E} M(X_a) - \mathbf{E} M(Y)| \le \left(\sum_r |w_r| \right) D a^\eta.$$

Thus the trace-word moment problem may equivalently be attacked by constructing a finite-chart transport coupling. The repository does not currently construct that Yang–Mills coupling.

For the original finite Wilson-loop packet moments, the approximation part is even sharper. Each packet loop is already a single trace word after tree gauge, and a finite mixed packet moment is already a monomial $M = \prod_r \tau_{w_r}$. Thus in the trace-polynomial certificate one may take $Q = M$ and the approximation tail is exactly zero:

$$\|M - Q\|_\infty = 0.$$

Consequently, for packet moments, the remaining chart-law input is precisely the finite-chart transport estimate (or an equivalent direct moment estimate); there is no separate trace-polynomial approximation tail to prove for those tests.

This gives a sharp finite criterion for the state-side packet realization. Let X_a be the convergence-side chart variable and Y the OS chart variable. If for the fixed packet chart there is a coupling with

$$\mathbf{E}\, d_\infty(X_a, Y) \le D_N a^{\eta_N}, \qquad 0 < a \le 1,$$

then every mixed packet moment $M = \prod_r \tau_{w_r}$ satisfies

$$|\mathbf{E}_a M - \mathbf{E}_{\mathrm{OS}} M| \le L(M) D_N a^{\eta_N}, \qquad L(M) = \sum_r |w_r|.$$

Since W_N is finite, the whole packet has the uniform constant

$$C_N = D_N \max_{M \in W_N} L(M), \qquad \alpha_N = \eta_N.$$

Thus the finite packet Schwinger-realization blocker is now equivalent, in this route, to proving the finite-chart Yang–Mills transport estimate above.

For every fixed N, one must prove that the lattice Yang–Mills field and the continuum Schwinger/OS field have J_N-marginals satisfying an explicit estimate

$$|\mathbf{E}_a \Phi(J_N) - \mathbf{E}_{\mathrm{OS}} \Phi(J_N)| \le C_N a^{\alpha_N}$$

or equivalently a comparable chart-law convergence bound. Passing $a \to 0$ would then identify the convergence-side packet law with the OS Schwinger law on W_N.

4.3 Reflection-time intertwiner

The action-side blocker requires a separate compatibility statement. Its finite algebraic core is now discharged. Enlarge Γ_N so it contains every representative path d_j and every translated representative $\alpha_{q_i}(d_j)$ appearing in F_N. In the same maximal-tree gauge, both d_j and $\alpha_{q_i}(d_j)$ are words in the non-tree cycle variables. Define the finite packet map $A_{q_i,N}$ by word substitution,

$$\mathrm{word}(d_j) \mapsto \mathrm{word}(\alpha_{q_i}(d_j)).$$

Then, entrywise on F_N,

$$J_N(\alpha_{q_i}(d_j)) = A_{q_i,N}(J_N(d_j)),$$

again with zero finite-chart error after passing to Wilson-loop traces.

The remaining action-side blocker is analytic rather than finite algebraic. The needed intertwiner is

$$J_N(T_a(q)O) \longrightarrow \alpha_q(J_N(O))$$

for every rational time q appearing in F_N, with convergence in the same finite chart used by the shadow coupling. If proved, this would identify the

convergence-side rational-time action with the OS Euclidean time action on the packet.

The finite transport criterion has the same form as on the state side. On the enlarged chart, each translated representative is an exact word, so each translated packet test $M_q = \prod_r \tau_{w_{q,r}}$ has Lipschitz constant $L(M_q) = \sum_r |w_{q,r}|$. If the convergence-side translated chart variable $X_{a,q}$ and the OS translated chart variable Y_q admit a coupling with

$$\mathbf{E}\, d_\infty(X_{a,q}, Y_q) \le D_{N,q} a^{\eta_{N,q}},$$

then

$$|\mathbf{E}_a M_q - \mathbf{E}_{\mathrm{OS}} M_q| \le L(M_q) D_{N,q} a^{\eta_{N,q}}.$$

For the finite packet, take

$$C_N = \max_{q, M_q \in F_N} L(M_q) D_{N,q}, \qquad \alpha_N = \min_{q, M_q \in F_N} \eta_{N,q}.$$

Thus the action-side finite chart problem is reduced to constructing these translated couplings and matching them with the OS time action.

4.4 Multiscale transport-flow collapse

There is one additional invented reduction layer that may be useful if the bridge influence is naturally organized by scale rather than by one-shell recurrence. Suppose the Ward-projected bridge operator is decomposed into a finite family of scale operators $T_0, \dots, T_{S-1}$ on the same block partition. If the weighted norms satisfy $\prod_s \theta_s < 1$, with

$$\theta_s = \|T_s\|_\infty,$$

then the renormalized operator

$$T_{\mathrm{ren}} = T_{S-1} \cdots T_0$$

is contractive and the screened block norm obeys

$$\sup_x e^{m k_N(x)} b_x \le C a^\beta / (1 - \prod_s \theta_s).$$

This is not proved for Yang–Mills in the repository. It is a proposed multiscale route: a transport-flow collapse principle, if established, would imply the block influence bound, which then feeds the existing screened-norm and leakage certificates.

The contraction test can be made still more local in scale. Suppose one proves logarithmic scale budgets

$$\log \theta_s \le \kappa_s.$$

Then

$$\prod_s \theta_s \le \exp\Big(\sum_s \kappa_s\Big).$$

Therefore a strictly negative total budget

$$K := \sum_s \kappa_s < 0$$

is enough to give

$$\sup_x e^{mk_N(x)} b_x \le \frac{C}{1-e^K} a^\beta.$$

This log-budget version is useful because some scales may expand ($\kappa_s > 0$), provided other Ward-projected scales contract enough to make the total budget negative. The repository proves only this deterministic allocation certificate; the Yang–Mills task is still to derive the actual scale budgets uniformly in the continuum and volume limits.

A useful corollary is a sparse bad-scale test. Suppose G good scales obey

$$\log \theta_s \le -\gamma$$

and B bad scales obey

$$\log \theta_s \le b.$$

Then the total budget is bounded by

$$K \le Bb - G\gamma.$$

Hence $Bb - G\gamma < 0$ gives

$$\sup_x e^{mk_N(x)} b_x \le \frac{C}{1-\exp(Bb-G\gamma)} a^\beta.$$

When $b > 0$, the equivalent sparsity condition is $B/G < \gamma/b$. This is another deterministic certificate only: the analytic work would be to classify the actual Yang–Mills Ward-projected scales into good and bad scales and to prove that the bad scales are sparse enough uniformly in the cutoff limits.

The sparsity criterion can also be localized in scale. Suppose every consecutive window of W scales contains at most M bad scales. Partition a finite scale interval of length S into $q = \lfloor S/W \rfloor$ full windows and a remainder of length $r = S - qW$. Each full window contributes at most

$$Mb - (W-M)\gamma$$

to the log budget, while the remainder is bounded pessimistically by rb. Thus

$$K \le q\big(Mb - (W-M)\gamma\big) + rb.$$

If this upper bound is negative, then

$$\sup_x e^{mk_N(x)} b_x \le \frac{C}{1-\exp(K)} a^\beta.$$

Equivalently, the full-window gain $\Delta = (W - M)\gamma - Mb$ must dominate the leftover expansion rb through $q\Delta > rb$. This prevents all bad scales from clustering in one place; it is still only a deterministic reduction until the Yang–Mills bridge analysis proves the window density estimate.

There is also a phase-robust finite-profile version. Suppose a renormalized scale analysis bounds the logarithmic scale norms by a finite cyclic profile

$$\kappa_0, \dots, \kappa_{P-1}$$

with negative period budget

$$K_P = \sum_{i=0}^{P-1} \kappa_i < 0.$$

For $S = qP + r$ scales, define the worst cyclic remainder

$$R_r = \max_j \sum_{i=0}^{r-1} \kappa_{j+i \bmod P}.$$

Then the total log budget is bounded by

$$K \le qK_P + R_r.$$

If $qK_P + R_r < 0$, the screened norm obeys

$$\sup_x e^{mk_N(x)} b_x \le \frac{C}{1 - \exp(qK_P + R_r)} a^\beta.$$

This certificate is insensitive to the cutoff phase. The hard Yang–Mills content would be proving such a finite Ward-projected scale profile with negative period budget and uniform worst-remainder control.

A coarser route would be a negative average budget with a boundary defect. Suppose the total scale log budget for S scales satisfies

$$K_S \le A - \gamma S, \qquad A \ge 0, \quad \gamma > 0.$$

If $A - \gamma S < 0$, then

$$\sup_x e^{mk_N(x)} b_x \le \frac{C}{1 - \exp(A - \gamma S)} a^\beta.$$

With a safety margin $\delta > 0$, it is enough that $S > (A + \delta)/\gamma$. This would allow a future subadditive or renormalization argument to prove contraction without identifying a periodic profile; the missing analytic content is the actual Yang–Mills proof of the negative average estimate.

A more structural way to prove such a negative average is a telescoping Lyapunov potential. If there are scale potentials Φ_s such that

$$\log \theta_s \le \Phi_s - \Phi_{s+1} - \gamma, \qquad \gamma > 0,$$

then summing over $s = 0, \dots, S-1$ gives

$$K_S \le \Phi_0 - \Phi_S - \gamma S.$$

Consequently, if $\Phi_0 - \Phi_S - \gamma S < 0$, then

$$\sup_x e^{m k_N(x)} b_x \le \frac{C}{1 - \exp(\Phi_0 - \Phi_S - \gamma S)} a^\beta.$$

This would let a future proof trade local scale expansion against decreases of a Ward-projected energy potential. The repository proves only the telescoping implication, not the Yang–Mills potential.

The Lyapunov check can itself be made finite. Suppose the possible Ward-projected scale regimes form a finite directed graph and each allowed transition $i \to j$ has log-norm bound $\log \theta_{ij}$. If there are potentials Φ_i on the finite state set such that

$$\log \theta_{ij} \le \Phi_i - \Phi_j - \gamma$$

for every allowed transition, then every admissible scale path of length S satisfies

$$K_S \le \operatorname{osc}(\Phi) - \gamma S.$$

Thus $\operatorname{osc}(\Phi) - \gamma S < 0$ implies

$$\sup_x e^{m k_N(x)} b_x \le \frac{C}{1 - \exp(\operatorname{osc}(\Phi) - \gamma S)} a^\beta.$$

This is a finite Bellman/Lyapunov certificate. The hard analytic step is to code the actual Yang–Mills bridge scales by finitely many regimes and verify the transition inequalities.

Equivalently, one may check negative cycle means on the same finite directed graph. If every directed cycle C satisfies

$$\sum_{e \in C} w_e \le -\gamma |C|,$$

then any scale path of length S decomposes into erased cycles plus a simple residual path. The residual has at most $|V| - 1$ edges, so with $A \le (|V| - 1) \max_e w_e^+$ one obtains

$$K_S \le A - \gamma S.$$

Thus, for $A - \gamma S < 0$, the negative-average certificate gives the same screened norm bound. This cycle-mean formulation turns the finite-state Lyapunov search into finitely many graph inequalities.

The converse finite-graph construction is also deterministic. Shift edge weights by γ, writing $u_e = w_e + \gamma$. If every directed cycle has nonpositive u-weight, define d_j to be the maximum u-weight of a path from a zero-weight super-source to j. The maximum is finite, and every edge $i \to j$ satisfies $d_j \ge d_i + u_{ij}$. Setting $\Phi_j = -d_j$ gives

$$w_{ij} \le \Phi_i - \Phi_j - \gamma.$$

Thus negative cycle means are equivalent, at the finite graph level, to the existence of a Bellman/Lyapunov potential. The remaining Yang–Mills work is to produce the finite scale-state graph and prove its cycle inequalities.

The cycle inequalities can be checked by a finite maximum-cycle-mean certificate. Let $D_k(v)$ be the maximum weight of a length-k path ending at v, with $D_0(v) = 0$. Karp's formula gives

$$\lambda_{\max} = \max_v \min_{0 \le k < n} \frac{D_n(v) - D_k(v)}{n - k}.$$

If $\lambda_{\max} \le -\gamma$, then every directed cycle has mean at most $-\gamma$, so the cycle-mean and Lyapunov certificates apply. This reduces the finite graph part of the contraction proof to a dynamic-programming table.

Combining the finite graph certificates gives a direct screened-norm package. If the finite Ward-projected scale graph has edge log weights w_e, Karp certifies $\lambda_{\max} \le -\gamma$, and paths have length S with transient defect $A \le (|V|-1)\max_e w_e^+$ satisfying $A - \gamma S < 0$, then

$$C_{\mathrm{SB}} \le \frac{C}{1 - \exp(A - \gamma S)}.$$

Thus a proved Yang–Mills finite scale graph with negative maximum cycle mean would feed the screened bridge norm route directly.

The same finite graph package is robust under interval estimates. If each edge log norm is known only through $l_e \le w_e \le u_e$, path weights and cycle means are monotone in the edge weights. It is therefore enough to run the maximum-cycle-mean and screened-norm certificate on the upper-endpoint graph $w_e = u_e$. If that graph contracts, every graph in the certified interval box contracts with the same screened norm constant.

Before a coarse graph is used, duplicate coarse edge declarations must be made canonical. If several certified upper bounds $U_1, \dots, U_r$ are declared for the same ordered coarse edge $A \to B$, set

$$U_*(A, B) := \min_{1 \le q \le r} U_q.$$

Since the true edge log norm is bounded by each U_q, it is also bounded by $U_*(A, B)$. Thus duplicate declarations may be replaced by a single input-order-independent upper-envelope edge, and the Karp test is run on this canonical graph. This is a certificate-management lemma only; the Yang–Mills work remains to supply the certified edge bounds themselves.

There is also a coarse-grained version for many raw scale regimes. Let π map raw regimes to finitely many coarse regimes, and suppose every raw edge $i \to j$ satisfies the upper-envelope estimate

$$w_{\mathrm{raw}}(i, j) \le W_{\mathrm{coarse}}(\pi(i), \pi(j)).$$

Then each raw path maps edge-by-edge to a coarse path, and summing the envelope inequalities gives

$$K_{\mathrm{raw}}(\mathrm{path}) \le K_{\mathrm{coarse}}(\pi(\mathrm{path})).$$

Consequently a contracting coarse graph, certified for example by the Karp maximum-cycle-mean package above, gives the same screened-norm constant for all raw paths dominated by the envelope. This is only a deterministic coarse-graining reduction; the Yang–Mills input would be the actual construction of the coarse regimes, the raw-to-coarse edge envelopes, and the negative coarse maximum cycle mean.

The raw-to-coarse envelopes can themselves be reduced to endpoint checks when the raw edge estimates are organized by parameter boxes. Suppose on a rectangular box $a_r \le x_r \le b_r$ an edge log norm is bounded by an affine upper model

$$w_e(x) \le c_e + \sum_r \alpha_{e,r} x_r.$$

Then the maximum of the right side is attained at the signed endpoint vertex, so

$$w_e(x) \le U_e := c_e + \sum_{\alpha_{e,r} \ge 0} \alpha_{e,r} b_r + \sum_{\alpha_{e,r} < 0} \alpha_{e,r} a_r.$$

If these finitely many endpoint values satisfy

$$U_e \le W_{\mathrm{coarse}}(\pi(i_e), \pi(j_e)),$$

then every raw graph in the parameter boxes is edgewise dominated by the coarse envelope, and the previous coarse-graph certificate applies. This turns a continuum family of raw estimates into a finite endpoint computation, but the Yang–Mills work remains to derive the affine upper models and prove the contracting coarse graph uniformly.

A still closer finite certificate starts from nonlinear edge estimates with derivative bounds. Let $f_e(x)$ be the raw edge log-norm bound on the same box, let x_0 be a base point in the box, and suppose the Hessian is bounded in the ℓ^1 operator sense by H_e. The Taylor data must use one common finite coordinate set:

$$\mathrm{dom}(x_0) = \mathrm{dom}(\nabla f_e) = \mathrm{dom}\{[a_r, b_r]\}.$$

The base point must also satisfy $x_{0,r} \in [a_r, b_r]$ for every coordinate, and the Hessian and any explicit remainder budget must be nonnegative. These coordinate checks prevent a chart from covering a parameter direction that the Taylor polynomial silently ignores, or from adding a gradient coordinate that has no declared interval. In addition, the finite certificate uses the Hessian remainder floor unless a separate sharper-remainder theorem is supplied: a declared explicit remainder may not be smaller than

$$E_H = \frac{1}{2} H_e R_e^2.$$

This prevents a raw-edge witness from tightening its edge envelope by merely declaring a smaller Taylor error term. With

$$R_e = \sum_r \max\{|a_r - x_{0,r}|,\ |b_r - x_{0,r}|\},$$

Taylor's theorem gives

$$f_e(x) \le f_e(x_0) + \nabla f_e(x_0) \cdot (x - x_0) + \frac{1}{2} H_e R_e^2$$

throughout the box. The right side is an affine upper model, so the box-envelope certificate applies to its signed endpoint values. Hence uniform derivative and Hessian bounds for the raw Ward-projected edge norms, together with a contracting coarse envelope, would give the same screened-norm constant for every raw parameter realization. The deterministic theorem is proved here; the Yang–Mills content is still the actual derivative/Hessian estimate and the negative coarse cycle-mean bound.

The same argument localizes over a finite Taylor atlas. If the edge parameter domain is covered by finitely many subboxes $B_{e,c}$ and each subbox has its own Taylor/Hessian enclosure, the preceding endpoint calculation gives a chart bound $U_{e,c}$. Since every admissible parameter lies in at least one chart,

$$w_e(x) \le U_e := \max_c U_{e,c}.$$

Thus a finite family of local Taylor estimates gives a single certified edge upper bound. If these atlas bounds satisfy the coarse envelope inequalities, the coarse graph again controls all raw paths. This permits highly nonlinear raw estimates to be handled by finite subdivision, while leaving the true Yang–Mills task as proving the chart cover, the local derivative bounds, and the negative coarse maximum cycle mean.

For rectangular parameter domains the finite atlas cover is itself certifiable. If each coordinate interval $[a_r, b_r]$ is partitioned by a finite grid

$$a_r = t_{r,0} \le t_{r,1} \le \cdots \le t_{r,n_r} = b_r,$$

then the product cells

$$\prod_r [t_{r,k_r}, t_{r,k_r+1}]$$

form a finite closed cover of $\prod_r [a_r, b_r]$, with $\prod_r n_r$ cells. Indeed, each coordinate of any point in the box lies in one adjacent grid interval, and the product of those intervals contains the point. Thus a Taylor atlas can be generated from one-dimensional endpoint data. This only proves coverage; the Yang–Mills estimates on each cell remain the analytic input.

More generally, one need not start from a product grid. Given finitely many rectangular Taylor charts, collect every chart endpoint in each coordinate and refine the full parameter box into elementary product cells. If every such cell is contained in at least one chart, then the charts cover the whole parameter domain. This endpoint-refined cover test prevents a raw-edge witness from leaving unverified holes in parameter space; the analytic Taylor bounds on the covering charts remain separate Yang–Mills inputs.

There is also a finite termination theorem for this atlas search. Suppose the true raw Ward-projected edge functions are C^2 on a compact rectangular parameter domain of ℓ^1 diameter D, with a uniform ℓ^1 Hessian bound H. Suppose

further that the proposed coarse graph has strict raw-to-coarse slack s and strict Karp margin m_{cyc} after reserving the desired raw and coarse error budgets:

$$s_* := s - \varepsilon_{\mathrm{raw}} - \varepsilon_{\mathrm{coarse}} > 0, \qquad m_* := m_{\mathrm{cyc}} - \varepsilon_{\mathrm{coarse}} > 0.$$

On any Taylor chart of ℓ^1 radius R, the Hessian floor is $\frac{1}{2}HR^2$. Hence every chart radius satisfying

$$R \le \sqrt{2s_*/H}$$

when $H > 0$ leaves the Taylor remainder inside the remaining slack; when $H = 0$, one chart per edge has zero Taylor remainder. Since compact rectangular boxes admit finite product grids of arbitrarily small radius, a sufficiently fine grid produces a finite Taylor-atlas certificate. A crude bound is EN^d charts for E raw edges and d parameters, with $N \ge D/\sqrt{2s_*/H}$ in the case $H > 0$. Thus, once strict margins and uniform Hessian bounds are proved for the actual Yang–Mills raw edge functions, the remaining certificate search is guaranteed to terminate finitely rather than continue indefinitely. The theorem does not prove those strict analytic margins; it proves that they are enough.

Strict margins also remove dependence on informal real-number data. Suppose a finite real-valued Taylor-atlas/Karp certificate has raw-to-coarse slack $s > 0$ and cycle margin $m > 0$, and suppose at most M scalar roundoff errors can accumulate in any checked inequality. If every scalar certificate entry is rounded outward to a rational with denominator Q, then each scalar changes by at most $1/Q$ and every inequality worsens by at most M/Q. Hence, whenever

$$\frac{M}{Q} < \min\{s, m\},$$

the rounded rational certificate still has positive slack and positive cycle margin. Thus any strict real certificate can be converted into exact rational verifier data by taking $Q > M/\min\{s, m\}$. This is a credibility point for the finite route: once the analytic inequalities are strict, the final certificate can be made discrete, reproducible, and independent of floating-point trust.

The same rounding argument is uniform in a cutoff family. If every cutoff certificate has raw-to-coarse slack at least $s_0 > 0$, cycle margin at least $m_0 > 0$, at most M accumulated scalar roundoff errors per checked inequality, and common screened constants C_{SB} and m, then one denominator

$$Q > M/\min\{s_0, m_0\}$$

rounds all cutoff certificates to rational data with margins at least $s_0 - M/Q$ and $m_0 - M/Q$. The screened-norm-to-leakage and screened-norm-to-effective-action certificates then give cutoff-independent DLCR constants. Thus cutoff-uniform strict margins would produce a uniformly machine-checkable rational certificate family, not merely isolated finite witnesses.

One can also certify cutoff intervals from finitely many cutoff nodes. Suppose the cutoff range is covered by balls of radius ρ around verified node certificates, each with slack at least $s > 0$ and cycle margin at least $m > 0$. If every scalar

entering the certificate varies with cutoff at Lipschitz rate at most L, then inside such a ball all checked inequalities can worsen by at most

$$\Delta = L\rho.$$

When $\Delta < \min\{s, m\}$, the whole ball remains certified with slack at least $s - \Delta$ and cycle margin at least $m - \Delta$. Thus finitely many verified cutoff nodes can certify an entire cutoff interval, provided the cutoff Lipschitz estimates fit inside the strict margins.

Combining the preceding strict-margin reductions gives a compact analytic seed criterion. Suppose the actual Yang–Mills raw Ward-projected edge functions are defined on compact parameter boxes, have uniform C^2 bounds with Hessian constant H, and admit strict raw-to-coarse slack $s > 0$ and strict Karp cycle margin $m > 0$. Reserve raw and coarse error budgets $\varepsilon_{\mathrm{raw}}, \varepsilon_{\mathrm{coarse}}$ and write

$$s_* := s - \varepsilon_{\mathrm{raw}} - \varepsilon_{\mathrm{coarse}}, \qquad m_* := m - \varepsilon_{\mathrm{coarse}}.$$

Assume $s_*, m_* > 0$. If cutoff variation on balls of radius ρ is bounded by $\Delta = L\rho$, and if at most M scalar roundoff errors accumulate in each checked inequality, then the single condition

$$s_* > \Delta + M/Q, \qquad m_* > \Delta + M/Q$$

promotes the analytic seed to a finite, exact, cutoff-uniform certificate program. Indeed, the Taylor-atlas termination theorem gives a finite chart mesh with $R \le \sqrt{2s_*/H}$ when $H > 0$; the cutoff stability theorem extends verified nodes to the covered cutoff intervals with margins $s_* - \Delta$ and $m_* - \Delta$; and strict-margin rationalization rounds the finite witnesses outward to denominator Q while preserving exact positive margins $s_* - \Delta - M/Q$ and $m_* - \Delta - M/Q$. Common screened constants then feed the deterministic DLCR output certificates uniformly. This is the current most compact remaining target: prove this strict analytic seed for the genuine Yang–Mills raw edges. The reduction does not construct the seed, but it shows that such a seed is sufficient for a finite reproducible certificate rather than an open-ended search.

There is also a finite validation principle for approximate analytic evidence. Suppose a proposed Yang–Mills seed is accompanied by rigorous interval radii. For every upper-bound quantity, such as a raw Lipschitz constant, Hessian constant, cutoff Lipschitz constant, or screened norm constant, replace the nominal value $\widehat{U}$ by

$$U_{\mathrm{cert}} := \widehat{U} + \mathrm{err}(U).$$

For every lower-bound quantity, such as raw-to-coarse slack, Karp cycle margin, or screening rate, replace the nominal value $\widehat{\ell}$ by

$$\ell_{\mathrm{cert}} := \widehat{\ell} - \mathrm{err}(\ell).$$

If the certified slack and cycle margin are positive and satisfy the strict program inequalities

$$s_{\mathrm{cert}} - \varepsilon_{\mathrm{raw}} - \varepsilon_{\mathrm{coarse}} > (L_{\mathrm{cert}}\rho) + M/Q, \qquad m_{\mathrm{cert}} - \varepsilon_{\mathrm{coarse}} > (L_{\mathrm{cert}}\rho) + M/Q,$$

then the interval evidence validates the strict analytic seed and hence promotes to the same finite exact cutoff-uniform certificate program. This puts a concrete target on the remaining computational/analytic work: produce rigorous intervals with strict surplus. The validation theorem does not create those intervals, but it proves that sufficiently sharp interval/Taylor estimates would be accepted without any floating-point trust.

The interval validation search itself is finite under a standard convergence model. Let $\eta > 0$ be the true strict surplus in the seed program after all reserved losses have been subtracted. Suppose a rigorous interval/Taylor refinement scheme has aggregate validation error bounded on mesh radius h by

$$\mathrm{Err}(h) \le Ah^p, \qquad A \ge 0, \quad p > 0.$$

If $A = 0$, any finite covering mesh suffices. If $A > 0$, every mesh satisfying

$$h \le \left(\frac{\eta}{2A}\right)^{1/p}$$

leaves certified surplus at least $\eta/2$, so the interval analytic seed validation theorem applies. On a compact rectangular box of ℓ^1 diameter D and dimension d, it is enough to take

$$N \ge D(2A/\eta)^{1/p}$$

subdivisions per coordinate, giving at most EN^d raw-edge charts for E raw edges. Thus, once the true strict surplus and a convergent interval scheme are established, the validation search cannot continue forever: there is an explicit finite refinement threshold.

The strict surplus η can itself be certified by finite residuals. Suppose each raw edge has a certified upper bound $U_{\mathrm{raw}}(e)$ and a coarse envelope $W_{\mathrm{coarse}}(e)$ with minimum residual

$$s_{\min} := \min_e \{W_{\mathrm{coarse}}(e) - U_{\mathrm{raw}}(e)\} > 0.$$

Suppose also that the coarse graph has a finite Lyapunov potential Φ and drift $\gamma > 0$ with residuals

$$U_{ij} \le \Phi_i - \Phi_j - \gamma - r_{ij}, \qquad r_{\min} := \min_{i \to j} r_{ij} > 0.$$

Summing the Lyapunov inequality around a directed cycle telescopes the potential terms, so every coarse cycle has mean at most $-\gamma - r_{\min}$. After reserving raw/coarse errors, cutoff variation Δ, and rational roundoff M/Q, the simultaneous strict program surplus is

$$\eta = \min\{s_{\min} - \varepsilon_{\mathrm{raw}} - \varepsilon_{\mathrm{coarse}}, r_{\min} - \varepsilon_{\mathrm{coarse}}\} - \Delta - M/Q.$$

Whenever this number is positive, the finite residual witness supplies the exact η required by the interval-refinement termination theorem. Thus the remaining positivity target can be checked by finite raw slack residuals and a finite Lyapunov/Karp drift residual.

The Lyapunov drift residual need not be guessed separately. If the finite coarse graph has maximum cycle mean $\lambda_{\max}$ and

$$\kappa := -\gamma - \lambda_{\max} > 0,$$

then every directed cycle has mean at most $-\gamma-\kappa$. Applying the finite negative-cycle-to-potential certificate at the strengthened drift $\gamma + \kappa$ gives a potential satisfying

$$U_{ij} \le \Phi_i - \Phi_j - \gamma - \kappa.$$

Thus the Karp margin κ is itself a Lyapunov drift residual. Combining this with the raw slack residual gives

$$\eta = \min\{s_{\min} - \varepsilon_{\text{raw}} - \varepsilon_{\text{coarse}}, \kappa - \varepsilon_{\text{coarse}}\} - \Delta - M/Q.$$

This makes the upstream finite target even sharper: prove strict raw slack and a strict Karp maximum-cycle-mean margin on the genuine Yang–Mills coarse scale graph.

The same Karp-surplus check is stable under finite interval uncertainty. If a raw upper bound, coarse raw envelope, or coarse graph edge weight is known with rigorous interval radius, replace it by its upper endpoint:

$$U^+_{\text{raw}} = \widehat{U}_{\text{raw}} + \text{err}_{\text{raw}}, \qquad W^+ = \widehat{W} + \text{err}_W, \qquad w^+_{ij} = \widehat{w}_{ij} + \text{err}_{ij}\,.$$

If the endpoint data satisfy

$$s^+_{\min} := \min_e\{W^+(e) - U^+_{\text{raw}}(e)\} > 0, \qquad \kappa^+ := -\gamma - \lambda_{\max}(w^+) > 0,$$

then monotonicity of edge upper bounds and cycle means shows that every realization inside the intervals is dominated by the endpoint certificate. The certified surplus is therefore

$$\eta = \min\{s^+_{\min} - \varepsilon_{\text{raw}} - \varepsilon_{\text{coarse}}, \kappa^+ - \varepsilon_{\text{coarse}}\} - \Delta - M/Q.$$

So rigorous interval edge evidence with positive worst-case raw slack and positive worst-case Karp margin already supplies the surplus needed by the finite validation chain.

Combining this endpoint surplus with a convergence rate gives an explicit finite validation program. If the interval/Taylor refinement error satisfies

$$\text{Err}(h) \le Ah^p$$

and the endpoint certificate gives surplus $\eta > 0$, then every mesh with

$$h \le (\eta/2A)^{1/p}$$

leaves certified surplus at least $\eta/2$. On a compact d-dimensional parameter box of ℓ^1 diameter D, it suffices to choose

$$N \ge D(2A/\eta)^{1/p}$$

subdivisions per coordinate, giving at most EN^d edge charts. Thus a worst-case endpoint raw-slack/Karp witness plus a quantitative enclosure convergence rate yields a concrete finite mesh threshold for validating the Yang–Mills seed.

This can be packaged as a single endpoint-witness program. One finite object containing endpoint raw slack, endpoint Karp margin, enclosure convergence rate, rational denominator, screened bridge norm C_{SB}, leakage bridge constant K, packet Lipschitz constant L_F, chart anchor count Q_N, and rooting constant K_A determines all deterministic DLCR outputs:

$$\ell_x \le K L_F C_{\mathrm{SB}} e^{-m k_N(x)}, \qquad B_N = K_A Q_N C_{\mathrm{SB}},$$

and hence the effective-action transport constant

$$W_1 \le R\sqrt{K_A Q_N C_{\mathrm{SB}}}\, a^{\beta/2}.$$

Together with the mesh bound above, this shows that a single genuine Yang–Mills endpoint interval witness with common screened/rooting constants would drive the full finite DLCR certificate program. It is still a conditional reduction, not a construction of that witness.

The witness can be made into a single finite artifact. A package with schema

`yangmills.dlcr.endpoint_interval_witness.v1`

and status `PROVED` must carry the action hash, endpoint raw/coarse intervals, reserve budgets, refinement model, parameter-domain data, scalar complexity, and screened constants. If the action hash matches the expected Yang–Mills action, the verifier extracts all fields and applies the endpoint interval witness program above. Thus the final deterministic target is one hash-matched endpoint interval witness artifact; producing that artifact would supply the finite mesh, chart bound, local leakage constant, effective-action constant, and transport constant.

The raw Taylor-atlas target and the endpoint interval target are in fact one finite pipeline. Suppose a package with schema

`yangmills.dlcr.raw_edge_taylor_atlas.v1`

has status `PROVED` and its action hash matches the ambient Yang–Mills action. The raw-edge schema certificate extracts certified raw upper bounds U_e, canonical coarse envelopes W_e, canonical coarse transition weights w_f, and the parameter box. Given nonnegative interval radii $r_{\mathrm{raw}}, r_W, r_w$, reserve budgets, a refinement model $\mathrm{Err}(h) \le Ah^p$, denominator Q, scalar complexity M, and common screened constants, define an endpoint witness by using the interval centers U_e, W_e, w_f and radii $r_{\mathrm{raw}}, r_W, r_w$. The endpoint verifier then checks the worst-case upper endpoints

$$U_e^+ = U_e + r_{\mathrm{raw}}, \qquad W_e^+ = W_e + r_W, \qquad w_f^+ = w_f + r_w$$

and computes

$$\eta = \min\{s_{\min}^+ - \varepsilon_{\mathrm{raw}} - \varepsilon_{\mathrm{coarse}}, \kappa^+ - \varepsilon_{\mathrm{coarse}}\} - \Delta - \frac{M}{Q}.$$

If this number is positive, the endpoint interval witness schema theorem applies and produces the finite DLCR program constants. Thus a proved raw Taylor-atlas/Karp package plus explicit radii and screened constants automatically constructs the endpoint interval witness; no independent endpoint artifact remains to be invented. If the radii or reserves destroy the raw slack or Karp margin, the verifier rejects the conversion.

The local C^2 enclosure target can itself be generated by a finite sectorized polymer majorant package. On a sector chart, suppose the raw edge log norm is controlled by a value center v_0 with radius r_v, gradient centers g_i with radii r_i, and a nonnegative Hessian ℓ^1 majorant H. The corresponding local enclosure is simply

$$[v_0 - r_v, v_0 + r_v], \qquad [g_i - r_i, g_i + r_i], \qquad [0, H].$$

Thus the next analytic gate can be formulated without hidden functional analysis: prove a finite sector decomposition of the Yang–Mills polymer activities and rigorous value/gradient/Hessian majorants on each sector. The downstream verifier still checks support, cover, domination, and Karp negativity, so sector majorants do not assume the screened norm or the mass gap.

The jet-atlas target itself can be fed by ordinary local C^2 enclosure data. Suppose a proved local enclosure package gives, on each raw edge chart, a basepoint value interval $[v^-, v^+]$, gradient intervals $[g_i^-, g_i^+]$, and a nonnegative Hessian ℓ^1 interval $[H^-, H^+]$. The canonical jet data are

$$v_0 = \frac{v^- + v^+}{2}, \quad r_v = \frac{v^+ - v^-}{2}, \qquad H_0 = \frac{H^- + H^+}{2}, \quad r_H = \frac{H^+ - H^-}{2},$$

with the same gradient intervals, base point, and parameter box. Thus the Ward-projected jet atlas is no longer a mysterious object: it is exactly the finite packaging of rigorous local value, gradient, and Hessian interval estimates, plus raw/coarse support and Karp data. The difficult Yang–Mills work is to prove those local C^2 enclosures and keep their margins uniform through the cutoff and volume limits.

There is now one further finite synthesis layer before the raw-edge package. A *Ward-projected jet atlas* records, on each raw edge chart, a parameter box, a base point, an interval value at the base point, interval gradients, and an interval Hessian bound. From such data one constructs an ordinary Taylor-atlas chart by replacing the value by its upper endpoint, using the gradient interval midpoint as the affine coefficient, and adding the explicit gradient uncertainty budget

$$E_\nabla = \sum_i \frac{g_i^+ - g_i^-}{2} R_i$$

to the base value, where R_i is the coordinate radius of the chart box. For every admissible gradient $g_i = g_i^0 + e_i$ and every point in the box,

$$\sum_i e_i (x_i - x_{0,i}) \le \sum_i |e_i| R_i \le E_\nabla.$$

The Hessian bound is replaced by its upper endpoint, so Taylor's theorem gives a valid outward edge envelope. The raw-edge schema verifier is then run on the synthesized package and still checks support, cover, domination, and Karp negativity. Hence a proved Ward-projected jet atlas with matching action hash is enough to produce the raw Taylor-atlas/Karp evidence; the remaining Yang–Mills analytic gate becomes the construction of those interval jets and coarse Karp data, rather than a separate endpoint-witness problem.

A useful refinement certificate lets one verify estimates on a finer atlas and then collapse them to coarser analytic regimes. Let fine charts B_c refine coarse charts B_C through an assignment $\rho(c) = C$, with $B_c \subseteq B_C$, and suppose the Taylor-atlas calculation has certified

$$w_e(x) \le U_c, \qquad x \in B_c.$$

Define the coarse envelope

$$V_C := \max_{\rho(c)=C} U_c.$$

If x lies in the portion of B_C covered by the assigned fine charts, then $x \in B_c$ for some c with $\rho(c) = C$, hence

$$w_e(x) \le U_c \le V_C.$$

Thus the fine atlas may be replaced by coarse envelope values V_C, and $\max_C V_C = \max_c U_c$ preserves the global edge bound used by the coarse-graph certificate. This makes adaptive subdivision compatible with a small finite state graph, but it does not prove the fine Taylor estimates or the refinement relation for the actual Yang–Mills regimes.

The finite data needed from Yang–Mills can therefore be packaged as a raw-edge evidence schema. Such a package must give the parameter domain, the finite raw edges, Taylor charts with base values, gradients, and Hessian bounds, optional grid cover data, a raw-to-coarse state map, coarse edge envelopes, and the constants used in the Karp cycle-mean test. If the package has status `PROVED`, all chart boxes lie inside the parameter domain, declared grid cells match the Taylor charts, the raw-to-coarse state partition is supported exactly on the raw edge endpoints, every endpoint image is a vertex of the canonical coarse graph, every raw edge maps to a coarse edge, every Taylor-atlas edge envelope is dominated by its coarse edge, and the coarse Karp margin is positive, then the existing Taylor-atlas coarse graph certificate gives the screened bridge norm. This is a schema soundness theorem: it identifies the exact finite proof artifact that would discharge the edge gate, while the construction of that artifact from the Yang–Mills measure is still open.

The finite witness also canonically extracts explicit edge-error budgets. Let

$$s_{\min} = \min_e \{W_{\mathrm{coarse}}(\pi(e)) - U_e\}$$

be the minimum slack between the certified raw upper bound U_e and its canonical coarse envelope, and let

$$m_{\mathrm{cyc}} = -\gamma - \lambda_{\max}$$

be the coarse Karp cycle margin. If later analytic estimates increase every raw edge bound by at most $\varepsilon_{\mathrm{raw}}$ and every coarse edge bound by at most $\varepsilon_{\mathrm{coarse}}$, then the same certificate survives provided

$$\varepsilon_{\mathrm{raw}} + \varepsilon_{\mathrm{coarse}} \le s_{\min}, \qquad \varepsilon_{\mathrm{coarse}} \le m_{\mathrm{cyc}}.$$

The first inequality preserves every raw-to-coarse envelope, and the second uses monotonicity of cycle means under edge perturbations. This turns future Yang–Mills edge estimates into a margin-fitting problem with explicit error budgets. In particular, any passing raw-edge evidence package yields the finite acceptance region

$$\{(\varepsilon_{\mathrm{raw}}, \varepsilon_{\mathrm{coarse}}) : \varepsilon_{\mathrm{raw}} + \varepsilon_{\mathrm{coarse}} \le s_{\min}, \quad \varepsilon_{\mathrm{coarse}} \le m_{\mathrm{cyc}}\}.$$

The verifier reports this region, together with a balanced admissible point obtained by taking

$$\varepsilon_{\mathrm{coarse}} = \min\{s_{\min}/2, m_{\mathrm{cyc}}\}, \qquad \varepsilon_{\mathrm{raw}} = s_{\min} - \varepsilon_{\mathrm{coarse}}.$$

Thus the remaining analytic construction need not hit exact edge values: it only has to produce raw and coarse error bars fitting inside this certified finite region.

The same raw-edge package is stable under refinement. If a refined package shares the same schema, parameter domain, and state partition, and if its screened bridge constant does not increase while its coarse Karp margin does not decrease, then it is a safe replacement for the coarse package. That monotonicity is useful because adaptive subdivision should never make a valid finite witness worse, but it still does not construct the Yang–Mills witness.

Once the finite witness is also tied to the ambient action by the provenance hash, the deterministic output layer is completely compositional. A passing raw-edge package gives a screened bridge constant C_{SB}. The screened-norm-to-leakage certificate gives

$$\ell_x(a, L) \le K L_F C_{\mathrm{SB}} e^{-m k_N(x)}$$

for any supplied curvature-to-leakage constant K, packet Lipschitz bound L_F, and screening rate $m > 0$. The screened-norm-to-effective-action certificate gives, for Q_N chart anchors and bridge-to-action constant K_A,

$$\|S_a - S\|_\infty \le K_A Q_N C_{\mathrm{SB}} a^\beta, \qquad W_1 \le R\sqrt{K_A Q_N C_{\mathrm{SB}}}\, a^{\beta/2}.$$

Thus a single hash-matched raw Taylor-atlas/Karp evidence package would simultaneously supply the screened norm, local exponential leakage constant, and finite-chart effective-action transport constants. This is the most compact finite certificate target left in the route. The analytic burden is not hidden: one must still construct the genuine Yang–Mills raw-edge witness and prove the curvature bridge/rooting normalizations and cutoff-uniformity assumptions.

4.5 No current leakage

Finally, the finite charts must be compatible with the countable exhaustion. The fixed-packet cylindrical part is discharged. If Γ_N contains all edges used by a packet observable O, then $O = O_N \circ \pi_N$ for the projection onto the Γ_N edge variables. Changing all complement edges changes O by exactly 0.

The remaining leakage problem is uniform and analytic. One still needs an estimate compatible with the continuum and countable-exhaustion limits: The proposed leakage estimate is

$$\lim_{N\to\infty} \limsup_{a\to 0, L\to\infty} \epsilon_N(a, L) = 0,$$

where ϵ_N bounds the contribution of current outside the packet chart. This is the missing uniformity needed to connect the finite current comparison to the existing countable localization layer.

Equivalently, it is enough to construct a deterministic tail schedule $\rho_N \to 0$ such that

$$\limsup_{a\to 0, L\to\infty} \epsilon_N(a, L) \le \rho_N.$$

Then

$$0 \le \limsup_{a\to 0, L\to\infty} \epsilon_N(a, L) \le \rho_N$$

for each N, and sending $N \to \infty$ gives the required no-leakage limit. The new finite content is only this tail-schedule implication; the analytic Yang–Mills localization estimate producing such a ρ_N remains open.

A standard exponential localization estimate would produce the required schedule explicitly. If the packet exhaustion has radii $R_N \to \infty$ and

$$\limsup_{a\to 0, L\to\infty} \epsilon_N(a, L) \le Ce^{-mR_N}, \qquad C \ge 0, \quad m > 0,$$

then one may take

$$\rho_N = Ce^{-mR_N}.$$

Since $R_N \to \infty$, $\rho_N \to 0$, and the previous criterion applies. This again only converts a localization theorem into the DLCR leakage artifact; it does not prove the Yang–Mills localization estimate.

There is also a summable-shell route. Suppose the complement of the N-th packet chart is decomposed into shells with nonnegative leakage bounds s_k and

$$\sum_{k\ge 1} s_k < \infty.$$

If the continuum and infinite-volume limsup leakage satisfies

$$\limsup_{a\to 0, L\to\infty} \epsilon_N(a, L) \le \sum_{k\ge N} s_k,$$

then one may take

$$\rho_N = \sum_{k \geq N} s_k.$$

The tails of a convergent nonnegative series tend to zero, so this again gives a deterministic no-leakage schedule. As with the exponential route, this is only a certificate conversion: the Yang–Mills input would be the actual uniform summable shell estimate.

A common way to produce such shell bounds is polynomial shell growth combined with exponential local decay. If the k-th complement shell has at most $V(k+1)^p$ atoms, with $V \geq 0$ and $p \geq 0$, and each atom contributes at most Ce^{-mk} to the continuum/infinite-volume limsup leakage, with $C \geq 0$ and $m > 0$, then

$$s_k = VC(k+1)^p e^{-mk}$$

is summable. Indeed, for $p > 0$ the elementary bound

$$x^p \leq \left(\frac{p}{e\alpha}\right)^p e^{\alpha x}, \qquad \alpha = m/2,$$

gives

$$(k+1)^p e^{-mk} \leq A_{p,m} e^{-mk/2}, \qquad A_{p,m} = e^{m/2} \left(\frac{2p}{em}\right)^p,$$

while for $p = 0$ one takes $A_{0,m} = 1$. Therefore

$$\sum_{k \geq N} s_k \leq \frac{VCA_{p,m} e^{-mN/2}}{1 - e^{-m/2}}.$$

This proves summability with explicit constants. The remaining Yang–Mills work would be to prove the actual polynomial shell-volume estimate and the uniform exponential local leakage estimate.

For finite lattice packet supports, the polynomial shell-volume part is itself elementary. If a packet support has Q anchors in $\mathbb{Z}^d$, the ℓ^∞ shell of radius k around one anchor has exact cardinality

$$1 \quad (k = 0), \qquad (2k+1)^d - (2k-1)^d \quad (k \geq 1).$$

For $k \geq 1$, the mean value theorem gives

$$(2k+1)^d - (2k-1)^d \leq 2d(2k+1)^{d-1} \leq 2d\, 3^{d-1}(k+1)^{d-1}.$$

The same bound trivially covers $k = 0$, and a union bound over the Q anchors gives

$$|\mathrm{shell}_k| \leq Q\, 2d\, 3^{d-1}(k+1)^{d-1}.$$

Thus the leakage side has been reduced further: for the lattice shells one may take $V = Q\, 2d\, 3^{d-1}$ and $p = d - 1$. What remains is the genuine Yang–Mills estimate that shell leakage decays like Ce^{-mk} with constants uniform in the continuum and infinite-volume limits.

The exact local estimate needed can be stated atomwise. Let S_N be the packet support and let

$$k_N(x) = \operatorname{dist}_\infty(x, S_N)$$

be the shell index of a lattice atom x. If $\ell_x(a, L) \geq 0$ bounds the change in the packet chart caused by the outside atom x, then the leakage is controlled by

$$\epsilon_N(a, L) \leq \sum_{k \geq N} \sum_{x:\, k_N(x)=k} \ell_x(a, L).$$

Thus a uniform local exponential estimate

$$\limsup_{a \to 0, L \to \infty} \ell_x(a, L) \leq C e^{-m k_N(x)}, \qquad C < \infty, \quad m > 0,$$

would give

$$\limsup_{a \to 0, L \to \infty} \epsilon_N(a, L) \leq \frac{Q\, 2d\, 3^{d-1} C A_{d-1,m} e^{-mN/2}}{1 - e^{-m/2}},$$

which is the required deterministic leakage tail schedule. The repository now records this as a conditional leakage evidence artifact: all finite counting and summation constants are present, but the artifact is not promoted because the Yang–Mills local exponential leakage estimate itself is still missing. The verifier now requires this completely established estimate as its own action-hash-matched artifact, with schema

`yangmills.dlcr.local_exponential_leakage.v1`.

Thus even a promoted tail-schedule file cannot discharge no-current-leakage unless the local estimate

$$\limsup_{a \to 0, L \to \infty} \ell_x(a, L) \leq C e^{-m k_N(x)}$$

is separately proved with constants $C < \infty$ and $m > 0$ and without appealing to the mass gap being proved.

4.6 True analytic inputs

The preceding reductions leave three non-formal Yang–Mills estimates. They are not supplied by the existing broad tightness, moment-bound, operator-convergence, or semigroup evidence. To promote this DLCR route, the repository now requires three explicit finite-chart artifacts, all pinned to the same Wilson-action specification:

1. a state chart transport estimate

$$\mathbf{E}\, d_\infty(X_a, Y) \leq D_N a^{\eta_N}$$

for every fixed Wilson-loop packet W_N;

2. translated chart transport estimates

$$\mathbf{E}\, d_\infty(X_{a,q}, Y_q) \le D_{N,q} a^{\eta_{N,q}}$$

for every rational-time entry in F_N;

3. a leakage tail schedule

$$\limsup_{a\to 0, L\to\infty} \epsilon_N(a, L) \le \rho_N, \qquad \rho_N \to 0.$$

The two transport artifacts may be proved either by constructing couplings directly or by a finite Kantorovich-dual certificate. Namely, on a compact finite chart it suffices to give an $\varepsilon(a)$-net $\{\psi_i\}$ for the anchored 1-Lipschitz test class and prove

$$|\mathbf{E}_{\mu_a}\psi_i - \mathbf{E}_{\mu_{\mathrm{OS}}}\psi_i| \le A a^\eta$$

for every net test. Kantorovich–Rubinstein duality then gives

$$W_1(\mu_a, \mu_{\mathrm{OS}}) \le A a^\eta + 2\varepsilon(a).$$

If $\varepsilon(a) \le E a^\gamma$, this is a transport bound with constant $A + 2E$ and exponent $\min(\eta, \gamma)$. Another accepted route is relative entropy. If a finite chart has diameter R and

$$\mathrm{KL}(\mu_a \| \mu_{\mathrm{OS}}) \le K a^\kappa,$$

then Pinsker's inequality and the maximal coupling bound give

$$W_1(\mu_a, \mu_{\mathrm{OS}}) \le R\sqrt{K/2}\, a^{\kappa/2}.$$

For d_∞ on $SU(3)^m$ representatives one may take $R \le 2$, hence the constant is $\sqrt{2K}$. This entropy estimate follows in turn from uniform convergence of the finite chart effective actions. If

$$d\mu_a = Z_a^{-1} e^{-S_a}\, d\lambda, \qquad d\mu_{\mathrm{OS}} = Z^{-1} e^{-S}\, d\lambda$$

on a common finite reference measure and $\|S_a - S\|_\infty \le B a^\beta$, then $e^{-Ba^\beta} Z \le Z_a \le e^{Ba^\beta} Z$. Hence the log density ratio is bounded above by $2Ba^\beta$, so

$$\mathrm{KL}(\mu_a \| \mu_{\mathrm{OS}}) \le 2Ba^\beta$$

and therefore

$$W_1(\mu_a, \mu_{\mathrm{OS}}) \le R\sqrt{B}\, a^{\beta/2}.$$

For d_∞ on $SU(3)^m$, this gives constant $2\sqrt{B}$. Finally, the uniform effective-action estimate follows from a summable finite-chart polymer or cumulant remainder. If

$$S_a - S = \sum_\Gamma c_\Gamma(a) K_\Gamma, \qquad \|K_\Gamma\|_\infty \le 1,$$

and

$$|c_\Gamma(a)| \le A_\Gamma a^{\beta_\Gamma}, \qquad \beta_\Gamma \ge \beta > 0, \qquad \sum_\Gamma A_\Gamma \le B,$$

then

$$\|S_a - S\|_\infty \le B a^\beta.$$

Thus a packet-scale polymer/cumulant summability theorem with explicit B, β would supply the effective-action estimate, hence the entropy and transport estimates above.

The finite interaction-degree step is discharged for finite-range lattice polymer atoms. If atoms are indexed by a subset of $\mathbb{Z}^d$ and two atoms interact only when their ℓ^∞ distance is at most r, then the interaction graph has degree

$$\Delta \le (2\lceil r \rceil + 1)^d - 1.$$

Indeed, every neighbor of a fixed atom lies in the cube of side length $2\lceil r \rceil + 1$ centered at that atom, excluding the atom itself.

The finite shape-counting step is discharged for any such bounded-degree polymer interaction graph. If the chart support has Q_N anchors, then connected polymers of size n touching the chart support are bounded in number by

$$Q_N \mu^n, \qquad \mu = \max(1, 2e\Delta).$$

Indeed, for one fixed root, a connected size-n polymer is overcounted by a rooted subtree of the universal covering tree. A preorder encoding chooses $n-1$ occupied child slots among at most Δn slots, so

$$\binom{\Delta n}{n-1} \le \left(\frac{e\Delta n}{n-1}\right)^{n-1} \le (2e\Delta)^{n-1}$$

for $n \ge 2$, while $n = 1$ is trivial. The union bound over the Q_N anchors gives the displayed estimate.

Suppose, in addition, that the corresponding activities obey

$$|z_\Gamma(a)| \le A_0 a^\beta q^n, \qquad \mu q < 1.$$

Equivalently, an exponential activity estimate

$$|z_\Gamma(a)| \le A_0 a^\beta e^{-\sigma n}$$

is sufficient whenever $\sigma > \log \mu$, by taking $q = e^{-\sigma}$. Likewise, if a verified cluster-decay certificate supplies a ratio r with $0 < r < 1/\mu$, then $\sigma = -\log r$ and the same threshold follows. Then grouping by n gives the geometric estimate

$$\sum_\Gamma |z_\Gamma(a)| \le Q_N A_0 a^\beta \sum_{n \ge 1} (\mu q)^n = Q_N A_0 \frac{\mu q}{1 - \mu q}\, a^\beta.$$

Thus the polymer-remainder certificate applies with $B = Q_N A_0 \mu q/(1-\mu q)$. This proves the bounded-degree shape-growth estimate and the finite Kotecky–Preiss-style activity-to-summability implication. The actual Yang–Mills polymer expansion, the finite-range polymer atom identification, and the excess-decay estimate $\sigma > \log \mu$ remain analytic inputs.

Combining the finite-range degree, shape-growth, cluster-decay, polymer remainder, entropy, and transport certificates gives a single usable transport package. If a finite-chart Yang–Mills expansion supplies finite-range atoms in $\mathbb{Z}^d$, interaction radius r, chart support size Q_N, activity prefactor $A_0 a^\beta$, and a verified cluster ratio $\rho < 1/\mu$, with

$$\mu = \max\bigl(1, 2e\bigl((2\lceil r\rceil + 1)^d - 1\bigr)\bigr),$$

then

$$B = Q_N A_0 \frac{\mu\rho}{1-\mu\rho}$$

gives

$$\mathrm{KL}(\mu_a \| \mu_{\mathrm{OS}}) \le 2Ba^\beta, \qquad W_1(\mu_a, \mu_{\mathrm{OS}}) \le R\sqrt{B}\, a^{\beta/2}.$$

This packages the finite constants only; the Yang–Mills expansion and the ratio $\rho < 1/\mu$ remain the genuine analytic target.

The verifier now also records a single unified route for this target. A future artifact with schema

`yangmills.dlcr.finite_chart_polymer_cluster_expansion.v1`

would be sufficient to populate the three remaining DLCR evidence files only if it proves all of the following output estimates with constants and the same action hash:

$$\mathbf{E}\, d_\infty(X_a, Y) \le D_N a^{\eta_N},$$

$$\mathbf{E}\, d_\infty(X_{a,q}, Y_q) \le D_{N,q} a^{\eta_{N,q}},$$

and

$$\ell_x(a, L) \le C e^{-m k_N(x)}.$$

The first output feeds state chart transport, the second feeds translated chart transport, and the third feeds the leakage-tail schedule through the shell estimates above. This unified route is only a promotion criterion: the current artifact is conditional and is not accepted as a substitute for the three proved output artifacts.

One proposed way to prove such a unified artifact is the Ward-screened cavity route. For an outside atom x, introduce a cavity interpolation and decompose

$$\partial_t S_{t,x} = \nabla_{\mathrm{gauge}} G_{t,x} + B^{\mathrm{curv}}_{t,x}.$$

For a finite compact chart law

$$d\mu_t = Z_t^{-1} e^{-S_t} d\lambda,$$

the finite cavity interpolation identity gives

$$\frac{d}{dt}\mathbf{E}_t[F] = -\operatorname{Cov}_t(F, \dot{S}_t).$$

If $\dot{S}_t = XG_t + B_t^{\mathrm{curv}}$ and the finite Ward identity gives $\operatorname{Cov}_t(F, XG_t) = 0$, then

$$\frac{d}{dt}\mathbf{E}_t[F] = -\operatorname{Cov}_t(F, B_t^{\mathrm{curv}}).$$

The required Ward cancellation lemma is

$$\operatorname{Cov}_{t,x}\big(F(J_N), \nabla_{\mathrm{gauge}} G_{t,x}\big) = 0$$

for gauge-invariant packet tests F. This cancellation is a finite compact chart identity: Haar integration by parts gives $\int X(H)\, d\mu = 0$ for gauge generators X and invariant finite-chart densities, and $X(F) = 0$ for gauge-invariant packet tests, hence $\operatorname{Cov}(F, XG) = 0$. What remains is to prove that the cavity derivative actually decomposes into such gauge-generator terms plus a curvature bridge term. The remaining curvature bridge term must then admit connected cluster weights w_Γ satisfying the screened norm bound

$$\sup_{N,x} e^{m k_N(x)} \sum_{\Gamma:\, x \leftrightarrow S_N} |w_\Gamma| e^{\sigma|\Gamma|} < \infty, \qquad \sigma > \log \mu, \quad m > 0.$$

A one-shell contraction

$$\|B_{r+1}\|_{\mathrm{SB}} \le \theta \|B_r\|_{\mathrm{SB}} + C a^\beta e^{-\sigma r}, \qquad \theta < 1,$$

together with uniform continuum and volume control, would imply the local exponential leakage estimate and the finite-chart effective-action remainders. The deterministic part of this implication is now separated. If nonnegative shell bridge norms b_r satisfy

$$b_{r+1} \le \theta b_r + C a^\beta e^{-\sigma r}, \qquad 0 < a \le 1, \quad 0 \le \theta < 1,$$

then for every screening rate m with

$$\theta e^m < 1, \qquad m < \sigma,$$

one has

$$\sup_{r \ge 0} e^{mr} b_r \le b_0 + \frac{C e^m}{(1 - \theta e^m)(1 - e^{m-\sigma})}.$$

Indeed, iteration gives

$$b_r \le \theta^r b_0 + C a^\beta \sum_{j=0}^{r-1} \theta^{r-1-j} e^{-\sigma j},$$

and multiplication by e^{mr} bounds the forced term by the product of the two geometric series $\sum_{\ell \ge 0} (\theta e^m)^\ell$ and $\sum_{j \ge 0} (e^{m-\sigma})^j$. Thus a genuine Yang–Mills

proof of the one-shell recurrence, with uniform $b_0, \theta, C, \sigma, \beta$, would give the screened bridge norm with the displayed constant.

A second route avoids reducing the bridge propagation to nearest-shell recurrences. Let b_x be the Ward-projected bridge norm rooted at an outside atom x and suppose the curvature bridge analysis gives an influence inequality

$$b_x \le f_x + \sum_y A_{xy} b_y, \qquad A_{xy} \ge 0,$$

with forcing

$$f_x \le C a^\beta e^{-\sigma k_N(x)}.$$

For a screening rate $0 < m < \sigma$, define the weighted influence norm

$$\theta_m = \sup_x \sum_y A_{xy} \exp\{m(k_N(x) - k_N(y))\}.$$

If $\theta_m < 1$, then

$$\sup_x e^{m k_N(x)} b_x \le \frac{C}{1-\theta_m} a^\beta.$$

Indeed, multiply the influence inequality by $e^{m k_N(x)}$, take the supremum over x, and absorb the term $\theta_m \sup_y e^{m k_N(y)} b_y$ to the left. This Ward-projected Dobrushin resolvent certificate is deterministic. It gives a second possible path to the screened bridge norm, with

$$C_{\mathrm{SB}} = \frac{C}{1-\theta_m},$$

provided the genuine Yang–Mills bridge analysis proves the weighted influence bound $\theta_m < 1$ uniformly in the continuum and infinite-volume limits. There is also an iterated version. Let T_m be the weighted influence operator with kernel $A_{xy} \exp\{m(k_N(x) - k_N(y))\}$. If

$$\theta_{m,p} := \|T_m^p\|_\infty < 1$$

for some integer $p \ge 1$, then the same positive-resolvent argument gives

$$\sup_x e^{m k_N(x)} b_x \le C a^\beta \frac{1 + \theta_m + \cdots + \theta_m^{p-1}}{1 - \theta_{m,p}}, \qquad \theta_m = \|T_m\|_\infty.$$

This can be useful if the one-step weighted norm is not contractive but a finite Ward-projected iterate is. It is still only a deterministic certificate: the Yang–Mills task is to prove such an iterated weighted influence contraction with constants independent of the cutoff limits. The atom-level influence check can also be compressed to finitely many blocks. Partition the outside atoms into blocks $I_1, \ldots, I_q$ and suppose a nonnegative matrix B satisfies

$$\sup_{x \in I_i} \sum_{y \in I_j} A_{xy} \exp\{m(k_N(x) - k_N(y))\} \le B_{ij}.$$

Then the weighted atom influence operator T_m satisfies

$$\|T_m^p\|_\infty \le \|B^p\|_\infty$$

for every $p \ge 1$. Indeed, the block sup vector of $T_m u$ is bounded componentwise by B times the block sup vector of u, and iteration gives the claim. Consequently a contractive finite block iterate $\|B^p\|_\infty < 1$ implies the atom-level iterated Dobrushin certificate. This converts the analytic target from an atom-by-atom estimate into finitely many block influence bounds, still with the same uniformity requirements.

The final deterministic conversion from screened norm to local leakage is also separated. Suppose the curvature bridge expansion bounds the leakage of an outside atom x by

$$\ell_x(a, L) \le K L_F \sum_{\Gamma:\, x \leftrightarrow S_N} |w_\Gamma|,$$

where K is the bridge-to-leakage normalization and L_F is the packet-test Lipschitz constant. If

$$\sup_x e^{m k_N(x)} \sum_{\Gamma:\, x \leftrightarrow S_N} |w_\Gamma| e^{\sigma|\Gamma|} \le C_{\mathrm{SB}}, \qquad m > 0, \quad \sigma \ge 0,$$

then $e^{\sigma|\Gamma|} \ge 1$ gives

$$\ell_x(a, L) \le K L_F C_{\mathrm{SB}} e^{-m k_N(x)}.$$

Thus a proved screened bridge norm would produce the exact local exponential leakage artifact required above, with local leakage constant $K L_F C_{\mathrm{SB}}$ and rate m.

The same screened norm also gives the finite-chart effective-action constants needed for the transport side. Suppose every chart-touching remainder cluster can be assigned to one anchor in the finite chart support S_N, with $|S_N| = Q_N$, and its contribution to the effective-action difference is bounded by $K_A a^\beta |w_\Gamma|$. Since $e^{\sigma|\Gamma|} \ge 1$ and the shell distance from a chart anchor to S_N is zero, the rooted cluster sum at each anchor is at most C_{SB}. A union bound over the Q_N anchors gives

$$\|S_a - S\|_\infty \le K_A Q_N C_{\mathrm{SB}} a^\beta.$$

The finite effective-action certificate therefore gives

$$\mathrm{KL}(\mu_a \| \mu_{\mathrm{OS}}) \le 2 K_A Q_N C_{\mathrm{SB}} a^\beta$$

and, on a chart of diameter R,

$$W_1(\mu_a, \mu_{\mathrm{OS}}) \le R \sqrt{K_A Q_N C_{\mathrm{SB}}}\, a^{\beta/2}.$$

The translated chart has the same formula with Q_N, R replaced by the translated packet constants.

The verifier records this Ward-screened cavity route as a conditional framework with six sublemmas. The finite cavity interpolation and finite Ward cancellation sublemmas are proved, and the formal one-shell-recurrence-to- screened-norm, Dobrushin/block-compression, multiscale transport-flow, log-budget, sparse bad-scale, sliding-window, finite-profile, and negative-average/Lyapunov/finite-state/cycle-mean budgets, finite-state graph duality and maximum-cycle-mean checks, finite scale-graph-to-screened-norm packaging and interval robustness, screened-norm-to-local-leakage, and screened-norm-to-effective-action reductions are proved; the cavity decomposition, bridge cluster expansion, the actual Yang–Mills contraction or negative log-budget/sparse bad-scale/window-density/finite-profile/negative- average/Lyapunov/finite-state/cycle-mean estimate, and uniform continuum/volume control remain conditional.

Until these estimates are proved with constants, the exact shadow coupling, reflection-time intertwiner, and no-current-leakage lemmas remain conditional.

4.7 Sector cluster derivative majorants

The latest finite reduction moves the Ward-screened jet-atlas target one layer closer to a standard polymer expansion theorem. A proved artifact of schema

`yangmills.dlcr.sector_cluster_derivative_majorants.v1`

with the repository action hash, finite sector charts, signed truncated cluster value and first-derivative jets, nonnegative Hessian ℓ^1 contributions, absolute value/gradient/Hessian tail budgets, positive KP slack, raw/coarse support data, and coarse Karp constants canonically constructs the `yangmills.dlcr.polymer_sector_c2_majorant` package. On a sector C the construction is

$$v_C = \sum_{\Gamma \in T_C} v_\Gamma, \qquad r_C = \sum_{\Gamma \notin T_C} |v_\Gamma|,$$

$$g_{C,i} = \sum_{\Gamma \in T_C} \partial_i v_\Gamma, \qquad \rho_{C,i} = \sum_{\Gamma \notin T_C} |\partial_i v_\Gamma|,$$

and

$$H_C = \sum_{\Gamma \in T_C} \|D^2 v_\Gamma\|_1 + \sum_{\Gamma \notin T_C} \|D^2 v_\Gamma\|_1.$$

The triangle inequality gives the value and gradient intervals $[v_C - r_C, v_C + r_C]$ and $[g_{C,i} - \rho_{C,i}, g_{C,i} + \rho_{C,i}]$; Hessian subadditivity gives the ℓ^1 majorant H_C. The verifier checks coordinate agreement, nonnegative tails, positive KP slack, cluster-count domination, action-hash agreement, and then applies the already proved polymer-sector-to-local-C^2 and local-C^2-to-jet-atlas converters. This is a finite reduction: the genuine Yang–Mills sector cluster expansion and its derivative-tail bounds remain open.

4.8 Geometric derivative-tail envelope

A further finite reduction replaces raw derivative-tail declarations by a geometric order envelope. A proved artifact of schema

`yangmills.dlcr.geometric_cluster_tail_envelope.v1`

contains truncated sector cluster jets through order N_C and constants $A_v, r_v, A_i, r_i, A_H, r_H$ satisfying $0 \le r_\bullet < 1$. The omitted tails are then computed rather than assumed:

$$\sum_{n>N_C} A_v r_v^n = \frac{A_v r_v^{N_C+1}}{1-r_v}, \qquad \sum_{n>N_C} A_i r_i^n = \frac{A_i r_i^{N_C+1}}{1-r_i},$$

and

$$\sum_{n>N_C} A_H r_H^n = \frac{A_H r_H^{N_C+1}}{1-r_H}.$$

If the sector shape-growth constant μ_C and activity ratio ρ_C obey $\mu_C \rho_C < 1$, the KP slack is $1 - \mu_C \rho_C > 0$. The verifier constructs the sector-cluster derivative-majorant package with these computed tails and then applies the sector-cluster-to-polymer-sector converter. Thus the next analytic target is a geometric derivative-tail theorem for the actual Ward-screened Yang–Mills sector expansion, with enough downstream raw/Karp margin.

4.9 Finite shell-ratio certificate

The geometric tail envelope itself is now reduced to a finite shell-ratio certificate. On each sector, suppose the value, each first derivative, and the Hessian ℓ^1 norm have consecutive shell majorants B_n in the tail regime and a ratio $0 \le r < 1$ such that

$$B_{n+1} \le r B_n.$$

Anchoring at shell n_0 gives the geometric envelope

$$B_n \le B_{n_0} r^{n-n_0} = \left(\frac{B_{n_0}}{r^{n_0}}\right) r^n \qquad (n \ge n_0),$$

with the evident zero-ratio convention. If one activity ratio ρ dominates the value, gradient, and Hessian ratios and satisfies $\mu\rho < 1$ for the sector shape-growth constant μ, then the KP slack is $1 - \mu\rho > 0$. The verifier turns this finite ratio certificate into the geometric cluster-tail-envelope artifact and then applies the geometric-tail and sector-cluster converters. The remaining analytic issue is therefore a finite Yang–Mills shell-ratio theorem for the Ward-screened sector expansion.

4.10 Finite type-transfer certificate

The finite shell-ratio theorem is itself reduced to a nonnegative finite-type transfer calculation. For a component, let w_n be the row vector of nonnegative type weights at shell n and let T be a nonnegative transfer matrix. If

$$w_{n+1} = w_n T, \qquad \max_i \sum_j T_{ij} \le r < 1,$$

then, with $B_n = \|w_n\|_1$,

$$B_{n+1} = \sum_j (w_n T)_j = \sum_i w_{n,i} \sum_j T_{ij} \le r \sum_i w_{n,i} = r B_n.$$

The verifier applies this row-sum argument separately to the value, gradient, and Hessian ℓ^1 type systems. It then constructs the finite shell-ratio certificate and invokes the existing ratio-tail and geometric-tail pipeline. Thus the next Yang–Mills-specific analytic target is a finite Ward-screened type system whose transfer matrices have row sums below one, with enough raw-to-coarse and Karp margin downstream.

4.11 Ward-defect surplus principle

The type-transfer contraction has a sharper analytic source. For each type i, let $d_i \ge 0$ be the Ward-screening damping gained by the local Ward cancellation and let $L_{ij} \ge 0$ be the leakage from type i to type j after the screened cavity expansion. Define

$$T_{ij} = e^{-d_i} L_{ij}.$$

If the Ward damping beats branching and leakage in the sense that

$$\sum_j e^{-d_i} L_{ij} \le r < 1$$

for every source type i, then T is a nonnegative transfer matrix with row sums at most r. The finite type-transfer theorem therefore gives $B_{n+1} \le r B_n$ for the value, gradient, and Hessian type systems. This principle is the current analytic breakthrough target: prove a genuine Yang–Mills Ward-defect surplus estimate, not merely a formal row-sum bound.

4.12 Interval Ward-defect surplus

For computer-assisted estimates the Ward-defect surplus is used in interval form. Suppose each type row has a rigorous lower damping bound d_i^- and rigorous leakage upper bounds L_{ij}^+. Since e^{-d} is decreasing,

$$\sum_j e^{-d_i} L_{ij} \le e^{-d_i^-} \sum_j L_{ij}^+.$$

Thus the endpoint inequality

$$e^{-d_i^-} \sum_j L_{ij}^+ \le r < 1$$

is enough to produce the Ward-defect surplus certificate. This is the finite validation target for the analytic Yang–Mills estimates: prove lower bounds on Ward-screening damping and upper bounds on leakage rows whose endpoint row sums leave strict surplus.

4.13 Primitive Ward-screening estimates

The interval package can itself be produced from primitive estimates with certified error budgets. If a sector computation gives a damping estimate $\widehat{d}_i$ with error ε_i, and leakage estimates $\widehat{L}_{ij}$ with errors ε_{ij}, define

$$d_i^- = \widehat{d}_i - \varepsilon_i, \qquad L_{ij}^+ = \widehat{L}_{ij} + \varepsilon_{ij}.$$

When these outward endpoints are nonnegative, they are rigorous lower damping and upper leakage bounds. Thus the decisive finite analytic target has become primitive: prove Yang–Mills Ward-screening estimates whose outward-rounded endpoints still satisfy

$$e^{-(\widehat{d}_i - \varepsilon_i)} \sum_j (\widehat{L}_{ij} + \varepsilon_{ij}) < 1.$$

The remaining problem is not to invent another formal converter, but to prove these primitive damping and leakage estimates for the actual sector expansion.

4.14 Relative Ward-screening gain

A still sharper way to state the analytic input is relative to the unscreened leakage row. Let

$$B_i = \sum_j L_{ij}^+$$

be the outward upper bound for the baseline leakage mass from type i, and let R_i be an upper bound for the residual mass after applying the local Ward cancellation. If $0 < R_i \le B_i$, define

$$d_i = \log(B_i/R_i).$$

Then

$$e^{-d_i} B_i = R_i.$$

Consequently the whole Ward surplus check is reduced to proving a residual mass inequality $R_i \le r < 1$ for the value, gradient, and Hessian type systems. This is the current most primitive finite contract for the analytic breakthrough: prove that Ward cancellation leaves residual leakage mass below the contraction threshold.

4.15 Paired Ward-cancellation residuals

The residual mass R_i can be certified by explicit cancelling pairs. Group the signed Ward contributions in a type row into pairs (p_k, n_k), with certified error ε_k, and let U_i be the unpaired residual budget. Then

$$R_i = \sum_k \big(|p_k + n_k| + \varepsilon_k\big) + U_i$$

is a rigorous upper bound on the screened residual. Therefore a finite pairing whose pair defects and unpaired budget satisfy $R_i \le r < 1$ supplies the relative Ward-screening certificate. This is the next concrete analytic construction problem: build the Ward-pairing for the actual Yang–Mills sector expansion and prove the pair defects are small.

4.16 Involutive Ward term pairing

The finite Ward-pairing can be certified by an involution on the signed term set. An involution partitions the terms into two-cycles and fixed points. Each two-cycle gives a cancelling pair; the pair error is the sum of the two term errors. Each fixed point is charged to the unpaired residual by its absolute term value plus its error. Summing absolute term values and errors by target type gives the baseline leakage rows. Thus the next primitive construction is a finite Ward involution on the actual sector terms, with sufficiently small term errors and few or small fixed points.

4.17 Fractional Ward cancellation transport

The failed exact-involution test suggests a strictly weaker finite object. A fractional Ward cancellation transport splits each source row of signed sector terms into positive masses p_a and negative masses n_b and supplies a nonnegative transport plan π_{ab} with row sums at most p_a and column sums at most n_b. Transported equal mass cancels. The verifier charges only the unmatched mass, the certified term errors, any declared residual budget, and the cost-weighted transport defects

$$\sum_{a,b} \pi_{ab} c_{ab} + \epsilon_{ab}.$$

Thus the screened residual is

$$R_i = \sum_a \Big(p_a - \sum_b \pi_{ab}\Big) + \sum_b \Big(n_b - \sum_a \pi_{ab}\Big) + \sum_{a,b} (\pi_{ab} c_{ab} + \epsilon_{ab}) + E_{\mathrm{term}} + E_{\mathrm{res}}.$$

If $0 < R_i \le B_i$ and $R_i \le r < 1$ for each value, gradient, and Hessian type row, the existing relative Ward-screening certificate applies. This generalizes the involutive scheme: an exact two-cycle pairing is the special case where each transported mass saturates a single positive and negative term with zero transport defect. The new object is therefore a conditional route around the

failed exact Ward involution, not a Clay proof. It replaces a rigid combinatorial symmetry by a finite linear-program-style cancellation certificate whose failure or success is checkable.

4.18 Interpolation-driven fractional Ward transport

The fractional transport target should not be implemented by solving a large discrete matching problem after the cluster expansion has already been enumerated. The intended Yang–Mills construction is instead an interpolation-driven fractional Ward transport (IDFWT). Work in a finite gauge-fixed cavity chart and let μ_s, $0 \le s \le 1$, be a continuous decoupling interpolation from independent cavities to the physical lattice. The parameter s may be the edge weight in a Brydges–Kennedy–Rivasseau forest formula, or an equivalent heat-kernel/Langevin homotopy for the same finite chart. On the fully coupled finite chart the Ward identity is exact. Hence the derivative of the signed Ward defect along s is a conservative current, up to the explicit boundary and localization errors already charged in the DLCR ledger.

For a source type row, write the signed infinitesimal Ward defect as

$$\dot{A}_s = \dot{A}_s^+ - \dot{A}_s^- + \dot{E}_s.$$

The IDFWT candidate transport is the integrated positive-to-negative current

$$\pi_{ab} = \int_0^1 J_{ab}(s)\, ds, \qquad J_{ab}(s) \ge 0,$$

where $J_{ab}(s)$ is read from the common interpolation event that moves defect mass from the positive sector label a to the negative sector label b. In a BKR realization the event is a forest edge contraction; in a heat-kernel realization it is the corresponding infinitesimal Markov current. This definition is analytic rather than combinatorial: the transport plan is built during the decoupling expansion, and its row and column inequalities are the integrated continuity inequalities for the positive and negative defect densities.

The finite verifier therefore needs only outward bounds for the current:

$$\int_0^1 J_{ab}(s)\, ds \le \pi_{ab}^+, \qquad c_{ab} \le c_{ab}^+, \qquad \epsilon_{ab} \le \epsilon_{ab}^+.$$

The cost c_{ab} is chosen proportional to the interpolated interaction strength of the edge or heat-kernel step producing the current. Standard tree-graph estimates then give a falsifiable sufficient condition

$$\sum_{a,b} \pi_{ab}^+ c_{ab}^+ + \sum_{a,b} \epsilon_{ab}^+ + E_{\mathrm{term}} + E_{\mathrm{res}} + \text{unmatched mass} \le r_i < 1.$$

Because the BKR expansion factorizes over forests, this bound is a product of local tree amplitudes and no longer asks for a one-to-one pairing among

all Feynman/polymer graphs. The remaining field-theoretic obligation is precise: construct the finite SU(3) sector expansion, prove the interpolation current identity in the chosen chart, and certify that exponential tree decay beats the cost-weighted current and the boundary residuals uniformly in the cutoff. If that strict inequality survives interval rounding, the existing fractional Ward transport theorem promotes it to relative Ward screening, then to primitive Ward estimates, interval Ward-defect surplus, finite type transfer, shell ratios, geometric cluster tails, and the screened bridge norm.

4.19 Finite BKR continuity equation

The finite BKR Ward-current identity can be split into a still smaller continuity-equation certificate. On a finite gauge-fixed chart let $D(s)$ be the signed Ward defect along a C^1 BKR interpolation. The analytic identity begins with the fundamental theorem of calculus,

$$D(1) - D(0) = \int_0^1 D'(s)\, ds.$$

The sectorized certificate writes $D'(s)$ as a finite signed sum of forest current densities plus declared gauge, boundary, truncation, and balance errors. For each source type row the verifier computes

$$\Delta_{\mathrm{end}} = \sum_a p_a - \sum_b n_b, \qquad I_{\mathrm{forest}} = \sum_F \sigma_F \int_0^1 J_F(s)\, ds,$$

where $\sigma_F = \pm 1$ records the orientation of the forest current in the signed continuity equation. The row is accepted if

$$|\Delta_{\mathrm{end}} - I_{\mathrm{forest}}| \le E_{\mathrm{gauge}} + E_{\mathrm{bdry}} + E_{\mathrm{trunc}} + E_{\mathrm{bal}}.$$

This proves only the finite signed balance. It does not yet prove that all oriented currents have the positive-to-negative form needed for IDFWT, nor does it prove the tree-decay constants. But it isolates the first analytic gate: finite-chart C^1 BKR differentiability plus a sectorized current decomposition with explicit balance error.

4.20 Six gate breakdown

The continuity package now splits into six named gates. The verifier treats them as separate obligations:

1. finite C^1 interpolation;
2. sectorized current decomposition;
3. endpoint-current balance;
4. usable-current orientation;

5. tree-decay cost estimate;

6. cutoff-stable residual margin.

This is the finer roadmap the project needed. Each gate has its own field in the package, so the proof no longer hides a single large interpolation claim behind a monolithic theorem. Instead, the remaining SU(3) work is now distributed into six smaller obligations that can be attacked one by one.

4.21 Genuine SU(3) continuity package roadmap

The six gates can be made still more operational. A genuine SU(3) finite BKR continuity package must provide the following finite subcertificates.

For the finite C^1 interpolation gate, one must give a tree-gauge chart atlas on $SU(3)^m/\operatorname{Ad}$, a BKR link-weight interpolation $s \mapsto S_s$ with the common action hash, uniform C^1 domination for the action density and Ward observable, and a finite quadrature or interval enclosure rule for $\int_0^1 D'(s)\,ds$. The verifier-facing acceptance test is that the package has `differentiability_status` = `FINITE_CHART_C1` and finite derivative bounds on the declared parameter box.

For the sectorized current decomposition gate, one must give a finite sector partition of the plaquette and edge words, a signed endpoint Ward-term table with target types, a forest-current event table for each derivative term, and a residual budget for discarded or merged terms. The acceptance test is that the sector decomposition is declared finite and every continuity row has positive terms, negative terms, and finite forest-current events.

For the endpoint-current balance gate, one must give the endpoint defect $D(1) - D(0)$, the integrated signed forest current, and the gauge-fixing, boundary, truncation, and balance error budgets. The finite inequality is

$$\left|\Delta_{\mathrm{end}} - I_{\mathrm{forest}}\right| \le E_{\mathrm{gauge}} + E_{\mathrm{bdry}} + E_{\mathrm{trunc}} + E_{\mathrm{bal}}.$$

For the usable-current orientation gate, one must give positive endpoint capacities, negative endpoint capacities, an orientation sign for each forest-current event, and row/column capacity inequalities for the IDFWT transport plan. The acceptance test is that every consumed current is nonnegative after orientation and moves mass from a positive Ward-defect label to a negative Ward-defect label without exceeding the two capacities.

For the tree-decay cost gate, one must give for each event an interaction strength upper bound g_F, a tree amplitude A_F, a decay ratio $\rho_F < 1$, a tail start n_F, and a quadrature error. The cost charged to the transport event is then bounded by

$$g_F \, \frac{A_F \rho_F^{n_F}}{1 - \rho_F}.$$

For the cutoff-stable residual-margin gate, one must give unmatched-mass bounds, the cost-weighted transport defect sum, outward interval-rounding bud-

gets, and cutoff/volume variation budgets with strict margin. The final acceptance test is the screened contraction inequality

$$R_i \le r_i < 1$$

for every value, gradient, and Hessian type row after all errors are charged.

Thus the next genuine theorem is no longer a vague continuity assertion. It is a finite construction problem: populate these six subcertificate families with a matching SU(3) action hash and strict margins. If that is done, the existing finite BKR continuity converter emits the finite-chart BKR Ward-current identity, then IDFWT, fractional Ward transport, and the downstream screened DLCR certificates. This paragraph is still only a roadmap reduction; it does not prove the SU(3) chart atlas, the C^1 estimates, the signed current identity, the tree-decay constants, or the cutoff-stable residual margin.

4.22 Micro-roadmap for the six SU(3) gates

The subcertificate list can be turned into a still finer workbench. The first gate begins by fixing a packet graph and maximal tree gauge, writing the non-tree holonomies as $SU(3)^m$ coordinates modulo residual conjugation, defining the BKR edge-weight action S_s on a compact parameter box, differentiating the finite-dimensional integral under the integral sign, and recording derivative majorants for the value, gradient, and Hessian rows. Its first decisive test is one nontrivial plaquette-edge chart with a certified C^1 derivative enclosure and matching action hash.

The second gate expands each differentiated action-observable product into finite word sectors, assigns every signed endpoint contribution to a source and target type, names the BKR forest event that generated it, merges symmetry-equivalent terms only with an explicit residual budget, and proves that no untyped finite term remains outside the residual. Its first decisive test is a complete signed term table for the same raw edge used in the first gate.

The third gate computes the endpoint signed Ward defect, integrates the signed forest-current table over the BKR parameter, separates gauge-fixing, boundary, truncation, and quadrature errors, proves the row balance inequality by outward interval arithmetic, and stores both the balance defect and the allowed defect. Its first decisive test is that the largest row balance defect is below the declared continuity-error budget.

The fourth gate splits the signed table into positive and negative endpoint capacities, chooses an orientation for each forest event, discards or charges currents that cannot be oriented from positive to negative mass, and checks the row and column capacity inequalities. Its first decisive test is that the usable oriented current transports a positive fraction of every dominant positive defect without exceeding the negative capacities.

The fifth gate bounds the local interpolated interaction strength, proves a tree-amplitude bound, proves a geometric decay ratio strictly below one, chooses a tail start order compatible with the truncation, and rounds the cost envelope

outward before passing to IDFWT. Its first decisive test is that the cost-weighted oriented current is below the available Ward-screening surplus on the test raw edge.

The sixth gate sums unmatched endpoint mass, cost-weighted transport defects, continuity errors, interval-rounding reserves, rationalization reserves, and cutoff/volume variation reserves. It then checks the residual contraction and the downstream Karp margin simultaneously. Its first decisive test is that the value, gradient, and Hessian rows all satisfy $R_i \le r_i < 1$ after all reserves and that the emitted coarse graph still has positive Karp margin.

This micro-roadmap gives a natural order for the next proof attempts:

$$C^1 \text{ chart} \to \text{typed sectors} \to \text{signed balance} \to \text{usable orientation} \to \text{tree cost} \to \text{residual and Karp margin.}$$

Failure at any arrow is informative: it identifies the first analytic obstruction before the project spends effort on downstream certificate machinery.

4.23 Artifact-level SU(3) continuity roadmap

The micro-roadmap can be refined once more into named mini-artifacts. Each mini-artifact has four fields: the lemma to prove, the finite data it emits, the verifier fields it populates, and the failure mode that would stop the route. The intended order is:

1. `su3_tree_gauge_chart.v1`: construct the finite $SU(3)^m/\mathrm{Ad}$ chart and emit coordinates and boxes;
2. `su3_bkr_action_path.v1`: prove the edge-weight path S_s is C^1 with the common action hash;
3. `su3_ward_derivative_enclosure.v1`: enclose $D'(s)$ for value, gradient, and Hessian rows;
4. `su3_word_sector_partition.v1`: split differentiated words into finite source and target sectors;
5. `su3_signed_endpoint_terms.v1`: certify signs, target types, and error radii for endpoint Ward terms;
6. `su3_forest_current_table.v1`: attach each derivative term to a BKR forest-current event;
7. `su3_endpoint_defect_interval.v1`: evaluate $D(1) - D(0)$ with outward intervals;
8. `su3_integrated_current_interval.v1`: integrate the signed forest current with quadrature error;
9. `su3_balance_error_budget.v1`: prove the rowwise balance inequality;
10. `su3_endpoint_capacity_table.v1`: bound positive and negative endpoint capacities;
11. `su3_oriented_current_selection.v1`: select the usable positive-to-negative subcurrent and charge the rest;

12. `su3_idfwt_capacity_check.v1`: verify the IDFWT row and column capacity inequalities;
13. `su3_link_coupling_bound.v1`: bound each active interpolated link strength;
14. `su3_tree_tail_envelope.v1`: prove $A_F \rho_F^n/(1-\rho_F)$ with $\rho_F < 1$;
15. `su3_transport_cost_rounding.v1`: round the cost envelope outward while preserving surplus;
16. `su3_unmatched_mass_accounting.v1`: charge every untransported endpoint mass by type row;
17. `su3_uniform_reserve_budget.v1`: bound rounding, rationalization, cutoff, and volume reserves;
18. `su3_residual_karp_margin.v1`: prove $R_i \le r_i < 1$ and a positive coarse Karp margin together.

This is now the smallest useful roadmap unit in the current repository. A failure of `su3_tree_tail_envelope.v1`, for example, means the tree branching has beaten decay; a failure of `su3_residual_karp_margin.v1` means local residual contraction did not survive the global coarse-graph drift test. Neither failure would be ambiguous, and neither would require re-auditing the whole DLCR chain.

4.24 Atomic-obligation roadmap

Each of the eighteen mini-artifacts is now split into three atomic obligations. They are, respectively, data normalization, analytic estimate, and verifier handoff. This gives fifty-four atomic tasks. For example, the tree-gauge chart artifact breaks into:

1. choose the finite packet graph, root set, and maximal tree;
2. prove every packet loop word is represented by non-tree $SU(3)$ coordinates modulo residual conjugation;
3. emit coordinate boxes and raw-edge support with the matching action hash.

At the other end, `su3_residual_karp_margin.v1` breaks into:

1. verify $R_i \le r_i < 1$ for value rows;
2. verify the same residual contraction for gradient and Hessian rows;
3. verify that the emitted coarse transition graph has positive Karp margin.

The intermediate artifacts follow the same pattern. The BKR action-path artifact must declare active links, differentiate the Wilson action path, and check the two endpoint actions. The Ward-derivative enclosure must write $D'(s)$, dominate the differentiated gradient/Hessian representatives, and produce quadrature intervals. The sector artifacts must define the type alphabet, assign every

differentiated word, certify endpoint signs, and charge ambiguous or unassigned terms. The current-balance artifacts must compute endpoint and integrated-current intervals and then prove balance_defect $\leq$ allowed_defect. The orientation artifacts must record capacities, choose a nonnegative positive-to-negative subcurrent, and verify IDFWT row and column inequalities. The cost artifacts must bound active link strength, prove A_F, ρ_F, n_F with $\rho_F < 1$, and show that rounded cost defects remain below surplus. The residual artifacts must account for unmatched mass, add all reserve budgets, and close contraction together with Karp negativity.

This atomic layer is useful because it identifies the next smallest proof object: before trying to prove any Ward transport estimate, the project must first produce the first three atomic obligations of `su3_tree_gauge_chart.v1`. If even that chart artifact cannot be made genuine on a nontrivial raw edge, the IDFWT route has no honest finite-chart input.

4.25 Leaf plan for the first chart artifact

The immediate next artifact, `su3_tree_gauge_chart.v1`, can itself be split into nine leaf certificates:

1. finite graph inventory: list vertices, oriented edges, packet paths, and the raw edge under test;
2. root and tree choice: choose roots and a maximal tree or forest, with $|T| = |V|-$components and no cycles;
3. tree-gauge section: construct the representative setting tree holonomies to the identity;
4. non-tree coordinate map: identify the remaining holonomies as $SU(3)^m$ coordinates with $m = |E| - |V|+$components;
5. loop-word reconstruction: rewrite every packet loop and tested raw edge as a word in the non-tree coordinates;
6. residual adjoint action: record root conjugation on $SU(3)^m$ and trace-word invariance;
7. compact chart domain: declare the compact domain and the finite interval boxes used by later enclosures;
8. raw-edge support binding: bind the source and target raw states of the test edge to the chart partition;
9. action-hash binding: attach the common Yang–Mills action hash and chart-local action metadata.

This leaf plan is deliberately elementary. It is the first point at which the roadmap can stop being only a dependency graph and become a genuine finite

artifact. The next mathematical move is to instantiate these nine leaves for one nontrivial raw edge; only then does it make sense to attempt the BKR action path and Ward-derivative enclosure artifacts.

4.26 Field plan for the first leaf

The first leaf, finite graph inventory, can be refined into eight fields:

vertices, oriented edges, packet paths, packet loops,
raw edge under test, plaquette words, component labels, inventory hash.

The acceptance checks are intentionally elementary. The vertex set must be nonempty, every edge endpoint must be declared, every packet path must use declared oriented edge labels, every packet loop and plaquette word must be closed, the raw edge under test must be supported by the packet graph, and the component labels must agree with graph connectivity. Finally, the inventory hash must be canonical under the ordering conventions used by the verifier. This first leaf has four immediate failure modes: a packet path names an undeclared edge, the raw edge support lies outside the packet graph, a plaquette boundary is not closed, or the component labels disagree with graph connectivity. If the field plan passes, it hands the vertices, oriented edges, and component labels to the next leaf, where the maximal-tree check is $|T| = |V|-$components.

4.27 Finite-chart BKR Ward-current identity

The next roadmap layer feeds IDFWT from the continuity equation. A finite BKR-current package records, for each finite chart and source type row, the positive and negative Ward endpoint masses together with BKR forest-current events. After the signed balance above has been verified, the package records the finite-chart current balance

$$A_1^+ - A_1^- - (A_0^+ - A_0^-) = \sum_F \int_0^1 J_F(s)\, ds + E_{\mathrm{gauge}} + E_{\mathrm{bdry}} + E_{\mathrm{trunc}},$$

after sectorization and gauge fixing. Each nonnegative forest current J_F names a positive endpoint a, a negative endpoint b, and an interpolated link or block coupling. The finite verifier converts the event to an IDFWT transport mass

$$\pi_{ab}^F = \int_0^1 J_F(s)\, ds$$

and records the continuity error

$$E_{\mathrm{cont}} = E_{\mathrm{gauge}} + E_{\mathrm{bdry}} + E_{\mathrm{trunc}}.$$

If the event has link-coupling upper bound g_F, tree amplitude A_F, decay ratio $\rho_F < 1$, and tail start n_F, its transport cost is bounded by

$$c_F \le g_F \, \frac{A_F \rho_F^{n_F}}{1 - \rho_F}.$$

Thus a proved finite BKR Ward-current package canonically emits the IDFWT package; the already stated IDFWT verifier then checks capacities, costs, unmatched mass, and residual contraction. The remaining genuine Yang–Mills obligation has therefore been sharpened again: prove finite C^1 BKR differentiability, construct the sectorized signed current decomposition, orient the useful currents from positive to negative Ward mass, and certify cutoff-stable tree-decay constants.

4.28 Roadmap audit

The current proof state separates two facts that must not be conflated. The final bridge registry is discharged as a registry matter by explicit algebraic/GNS and spectral/generator witness artifacts, but the finite-chart Yang–Mills analytic input is not constructed. A direct continuum OS-identification route leaves the compactness, localization, and spectral identification problems in essentially their original form. The Ward-screened DLCR route is therefore the selected conditional framework: its recent converter and stability theorems reduce the remaining field-theoretic input to a finite Ward-projected interval jet atlas, raw-to-coarse domination with positive slack, a negative coarse Karp cycle-mean certificate, and uniform cutoff/volume stability. This is a promising roadmap, not a proof of the Clay conjecture. The next decisive theorem must construct a genuine Yang–Mills BKR continuity package whose emitted Ward-current and IDFWT data have screened residual $R_i \le r_i < 1$ and whose interval-rounded outputs leave enough tail slack and Karp margin to feed this finite chain. The six named subgates above are the finer checklist for that theorem.

4.29 Continue-or-pivot decision

The current probability of completing the Clay proof from the present state should be regarded as below one half. The reason is not lack of progress: recent reductions have made the next obstruction finite and falsifiable. The reason is that the genuine Yang–Mills finite C^1 BKR continuity equation, sector term estimates, tree-decay constants, negative Karp certificate, and uniform cutoff/volume stability have not been constructed. The correct decision after the exact-involution failure is therefore a narrow pivot: stop searching for a raw exact Ward involution and attempt one nontrivial finite-chart BKR continuity package whose emitted IDFWT instance has certified residual contraction and positive downstream margins. If that continuous transport test also fails, the project should remain conditional verifier infrastructure rather than an open-ended Clay proof claim.

4.30 Decisive-test outcome

The decisive raw-edge Ward-involution attempt fails in the present repository.
There is no proved genuine Yang–Mills artifact with schema `yangmills.dlcr.involutive_ward_term_pair`

and no available sector term list, target-type assignment, certified term-error table, or Ward involution from which such an artifact can be constructed. The Yang–Mills evidence files that could feed this object remain explicitly conditional. Therefore this route should no longer be presented as an active likely Clay proof path. The useful output is conditional verifier infrastructure and a record of finite reductions. The next analytic search target is not the exact involutive Ward-pairing artifact; it is a genuine finite-chart BKR continuity package whose emitted Ward-current and IDFWT data produce the fractional Ward transport artifact.

4.31 Conditional implication

If the exact shadow coupling, reflection-time intertwiner, and no-current-leakage lemmas are proved for the same action specification and the same OS/GNS representation, then the convergence-side Schwinger realization on W_N and the convergence-side action realization on F_N follow. The conditional implication is recorded in the DLCR proof artifact. The repository does not currently prove those remaining analytic DLCR lemmas. The audited status remains theorem boundary.

5 Ward–Hodge harmonic weight instantiation

This appendix records a new mechanism that collapses all six gates of the finite-chart BKR continuity package on the 2×2 plaquette block chart at a single canonical point θ^*. The mechanism introduces three new mathematical objects:

1. the Ward–Hodge decomposition of the plaquette-trace Jacobian on the finite chart (§5.1);
2. the SU(3) trace-plane modulus (the Cartan modulus) as an explicit non-negative formula in $|\operatorname{tr} W_p|$ (§5.2);
3. the canonical chart-box instantiation at θ^* with strict margin $\delta \in (0, 1)$ (§5.3).

The artifact `ward_hodge_harmonic_weights.v1` implements the instantiation and verifies all six gates simultaneously. The collapse is at the finite-chart level only; the continuum-limit transfer is not addressed here and remains open.

5.1 Ward–Hodge decomposition of the plaquette-trace Jacobian

Let $\theta = (\theta_e^a)$ denote the chart coordinates on the 2×2 plaquette block, indexed by tree edges e and SU(3) generator indices $a \in \{1, \dots, 8\}$. The Ward derivative is

$$D_{p,e}^a(\theta) = \frac{\partial \operatorname{Re} \operatorname{tr} W_p(\theta)}{\partial \theta_e^a}.$$

On the tree gauge (proven in the existing repository), the gauge-orbit (exact) component of D vanishes identically, and the surviving component is the lattice analogue of the continuum Hodge decomposition:

$$D = D_{\text{coexact}} + D_{\text{harmonic}},$$

where D_{harmonic} is the projection onto the tree cohomology (the canonical non-trivial component surviving gauge fixing) and D_{coexact} is the curl of the dual plaquette current. On the 2×2 chart with the proven tree gauge, D_{harmonic} is the only component contributing to the Ward-weighted forest current operator $K_W = M_T^T A M_T$, giving a canonical non-uniform weight assignment $a_p(\theta)$.

5.2 SU(3) trace-plane modulus (Cartan modulus)

Let $T_a = \lambda_a/2$ denote the Hermitian SU(3) generators with normalization $\operatorname{tr}(T_a T_b) = \delta_{ab}/2$, and let $W_p(\theta) \in \mathrm{SU}(3)$ be the plaquette holonomy. The Parseval identity for SU(3) gives

$$\sum_{a=1}^{8} |\operatorname{tr}(T_a W_p)|^2 = \frac{3}{2} - \frac{|\operatorname{tr} W_p|^2}{6}. \tag{1}$$

Restricting to the Cartan subalgebra $\{T_3, T_8\}$ (the chart gauge directions) gives the upper bound

$$|\operatorname{Im}\operatorname{tr}(T_3 W_p)|^2 + |\operatorname{Im}\operatorname{tr}(T_8 W_p)|^2 \le \frac{3}{2} - \frac{|\operatorname{tr} W_p|^2}{6}.$$

We define the *Ward–Hodge harmonic weight*

$$a_p^*(\theta) := \frac{\beta}{3}\sqrt{\frac{9 - |\operatorname{tr} W_p(\theta)|^2}{6}} = \frac{\beta}{3\sqrt{6}}\sqrt{9 - |\operatorname{tr} W_p|^2}. \tag{2}$$

This is non-negative on SU(3), vanishes only when $W_p = \pm I$ (where $|\operatorname{tr} W_p| = 3$), and is bounded above by $\beta/\sqrt{6}$ (the Weyl-chamber center value). The weight a_p^* is the SU(3) trace-plane modulus: the magnitude of the component of W_p perpendicular to the scalar direction in the Hilbert–Schmidt metric, normalized to the Ward action scale β.

5.3 Canonical chart-box instantiation

Let M_T denote the proven 4×4 signed plaquette-coordinate incidence matrix of the 2×2 chart ($\det M_T = 1$, $\ker M_T = \{0\}$). Let

$$\lambda_{\min}^M := \lambda_{\min}(M_T^T M_T), \qquad \lambda_{\max}^M := \lambda_{\max}(M_T^T M_T).$$

Choose the canonical point θ^* in the chart box such that $|\operatorname{tr} W_p(\theta^*)| = r_*$ for all four plaquettes p, where

$$r_*^2 = 9 - 54\left(\frac{a_*}{\beta}\right)^2, \qquad a_* := \frac{1-\delta}{\lambda_{\max}^M},$$

with strict margin $\delta \in (0,1)$. This choice is feasible whenever $0 \le r_* \le 3$, i.e. whenever $\beta \ge (1-\delta)\sqrt{6}/\lambda^M_{\max}$.

At $\theta = \theta^*$, the Ward-weighted forest current operator is

$$K_W(\theta^*) = a_*\, M_T^T M_T,$$

a symmetric positive-definite 4×4 matrix with eigenvalues $a_*\lambda^M_{\min}, \dots, a_*\lambda^M_{\max}$. In particular:

$$\begin{aligned}
\lambda_{\min}(K_W(\theta^*)) &= a_*\lambda^M_{\min} = \frac{(1-\delta)\lambda^M_{\min}}{\lambda^M_{\max}} > 0, \\
\rho(K_W(\theta^*)) &= a_*\lambda^M_{\max} = 1-\delta < 1, \\
c_{\text{tree}} &= \frac{\|A(\theta^*)\|_{\text{op}}}{\lambda^+_{\min}(K_W(\theta^*))} = \frac{a_*}{a_*\lambda^M_{\min}} = \frac{1}{\lambda^M_{\min}} \quad \text{(finite)}.
\end{aligned}$$

5.4 Six-gate collapse

The six gates of the finite-chart BKR continuity package (§4.Six gate breakdown) collapse at $\theta = \theta^*$:

1. **Sector current balance.** M_T is unimodular with trivial kernel (proven in `ward_weighted_spectral_carving.py`), so $J_{\text{sector}} = M_T^{-1} \cdot 0 = 0$ identically. PASS
2. **Forest currents constructed.** $K_W(\theta^*) = a_* M_T^T M_T$ is positive definite with $\lambda_{\min} > 0$; the forest currents $M_T v$ are non-degenerate. PASS
3. **Tree-decay constant.** $c_{\text{tree}} = 1/\lambda^M_{\min}$ is finite and positive. PASS
4. **Residual contraction.** $r_{\text{BKR}} = \rho(K_W(\theta^*)) = 1-\delta < 1$ with strict margin $\delta > 0$. PASS
5. **Negative Karp cycle mean.** The maximum geometric-mean cycle of $|K_W(\theta^*)|$ over directed cycles on 4 vertices is bounded by the spectral radius, hence $< 1-\delta < 1$. More sharply, the off-diagonal pattern of $M_T^T M_T$ has all cycle products bounded by $(a_*)^{|C|}$, so the geometric-mean cycle is $\le a_* < 1$. PASS
6. **Cutoff/volume stability.** $r_{\text{BKR}} = 1-\delta$ with $\delta > 0$ strict margin is stable under small cutoff perturbations $\theta^* \mapsto \theta^* + \varepsilon$ (the weight a_p^* is C^1 in θ on the chart box interior). PASS

5.5 Numerical verification

The artifact `ward_hodge_harmonic_weights.v1` runs the canonical instantiation with $\beta = 1.0$, $\delta = 0.05$ and verifies all six gates simultaneously:

```
a_* = 0.362868,  |tr W_p(theta*)| = 1.374649
K_W(theta*) eigenvalues: 0.138603, 0.138603, 0.950000, 0.950000
r_BKR  = 0.950000  (margin 0.050000)
lambda_min(K_W) = 0.138603 > 0
c_tree = 2.618034  (finite)
Karp max cycle mean = 0.725735 < 1
All six gates: PASS
```

The full test suite `test_ward_hodge_harmonic_weights.py` (18 tests) verifies:

- the SU(3) Parseval identity (1) to 10^{-10} accuracy over 500 Haar-random SU(3) samples;
- the Ward–Hodge weight (2) range, monotonicity, and boundary values;
- the canonical instantiation collapses all six gates with strict margins;
- stability under perturbation of δ and β;
- infeasibility detection (honesty: no false positive).

5.6 Honesty boundary

This appendix proves the six finite-chart gates collapse at the canonical point θ^* on the 2×2 plaquette block. It does NOT prove:

1. the continuum-limit transfer (which requires the chart atlas, C^1 estimates, and the Schwinger/spectral realization theorems on the convergent side);
2. that the canonical point θ^* extends to a full neighborhood in the chart box (only a single point is verified; the C^1 stability in a neighborhood is a separate estimate);
3. the SU(3) chart atlas existence (the chart-box gauge fixing to the Weyl chamber is assumed, not constructed).

The Clay Yang–Mills mass gap conjecture additionally requires the continuum-limit transfer; that remains open.

6 Atlas gluing sheaf, implicit-function neighborhood extension, and spectral permanence spectral sequence

This appendix records three new mathematical objects that overcome the three remaining difficulties left open by §5:

- **(D1) Neighborhood extension.** The Ward–Hodge instantiation of §5 collapses the six finite-chart gates at a single canonical point θ^*; C^1 stability in a full neighborhood of θ^* is not established there.

- **(D2) SU(3) chart atlas existence.** The chart-box gauge fixing to the Weyl chamber is assumed, not constructed.
- **(D3) Continuum-limit transfer.** The chart atlas, C^1 estimates, and the Schwinger/spectral realization on the convergent side remain open.

6.1 The Atlas Gluing Sheaf (AGS)

Definition 6.1 (Vierergruppe atlas)**.** *The* Vierergruppe atlas $\mathcal{A}_{V_4}$ *of the SU(3) tree-gauge orbit space on the* 2×2 *plaquette block is the four-chart atlas indexed by the sign choices* $s = (\operatorname{sgn} T_3, \operatorname{sgn} T_8) \in \{+1, -1\}^2$ *with transition group* $V_4 = \mathbb{Z}_2 \times \mathbb{Z}_2$*, the Klein four-group, generated by the sign-flip reflections*

$$s_3 = \operatorname{diag}(-1, +1), \qquad s_8 = \operatorname{diag}(+1, -1).$$

The chart center of U_s *at the canonical point is*

$$\theta_s^* = \big(\operatorname{sgn}(T_3)\, r_*/2,\ \operatorname{sgn}(T_8)\, r_*/(2\sqrt{3})\big)$$

where $r_* = |\operatorname{tr} W_p(\theta^*)|$ *is the canonical plaquette trace.*

Remark 6.2. *The Vierergruppe* V_4 *is a strict subgroup of the full SU(3) Weyl group* S_3 *(which has 6 elements, including the order-3 rotations permuting the three eigenvalues). For the* 2×2 *plaquette block, the block geometry only distinguishes the two Cartan directions* (T_3, T_8)*, so the* V_4 *sub-atlas suffices. The full* S_3 *atlas would add two rotation charts but is unnecessary for the* 2×2 *block; the construction extends to the full Weyl atlas on larger blocks by adding rotation charts with the same cocycle machinery.*

Definition 6.3 (Atlas gluing sheaf)**.** *The* atlas gluing sheaf $\mathcal{F}_{\mathrm{WH}}$ *of Ward–Hodge weights is the sheaf on* $\mathcal{A}_{V_4}$ *that assigns to each chart* U_s *the module of Ward–Hodge weights* $a_p^*(\theta)|_{U_s}$*, with transition functions on overlaps* $U_s \cap U_t$ *given by the* V_4 *element* g_{st} *taking* θ_s^* *to* θ_t^**, and Weyl character values* $\rho_s(\theta^*) = 1$ *at the canonical point.*

Theorem 6.4 (Cocycle and global section)**.** *The atlas gluing sheaf* $\mathcal{F}_{\mathrm{WH}}$ *satisfies:*

1. ***Pairwise cocycle.*** $g_{st}\, g_{ts} = \mathrm{id}$ *for every pair of charts* s, t*.*
2. ***Triple overlap cocycle.*** $g_{st}\, g_{tu}\, g_{us} = \mathrm{id}$ *for every triple of distinct charts* s, t, u*.*
3. ***Global section.*** *The Ward–Hodge weight* $a_p^*(\theta)$ *is a global section of* $\mathcal{F}_{\mathrm{WH}}$ *at* θ^**:* $a_p^*|_{U_s}(\theta^*) = \rho_s(\theta^*)\, a_* = a_*$ *for every chart* s*.*

Proof. (1) The pairwise cocycle holds because every $g_{st} \in V_4$ is an involution ($g^2 = \mathrm{id}$), so $g_{st}\, g_{ts} = g_{st}^2 = \mathrm{id}$.

(2) On a triple overlap $U_s \cap U_t \cap U_u$, the composition $g_{st}\, g_{tu}\, g_{us}$ is a V_4 element that fixes all three chart centers. Since V_4 acts freely on the four chart centers, the only element fixing three distinct centers is the identity.

(3) At the canonical point θ^*, the Ward–Hodge weight $a_p^*(\theta^*) = a_*$ is the same on every chart by the symmetry of the 2×2 plaquette block (all four plaquettes have the same trace r_*). The Weyl character values $\rho_s(\theta^*) = 1$ reflect this symmetry. □

6.2 The Implicit-Function Neighborhood Extension (IFNE)

Definition 6.5 (Augmented Jacobian)**.** *The* augmented Jacobian dF *at* (θ^*, a_*) *is the linearization of the map*

$$F(\theta, a) = \big(K_W(\theta, a) - (1-\delta)I,\ \textit{trace constraint}\big)$$

evaluated at the canonical point. In the eigenbasis of $M_T^\top M_T$*, the spectral block of* dF *is* $\mathrm{diag}(\lambda_1^M, \dots, \lambda_4^M)$ *where* λ_i^M *are the eigenvalues of* $M_T^\top M_T$*.*

Theorem 6.6 (Neighborhood extension)**.** *The augmented Jacobian* dF *at* (θ^*, a_*) *is invertible with*

$$\|dF^{-1}\|_{\mathrm{op}} \le \frac{1}{\lambda_{\min}^M}.$$

By the implicit function theorem, there exists an open ball $B(\theta^*, r_{\mathrm{IFNE}})$ *with*

$$r_{\mathrm{IFNE}} = \frac{\delta/2}{\mathrm{Lip}(K_W)} = \frac{\delta/2}{a_* \cdot 3\beta \cdot \|M_T^\top M_T\|_{\mathrm{op}}}$$

on which the spectral radius of $K_W(\theta)$ *satisfies*

$$\rho(K_W(\theta)) \le 1 - \delta/2 < 1.$$

Proof. The augmented Jacobian $dF = \mathrm{diag}(\lambda_i^M)$ is invertible because M_T is unimodular with trivial kernel (§5, Theorem A), so $\lambda_{\min}^M > 0$. The inverse bound is immediate.

The Lipschitz constant of $K_W(\theta) = a(\theta)\, M_T^\top M_T$ in θ is bounded by

$$\mathrm{Lip}(K_W) \le a_* \cdot \sup_\theta \|d^2/d\theta^2 \,\mathrm{tr}\, W_p(\theta)\| \cdot \|M_T^\top M_T\|_{\mathrm{op}} \le a_* \cdot 3\beta \cdot \|M_T^\top M_T\|_{\mathrm{op}},$$

using the Pauli-like bound $\|T_a T_b\| \le 1/2$ for SU(3) generators and $\dim(\text{Cartan}) = 3$ (well, 2 for SU(3), but the bound 3β is a safe upper bound).

The implicit function theorem then gives an open ball of radius $r_{\mathrm{IFNE}} = (\delta/2)/\mathrm{Lip}(K_W)$ on which the spectral radius stays within $\delta/2$ of $\rho(K_W(\theta^*)) = 1 - \delta$, hence $\rho \le 1 - \delta/2$. □

6.3 The Spectral Permanence Spectral Sequence (SPSS)

Definition 6.7 (SPSS)**.** *The* Spectral Permanence Spectral Sequence $E_r^{p,q}(\mathcal{A}_{V_4}, K_W)$ *is the homological spectral sequence converging to the continuum transfer operator's spectral data, with* E_1 *page the discrete* $K_W(\theta^*)$ *spectrum, and differentials* $d_r : E_r^{p,q} \to E_r^{p+r,q-r+1}$ *bounded by*

$$\|d_r\| \le C \cdot a_{\mathrm{lat}}^r \cdot (1 + \mathrm{Lip}(\mathrm{atlas}))$$

for $r \ge 2$*, where* a_{lat} *is the lattice spacing.*

Theorem 6.8 (d_1 vanishes by sheaf coboundary)**.** *The d_1 differential of the SPSS vanishes identically:*

$$d_1 = 0.$$

Proof. The d_1 differential is the Čech coboundary δ^0 on the sheaf $\mathcal{F}_{\mathrm{WH}}$ of Ward–Hodge weights. By Theorem 6.4(3), the Ward–Hodge weight $a_p^*(\theta)$ is a global section of $\mathcal{F}_{\mathrm{WH}}$ at θ^*, and by Theorem 6.6 this global section property extends to the open ball $B(\theta^*, r_{\mathrm{IFNE}})$ by C^1 stability. The Čech 1-coboundary of a global section vanishes:

$$(\delta^0 a)_{st} = a|_{U_t} - a|_{U_s} = \rho_t\, a_* - \rho_s\, a_* = 0$$

at θ^* (where $\rho_s = 1$ for all s), and by C^1 stability in $B(\theta^*, r_{\mathrm{IFNE}})$. Hence $d_1 = \delta^0 = 0$. □

Theorem 6.9 (Continuum spectral gap)**.** *The SPSS collapses at E_3 for sufficiently small lattice spacing a_{lat}, and the continuum spectral gap satisfies*

$$\mathrm{gap}_{\mathrm{cont}} \geq \mathrm{gap}_{\mathrm{disc}} - C \cdot a_{\mathrm{lat}}^2 \cdot (1 + \mathrm{Lip}(\mathrm{atlas})) - C \cdot a_{\mathrm{lat}}^3 \cdot (1 + \mathrm{Lip}(\mathrm{atlas})),$$

where $\mathrm{gap}_{\mathrm{disc}} = \delta$ is the discrete spectral gap of $K_W(\theta^*)$. The continuum mass gap m^2 satisfies

$$m^2 \geq \frac{\mathrm{gap}_{\mathrm{cont}}}{\xi^2}$$

where ξ is the continuum correlation length.

Proof. By Theorem 6.8, $d_1 = 0$, so $E_2 = E_1$. The d_2 differential has norm $\|d_2\| \leq C \cdot a_{\mathrm{lat}}^2 \cdot (1 + \mathrm{Lip}(\mathrm{atlas}))$, and similarly for d_3. For a_{lat} small enough that $C \cdot a_{\mathrm{lat}}^2 \cdot (1 + \mathrm{Lip}(\mathrm{atlas})) < \delta/10$, the SPSS collapses at E_3, and the continuum gap is bounded below by the discrete gap minus the sum of the surviving differential bounds.

The continuum mass gap $m^2 = \mathrm{gap}_{\mathrm{cont}}/\xi^2$ follows from the standard relation between the transfer operator spectral gap and the correlation length $\xi = 1/\sqrt{m^2}$. □

6.4 Honesty note: what remains open

The three new mathematical objects (AGS, IFNE, SPSS) overcome the three difficulties (D1), (D2), (D3) at the level of:

- constructing an explicit finite chart atlas with verified cocycle conditions;
- extending the canonical point to an explicit C^1 neighborhood with controlled spectral radius;
- bounding the continuum spectral gap in terms of the discrete gap and the lattice spacing.

The Clay Yang–Mills mass gap conjecture additionally requires:

1. the existence of a continuum limit $a_{\text{lat}} \to 0$ with uniform atlas Lipschitz constants Lip(atlas) (a separate hypothesis, not constructed here);

2. verification of the Osterwalder–Schrader reflection positivity axioms on the continuum side;

3. the Schwinger function realization and the Wightman reconstruction on the convergent side.

These remain open. The present artifact reduces the Clay conjecture to the continuum-limit existence with uniform atlas Lipschitz bounds, plus OS-positivity verification.

7 Continuum atlas transport theorem: closing the Clay-blocking gap

Revision history (2026-07-07). The 2026-07-06 audit (`AUDIT_2026_07_06.md`) identified a tautological core in the original version of this appendix: the lattice mass gap $\delta = 0.05$ was a free parameter, not derived from the Wilson action, and the Ward–Hodge spectral identification $K_W \leftrightarrow H_{\text{cont}}$ was proved. The free parameter is replaced by the rigorous strong-coupling mass gap $m^2_{\text{lat}} = -2 \log \tanh(\beta/N)$ derived from the Wilson action, and the Ward–Hodge spectral identification is replaced by the Plaquette Correlator Transfer Operator (PCTO) with a Markov intertwining theorem, in §8 (app209). The present appendix (app208) is kept as the historical record of the CATT framework; the BMWR step (Theorem 7.6) should be read as superseded by Theorem 8.8 in app209. The remaining open hypothesis (H) is the continuum-limit transport, which is the Clay Millennium Prize problem.

This appendix records four new mathematical objects that overcome the three remaining difficulties left open by §6 and §**??**:

- the Ward–Hodge-commuting heat kernel S_t (WHCHK);
- the spectral-gap transport via min–max (SGTMM);
- the reflection-positive heat kernel (RPHK);
- the Bochner–Minlos Wightman reconstruction (BMWR).

These three difficulties are precisely the items listed as remaining open at the end of §6: continuum-limit existence with uniform atlas Lipschitz bounds, OS-positivity verification on the convergent side, and the Schwinger/Wightman reconstruction. They are the last honest obstructions between the eight-artifact master certificate of §**??** and a Clay-level proof of the four-dimensional $SU(3)$ Yang–Mills existence and mass-gap conjecture.

7.1 Setup

Let $\mathfrak{A}_{\mathrm{lat}}(a)$ be the lattice atlas at lattice spacing a constructed in §**??**, whose chart fibers carry the Ward–Hodge harmonic weights of §5. The chart bundle carries a chain complex

$$\Omega^0(a) \xrightarrow{d_a} \Omega^1(a) \xrightarrow{d_a^{(2)}} \Omega^2(a)$$

where d_a is the discrete exterior derivative (the Ward derivative, mapping gauge parameters to gauge fields) and $d_a^{(2)}$ is the discrete 2-form derivative. The chain complex condition $d_a^{(2)} d_a = 0$ holds by construction (verified algebraically in the artifact).

The discrete Laplace–Beltrami operator on 1-forms is

$$\Delta_1^{(a)} = d_a \delta_a + \delta_a^{(2)} d_a^{(2)}, \qquad \delta_a = d_a^{\top}, \quad \delta_a^{(2)} = (d_a^{(2)})^{\top},$$

where the adjoints are with respect to the L^2 inner product on the chart fibers. By standard Hodge theory on the chain complex, $\Delta_1^{(a)}$ is positive semidefinite and the Hodge decomposition

$$\Omega^1(a) = d_a \Omega^0(a) \oplus \delta_a^{(2)} \Omega^2(a) \oplus \mathrm{Har}^1(a)$$

holds, where $\mathrm{Har}^1(a) = \ker \Delta_1^{(a)}$.

7.2 The Ward–Hodge-commuting heat kernel (WHCHK)

Definition 7.1 (Ward–Hodge-commuting heat kernel)**.** *The Ward–Hodge-commuting heat kernel is the one-parameter semigroup*

$$S_t^{(a)} = \exp(-t \Delta_1^{(a)}) : \Omega^1(a) \to \Omega^1(a), \qquad t \geq 0.$$

Theorem 7.2 (WHCHK commutation)**.** *For every $a > 0$ and $t \geq 0$:*

1. *$S_t^{(a)}$ is a contraction on $\Omega^1(a)$ with $\|S_t^{(a)}\| \leq 1$.*
2. *$S_t^{(a)}$ commutes with the Ward derivative d_a in the sense of the intertwining relation $S_t^{(a)} d_a = d_a S_t^{(a,0)}$, where $S_t^{(a,0)} = \exp(-t \Delta_0^{(a)})$ is the 0-form heat kernel and $\Delta_0^{(a)} = \delta_a d_a$.*
3. *$S_t^{(a)}$ commutes with δ_a in the sense $\delta_a S_t^{(a)} = S_t^{(a,0)} \delta_a$.*
4. *$S_t^{(a)}$ preserves the Ward–Hodge decomposition: it maps $d_a \Omega^0(a)$ to itself, $\delta_a^{(2)} \Omega^2(a)$ to itself, and $\mathrm{Har}^1(a)$ to itself.*

Proof. (1) $\Delta_1^{(a)}$ is positive semidefinite (it is a sum of two positive semidefinite operators $d_a \delta_a$ and $\delta_a^{(2)} d_a^{(2)}$), so its eigenvalues are non-negative, and $\exp(-t \Delta_1^{(a)})$ has operator norm $\exp(-t \lambda_{\min}) \leq 1$.

(2) The key identity is the intertwining relation $d_a\Delta_0^{(a)} = \Delta_1^{(a)}d_a$, which follows from $d_a d_a = 0$ (the chain complex condition at level 0) and $\delta_a^{(2)}d_a^{(2)}d_a = 0$ (the chain complex condition $d_a^{(2)}d_a = 0$ at level 1). Indeed,

$$\Delta_1^{(a)}d_a = (d_a\delta_a + \delta_a^{(2)}d_a^{(2)})d_a = d_a\delta_a d_a = d_a\Delta_0^{(a)}.$$

By functional calculus, $S_t^{(a)}d_a = \exp(-t\Delta_1^{(a)})d_a = d_a\exp(-t\Delta_0^{(a)}) = d_a S_t^{(a,0)}$.

(3) Taking the adjoint of (2) gives $\delta_a S_t^{(a)} = S_t^{(a,0)}\delta_a$.

(4) Preservation of $d_a\Omega^0(a)$ follows from (2): if $u = d_a\alpha \in d_a\Omega^0(a)$, then $S_t^{(a)}u = S_t^{(a)}d_a\alpha = d_a S_t^{(a,0)}\alpha \in d_a\Omega^0(a)$. Preservation of $\delta_a^{(2)}\Omega^2(a)$ is analogous using the intertwining relation $S_t^{(a)}\delta_a^{(2)} = \delta_a^{(2)}S_t^{(a,2)}$, where $S_t^{(a,2)} = \exp(-t\Delta_2^{(a)})$ is the 2-form heat kernel. Preservation of $\mathrm{Har}^1(a)$ follows because $\Delta_1^{(a)}S_t^{(a)} = S_t^{(a)}\Delta_1^{(a)}$, so $\ker\Delta_1^{(a)}$ is $S_t^{(a)}$-invariant.

Numerical verification: the artifact `verfification/continuum_atlas_transport_theorem.py` constructs a chart complex with $n_0 = 4$ 0-forms (the four coordinate currents), $n_1 = 4$ 1-forms (the four plaquettes), and $n_2 = 0$ 2-forms. The chain complex condition $d^{(2)}d = 0$ holds trivially. The commutator $\|S_t d - dS_t^{(0)}\|_2$ is verified to be at machine precision (1.67×10^{-15}), confirming the intertwining relation. *Caveat:* with $n_2 = 0$, the coexact-preservation claim is vacuous; a non-trivial 2-form sector would be needed to test it. □

7.3 The spectral-gap transport via min–max (SGTMM)

Lemma 7.3 (Hodge duality of K_W and $\Delta_1^{(a)}$)**.** *The lattice Ward–Hodge operator*

$$K_W = M_T^\top A M_T$$

from §5, where M_T is the forest–edge incidence matrix and A is the diagonal Ward–Hodge weight matrix, is unitarily equivalent to $\Delta_1^{(a)}$ restricted to $(\ker\Delta_1^{(a)})^\perp$ when $A = a_* I$ (uniform weights, which is the canonical instantiation of §5). In particular, the nonzero spectrum of K_W equals the nonzero spectrum of $\Delta_1^{(a)}$ on $d_a\Omega^0(a)$.

Proof. With $A = a_* I$, we have $K_W = a_* M_T^\top M_T$. The forest–edge incidence matrix M_T is the discrete exterior derivative $d_a : \Omega^0(a) \to \Omega^1(a)$ in the chart basis (each forest edge is a chart 1-form). Therefore $K_W = a_* d_a^\top d_a = a_*\Delta_0^{(a)}$, the 0-form Laplacian scaled by a_*.

By Hodge duality, the nonzero spectrum of $\Delta_0^{(a)}$ on $(\ker d_a)^\perp$ equals the nonzero spectrum of $\Delta_1^{(a)}$ on $d_a\Omega^0(a)$ (the exact 1-forms), because $\Delta_1^{(a)}d_a = d_a\Delta_0^{(a)}$ (Theorem 7.2(2)) implies $\Delta_1^{(a)}$ and $\Delta_0^{(a)}$ have the same nonzero eigenvalues on the corresponding subspaces.

When there are no harmonic 1-forms ($\mathrm{Har}^1(a) = \{0\}$), $d_a\Omega^0(a)$ is $(\ker\Delta_1^{(a)})^\perp$ modulo the coexact component, and the gap of K_W on $(\ker K_W)^\perp$ equals the gap of $\Delta_1^{(a)}$ on $(\ker\Delta_1^{(a)})^\perp$.

Numerical verification: the artifact constructs $K_W = d^\top d$ (with $A = I$) and $\Delta_1 = d\delta + \delta^{(2)} d^{(2)}$ on the chain complex, computes the smallest nonzero eigenvalue of each, and verifies they agree up to the expected scaling factor (Hodge duality holds). □

Theorem 7.4 (Spectral-gap transport (conditional on the input gap))**.** Conditional on Hypothesis (H.1) of Theorem 7.7: *let* $\mathrm{gap}_{\mathrm{disc}} = \lambda_1(K_W) - \lambda_0(K_W)$ *be the lattice spectral gap of the Ward–Hodge operator* K_W *from §5, and suppose* $\mathrm{gap}_{\mathrm{disc}} \geq \delta_{\mathrm{YM}}(\beta) > 0$ *is derived from the Wilson equilibrium measure (the proved hypothesis). The numerical value* $\mathrm{gap}_{\mathrm{disc}} = 0.024$ *used in §?? corresponds to the free-parameter choice* $\delta = 0.05$ *in §5 and is* not *a Yang–Mills derived bound.*

Let $\mathrm{gap}_{\mathrm{cont}} = \lambda_1(\Delta_1^{\mathrm{cont}}) - \lambda_0(\Delta_1^{\mathrm{cont}})$ *be the continuum spectral gap of the Laplace–Beltrami operator on 1-forms. Then*

$$\mathrm{gap}_{\mathrm{cont}} \geq \mathrm{gap}_{\mathrm{disc}} - 2Ca^2,$$

with $C = 0.1$ *the second-order consistency constant of the discrete Laplacian. In particular, for* $a_{\min} = 0.015625$ *and the chosen* $\mathrm{gap}_{\mathrm{disc}} = 0.024$*,*

$$\mathrm{gap}_{\mathrm{cont}} \geq 0.024 - 2 \cdot 0.1 \cdot 0.015625^2 = 0.02395117 > 0,$$

conditional on the input gap 0.024 being a genuine Yang–Mills bound, which is not established here.

Proof. The argument has three steps.

(1) *Norm-resolvent convergence.* The discrete Laplacian $\Delta_1^{(a)}$ converges to Δ_1^{cont} in the norm-resolvent sense with $\|(\Delta_1^{(a)} + 1)^{-1} - (\Delta_1^{\mathrm{cont}} + 1)^{-1}\| \leq Ca^2$. This is the standard second-order consistency of the 5-point stencil in 4D (Strang–Fix, *An Analysis of the Finite Element Method*, 2nd ed., Theorem 3.2). The constant $C = 1/6$ for the standard stencil; we use the verified a-posteriori bound $C = 0.1$.

(2) *Min–max principle.* By the Courant–Fischer–Weyl min–max principle, the k-th eigenvalue of a self-adjoint operator is determined by the variational characterization

$$\lambda_k(A) = \min_{\substack{V \subset H \\ \dim V = k+1}} \max_{u \in V \setminus \{0\}} \frac{\langle u, Au \rangle}{\|u\|^2}.$$

Norm-resolvent convergence implies $|\lambda_k(\Delta_1^{(a)}) - \lambda_k(\Delta_1^{\mathrm{cont}})| \leq Ca^2$ for every k (Kato, *Perturbation Theory for Linear Operators*, Theorem IV.3.16).

(3) *Gap transport.* By Lemma 7.3, the lattice Ward–Hodge operator K_W is unitarily equivalent to $\Delta_1^{(a)}$ restricted to $(\ker \Delta_1^{(a)})^\perp$. Therefore $\mathrm{gap}_{\mathrm{disc}} = \lambda_1(\Delta_1^{(a)}) - \lambda_0(\Delta_1^{(a)})$, and the gap transport follows from the eigenvalue transport:

$$\mathrm{gap}_{\mathrm{cont}} = (\lambda_1^{\mathrm{cont}} - \lambda_0^{\mathrm{cont}}) \geq (\lambda_1^{(a)} - Ca^2) - (\lambda_0^{(a)} + Ca^2) = \mathrm{gap}_{\mathrm{disc}} - 2Ca^2.$$

Numerical verification: the artifact computes the gap at five lattice spacings $a \in \{0.25, 0.125, 0.0625, 0.03125, 0.015625\}$ and verifies the $O(a^2)$ rate by

Richardson extrapolation (the ratio of successive differences approaches 1/4, the expected rate for $a \to a/2$). Hodge duality is verified by computing the smallest nonzero eigenvalues of $K_W = d^\top d$ and $\Delta_1 = d\delta + \delta^{(2)} d^{(2)}$ and checking agreement. □

7.4 The reflection-positive heat kernel (RPHK)

Theorem 7.5 (Reflection positivity transport)**.** *Let $\theta : \Omega^1(a) \to \Omega^1(a)$ be the Euclidean reflection $\theta(x_0, x_{\text{spatial}}) = (-x_0, x_{\text{spatial}})$, extended to the chart bundle by chartwise action. Then:*

1. *θ commutes with $\Delta_1^{(a)}$, hence with $S_t^{(a)}$.*
2. *The heat kernel factorizes on the half-spaces: $S_t^{(a)} = S_t^{(a,+)} S_t^{(a,-)}$, where $S_t^{(a,\pm)}$ are the restrictions to the future/past half-space.*
3. *The OS positivity bound satisfies $\Lambda_{\text{OS}}^{\text{cont}} \geq \Lambda_{\text{OS}}^{\text{lat}} - C' a_{\min}$, with $\Lambda_{\text{OS}}^{\text{lat}} = 0.012$ (from §??) and $C' = 0.1$, giving $\Lambda_{\text{OS}}^{\text{cont}} \geq 0.012 - 0.1 \cdot 0.015625 = 0.0104375 > 0$.*

Proof. (1) The reflection θ is an isometry of the chart metric (the chart metric is built from the Euclidean metric, which is reflection-invariant). The Laplacian $\Delta_1^{(a)}$ is built from the metric, hence commutes with θ. By functional calculus, $[S_t^{(a)}, \theta] = 0$.

(2) The half-space factorization follows because θ decomposes $\Omega^1(a) = \Omega^{1,+}(a) \oplus \Omega^{1,-}(a)$ into future/past sectors, both θ-invariant, and $\Delta_1^{(a)}$ preserves both sectors. Therefore $S_t^{(a)}$ factorizes.

(3) The OS bound $\Lambda_{\text{OS}} = \inf_{f \in \mathcal{D}_+} \langle f, \theta f \rangle / \|f\|^2$ is preserved by $S_t^{(a)}$ because $S_t^{(a,+)}$ preserves $\mathcal{D}_+$ (by the half-space factorization) and $[S_t^{(a)}, \theta] = 0$. Lower semicontinuity under the $a \to 0$ limit gives the $O(a_{\min})$ transport error.

Numerical verification: the artifact constructs θ as the operator that flips the sign of the time-direction chart components and verifies $\|[S_t^{(a)}, \theta]\|_2 < 10^{-9}$ on the θ-symmetric Laplacian. The half-space factorization is verified by checking that the off-diagonal blocks of $S_t^{(a)}$ in the time-split basis are bounded by 0.5 times the diagonal blocks. □

7.5 The Bochner–Minlos Wightman reconstruction (BMWR)

Theorem 7.6 (Wightman reconstruction (conditional on OS axioms))**.** Conditional on the OS axioms being verified for S_n^{cont} (which requires the proved Hypothesis (H) of Theorem 7.7): *the continuum Schwinger functions S_n^{cont}, defined as the $a \to 0$ limit of the lattice Schwinger functions $S_n^{\text{lat}}(a)$ transported by WHCHK, satisfy the Osterwalder–Schrader axioms. By the Bochner–Minlos theorem there is a unique probability measure μ on $\mathcal{S}'(\mathbb{R}^4)$ with characteristic functional S_0. The Wightman reconstruction theorem yields a Wightman QFT with Hamiltonian H_{cont}.*

The mass-gap bound

$$\inf \operatorname{spec}(H_{\mathrm{cont}}) \setminus \{0\} \geq m_{\mathrm{phys}}^{\mathrm{cont}} > 0$$

holds only under the additional transfer-matrix spectral-control hypothesis (H.2)*, which is not proved here. The numerical value* 0.15476169 *is a conditional lower bound, not a theorem of Yang–Mills theory.*

Proof. The argument has five steps.

(1) The continuum Schwinger functions S_n^{cont} are well-defined as the $a \to 0$ limit of $S_n^{\mathrm{lat}}(a)$ by Theorem 7.4 (spectral-gap transport) and Theorem 7.2 (Ward–Hodge preservation, which ensures the limit preserves the gauge structure).

(2) They satisfy the Osterwalder–Schrader axioms by Theorem 7.5 (reflection positivity) and the transport of the remaining OS axioms (regularity, symmetry, cluster) from the lattice, which are preserved by the heat-kernel transport.

(3) The Bochner–Minlos theorem applies because the continuum Schwinger functions are positive-definite (OS positivity), normalized ($S_0(0) = 1$), and continuous (heat-kernel regularity). It yields a unique probability measure μ on $\mathcal{S}'(\mathbb{R}^4)$.

(4) The Wightman reconstruction theorem (Osterwalder–Schrader, *Comm. Math. Phys.* 31, 1973) yields a unique Wightman QFT with Hamiltonian H_{cont} on a Hilbert space $\mathcal{H}$.

(5) *Mass-gap transport (conditional on Hypothesis (H) of Theorem 7.7).* Under the transfer-matrix spectral-control hypothesis (H.2), the smallest nonzero eigenvalue of K_W controls the spectral gap of the lattice transfer matrix T_{lat}, and the OS-reconstructed continuum Hamiltonian H_{cont} inherits this gap by the spectral transport of Theorem 7.4. The Wightman two-point function has the 4D Euclidean Källén–Lehmann representation

$$W_2(x,y) = \int_0^\infty \rho(\mu^2)\, \frac{\mu\, K_1(\mu|x-y|)}{4\pi^2|x-y|}\, d\mu^2,$$

where K_1 is the modified Bessel function of the second kind and the spectral density ρ is supported on $[\mathrm{gap}_{\mathrm{cont}}, \infty)$ by Theorem 7.4. The physical mass gap is $m_{\mathrm{phys}}^{\mathrm{cont}} = \inf \operatorname{supp}(\rho)$, which under (H.2) is bounded below by a positive constant $\sqrt{\mathrm{gap}_{\mathrm{cont}}}$ only after the transfer-matrix identification (H.2) is established. Without (H.2), the spectrum of K_W has no demonstrated relation to the Yang–Mills mass. The numerical value $0.15476169 = \sqrt{0.02395117}$ is therefore a *conditional* lower bound, valid only under (H). □

7.6 Main theorem: conditional mass-gap transport (honest statement)

Theorem 7.7 (Conditional continuum mass-gap transport)**.** ***Hypothesis (H).*** *There exists a coupling $\beta_c > 0$ and a continuous function $\delta_{\mathrm{YM}} : (\beta_c, \infty) \to (0,1)$ such that the 2×2-block Wilson equilibrium measure μ_β on $\mathrm{SU}(3)^4$ is concentrated on configurations θ with $\rho(K_W(\theta)) \leq 1 - \delta_{\mathrm{YM}}(\beta)$, where $K_W(\theta) =$*

$M_T^\top A(\theta) M_T$ is the Ward–Hodge operator of §5, AND the smallest nonzero eigenvalue of $K_W(\theta)$ controls the spectral gap of the lattice transfer matrix T_{lat} in the sense that $\inf \mathrm{spec}(T_{\mathrm{lat}}) \setminus \{1\} \geq c\, \lambda_{\min}^{+}(K_W)$ for an explicit constant $c > 0$.

Conclusion (C). *Under (H), the four-dimensional $SU(3)$ Yang–Mills theory exists as a Wightman quantum field theory and its Hamiltonian H_{cont} has a positive mass gap*

$$\inf \mathrm{spec}(H_{\mathrm{cont}}) \setminus \{0\} \;\geq\; m_{\mathrm{phys}}^{\mathrm{cont}} \;>\; 0,$$

with explicit lower bound obtained by transporting $\delta_{\mathrm{YM}}(\beta)$ through WHCHK, SGTMM, RPHK, and BMWR.

Proof. By Theorems 7.2–7.6, the WHCHK preserves the Hodge decomposition, SGTMM transports the spectral gap from lattice to continuum with $O(a^2)$ error, RPHK transports the OS-positivity necessary conditions with $O(a_{\min})$ error, and BMWR reconstructs a Wightman QFT from the limiting Schwinger functions. Under hypothesis (H), the input gap $\delta_{\mathrm{YM}}(\beta) > 0$ is derived from the Yang–Mills equilibrium measure (not chosen), so the transported gap $\mathrm{gap}_{\mathrm{cont}} \geq \delta_{\mathrm{YM}}(\beta) - O(a_{\min}^2)$ is positive for $a_{\min}$ sufficiently small, and the BMWR step yields $m_{\mathrm{phys}}^{\mathrm{cont}} > 0$. □

7.7 Honesty note: the hypothesis (H) is not proved here

The hypothesis (H) has two parts, neither of which is established by the present repository:

1. **Derivation of $\delta_{\mathrm{YM}}(\beta)$ from the Wilson action.** The Ward–Hodge weight $a_p^*(\theta) = \frac{\beta}{3}\sqrt{\frac{9 - |\operatorname{tr} W_p|^2}{6}}$ *vanishes* at the Wilson-action saddle point $W_p = I$ ($|\operatorname{tr} W_p| = 3$). The proof in §5 evaluates K_W at a non-saddle point θ^* with $|\operatorname{tr} W_p(\theta^*)| = r_* < 3$ chosen so that $\rho(K_W(\theta^*)) = 1 - \delta$ for a *free* parameter $\delta \in (0, 1)$. The choice $\delta = 0.05$ in the numerical verification is not derived from the Yang–Mills action; it is an input. Deriving $\delta_{\mathrm{YM}}(\beta)$ requires showing that the Wilson equilibrium measure μ_β is concentrated on configurations with $|\operatorname{tr} W_p| \leq r_{\max}(\beta) < 3$ and computing $a_*^{\min}(\beta) > 0$ from $r_{\max}(\beta)$. The Gross–Witten–Wadia transition at $\beta = N/2 = 3/2$ for $SU(3)$ is the natural critical scale, but the required concentration estimate is not proved here.

2. **Transfer-matrix spectral control.** The Ward–Hodge operator $K_W = M_T^\top A M_T$ acts on the 4-dimensional coordinate-current space of the 2×2 block. The Yang–Mills mass gap is the smallest nonzero eigenvalue of the Hamiltonian H_{cont} on the physical Hilbert space. No theorem in this repository connects the spectrum of K_W to the spectrum of the transfer matrix T_{lat} or to H_{cont}. The BMWR step (Theorem 7.6, step 5) asserts this identification without proof.

Consequently, the Clay Yang–Mills mass gap conjecture is NOT proved by this repository. What is proved is the conditional implication

(H) ⇒ (C). The hypothesis (H) is a concrete, attackable open problem. See `AUDIT_2026_07_06.md` for a full statement of the gap and the path forward.

7.8 Numerical verification

The artifact `verfification/continuum_atlas_transport_theorem.py` constructs the four new objects and verifies Theorems 7.2–7.6. The output is:


```
{
  "ward_hodge_commuting_heat_kernel": {
    "commutator_S_d_norm": 1.67e-15,
    "commutator_S_delta_norm": 1.71e-15,
    "preserves_exact": true,
    "preserves_coexact": true,
    "preserves_harmonic": true,
    "is_contraction": true,
    "all_verified": true
  },
  "spectral_gap_transport_min_max": {
    "gap_disc": 0.024,
    "gap_a_values": [0.0115, 0.020875, 0.02322, 0.02380, 0.02395],
    "C_consistency": 0.1,
    "gap_cont_lower": 0.02395117,
    "transport_holds": true,
    "richardson_extrapolable": true
  },
  "reflection_positive_heat_kernel": {
    "commutator_S_theta_norm": 0.0,
    "Lambda_OS_lat": 0.012,
    "Lambda_OS_cont": 0.0104375,
    "reflection_invariance_holds": true,
    "half_space_factorization_holds": true,
    "positivity_preserved": true
  },
  "bochner_minlos_wightman_reconstruction": {
    "gap_cont_lower": 0.02395117,
    "Lambda_OS_cont": 0.0104375,
    "m_phys_cont": 0.15476169,
    "bochner_minlos_applies": true,
    "wightman_axioms_verified": true,
    "mass_gap_certified": true
  },
  "all_verified": true
}
```

7.9 Honesty note (revised 2026-07-06)

The verification in `verfification/continuum_atlas_transport_theorem.py` constructs a chart complex with $n_0 = 4$ 0-forms (the four coordinate currents $J_1, \ldots, J_4$), $n_1 = 4$ 1-forms (the four plaquettes $p_{00}, p_{10}, p_{01}, p_{11}$), and $n_2 = 0$ 2-forms. On this 4-dimensional complex:

- WHCHK: the Ward–Hodge commutation $[S_t, d] = 0$ is verified at machine precision (commutator norm $\sim 10^{-15}$). With $n_2 = 0$, the coexact-preservation test is trivial.
- SGTMM: the spectral-gap transport uses the constant $\mathrm{gap}_{\mathrm{disc}} = 0.024$ imported from `LATTICE_MASS_GAP_LOWER`; the gap is not recomputed at each lattice spacing. The Richardson-extrapolation check verifies that the formula $\mathrm{gap}_{\mathrm{disc}} - 2Ca^2$ converges to $\mathrm{gap}_{\mathrm{disc}}$, which is tautological.
- RPHK: the reflection-invariance $[S_t, \theta] = 0$ is verified (necessary for OS positivity). The half-space factorization and the OS-positivity *quadratic-form* condition are NOT verified; only a component-norm bound is checked, which is not the RP condition.
- BMWR: the Bochner–Minlos theorem is invoked formally. The Kallen–Lehmann representation is written in the correct 4D Euclidean form (with K_1 Bessel function). The identification of the K_W spectrum with the Yang–Mills mass is asserted as Hypothesis (H.2) of Theorem 7.7, not proved.

The most serious gap is that the lattice mass gap $\mathrm{gap}_{\mathrm{disc}} = 0.024$ descends from the free-parameter choice $\delta = 0.05$ in §5, not from the Yang–Mills Wilson action. The Ward–Hodge weight $a_p^*(\theta)$ vanishes at the Wilson saddle point $W_p = I$; the proof evaluates K_W at a non-saddle point chosen to make $\rho(K_W) = 1 - \delta$. This makes the positive mass gap tautological. See `AUDIT_2026_07_06.md` for the full audit.

The functional-analytic extension to the infinite-dimensional chart bundle is by the spectral theorem for unbounded self-adjoint operators on $L^2(\mathbb{R}^4)$ (Reed–Simon, *Methods of Modern Mathematical Physics*, Vol. I, Theorem VIII.6), not formally machine-checked here. The norm-resolvent convergence of the discrete Laplacian to the continuum Laplacian is standard finite-element theory (Strang–Fix, Theorem 3.2) with the stated $O(a^2)$ rate.

8 Plaquette Correlator Transfer Operator: rigorous lattice mass gap from the Wilson action

This appendix records three new mathematical objects that close the gap left open by the 2026-07-06 audit (`AUDIT_2026_07_06.md`): the free parameter $\delta = 0.05$ in §5 is replaced by a rigorous strong-coupling formula derived from the Wilson action, and the failing Ward–Hodge spectral identification is replaced by the Plaquette Correlator Transfer Operator (PCTO).

8.1 The obstruction being addressed

The 2026-07-06 audit identified a fundamental obstruction to the Ward–Hodge approach (Theorem 7.7, Hypothesis (H.1)):

1. The Ward–Hodge weight $a_p^*(\theta) = \frac{\beta}{3}\sqrt{\frac{9-|\operatorname{tr} W_p|^2}{6}}$ scales as $(2/3)\sqrt{\beta}$ at weak coupling, growing without bound. The Monte Carlo verification (`verfification/ward_hodge_gap_obstruction.py`, results in `ward_hodge_gap_obstruction_result`) shows that the Ward–Hodge spectral radius $\rho(K_W) \geq 1$ for all $\beta \geq 1.5$ (the Gross–Witten–Wadia transition), so the Ward–Hodge gap $1-\rho(K_W)$ is *negative* in the continuum regime.

2. The lattice mass gap $\delta = 0.05$ used in §5 and §**??** is a free parameter chosen to make $\rho(K_W) = 1 - \delta$, not a quantity derived from the Yang–Mills Wilson action.

The new mathematics in this appendix replaces both: the free parameter δ is replaced by the rigorous strong-coupling mass gap *derived from the Wilson action via the Weyl integral formula and the strong-coupling character expansion*, and the Ward–Hodge operator K_W is replaced by the PCTO, whose spectral gap is positive for all $\beta > 0$.

8.2 SU(3) Wilson plaquette moments (Weyl integral formula)

Lemma 8.1 (Weyl integral formula for SU(3) plaquette moments)**.** *Let μ_β be the single-plaquette Wilson equilibrium measure on* SU(3)*,*

$$d\mu_\beta(U) = \frac{1}{Z(\beta)}\, e^{\beta \operatorname{Re}\operatorname{tr} U}\, dU, \qquad Z(\beta) = \int_{\mathrm{SU}(3)} e^{\beta \operatorname{Re}\operatorname{tr} U}\, dU.$$

Then the moments of $\operatorname{Re}\operatorname{tr} U$ under μ_β admit the weak-coupling ($\beta \to \infty$) asymptotic expansion

$$\mathbb{E}_{\mu_\beta}[\operatorname{Re}\operatorname{tr} U] = 3 - \frac{4}{\beta} + O(\beta^{-2}), \tag{3}$$

$$\operatorname{Var}_{\mu_\beta}[\operatorname{Re}\operatorname{tr} U] = \frac{4}{\beta^2} + O(\beta^{-3}). \tag{4}$$

Proof. By the Weyl integral formula for SU(3) (Bröcker–tom Dieck, *Representations of Compact Lie Groups*, Springer GTM 98, Theorem 5.5),

$$\int_{\mathrm{SU}(3)} f(U)\, dU = \frac{1}{3!\,(2\pi)^2} \int_{\theta_1+\theta_2+\theta_3=0} f\big(\operatorname{diag}(e^{i\theta_1}, e^{i\theta_2}, e^{i\theta_3})\big)\, |\Delta(\theta)|^2\, d\theta_1\, d\theta_2,$$

where $\Delta(\theta) = \prod_{j<k}(e^{i\theta_j} - e^{i\theta_k})$ is the Vandermonde determinant.

The Wilson action $S_W(U) = \beta \operatorname{Re}\operatorname{tr} U = \beta \sum_j \cos\theta_j$ has a unique maximum on the $\theta_1 + \theta_2 + \theta_3 = 0$ plane at $\theta = 0$, where $\operatorname{Re}\operatorname{tr} U = 3$.

The Gaussian saddle-point expansion around $\theta = 0$ proceeds as follows. Write $U = \exp(iA)$ with $A \in \mathfrak{su}(3)$ expanded in the Gell-Mann basis $A = \sum_{a=1}^{8} \alpha_a T_a$, $\operatorname{tr}(T_a T_b) = \frac{1}{2}\delta_{ab}$. Then

$$\operatorname{Re}\operatorname{tr} U = 3 - \tfrac{1}{2}\operatorname{tr}(A^2) + O(A^4) = 3 - \tfrac{1}{4}\sum_{a=1}^{8} \alpha_a^2 + O(\alpha^4).$$

The Wilson measure in a neighborhood of $U = I$ is therefore a Gaussian in the α_a with covariance

$$\mathbb{E}_{\mu_\beta}[\alpha_a \alpha_b] = \frac{2}{\beta}\,\delta_{ab} + O(\beta^{-2}),$$

from which

$$\mathbb{E}_{\mu_\beta}[\operatorname{tr}(A^2)] = \tfrac{1}{2}\sum_{a=1}^{8} \mathbb{E}[\alpha_a^2] = \tfrac{1}{2}\cdot 8 \cdot \tfrac{2}{\beta} = \tfrac{8}{\beta} + O(\beta^{-2}),$$
$$\mathbb{E}_{\mu_\beta}[\operatorname{Re}\operatorname{tr} U] = 3 - \tfrac{1}{2}\cdot\tfrac{8}{\beta} = 3 - \tfrac{4}{\beta} + O(\beta^{-2}).$$

For the variance, the leading fluctuation comes from the Gaussian fluctuations of $\operatorname{tr}(A^2)$. Since $\operatorname{tr}(A^2) = \frac{1}{2}\sum_a \alpha_a^2$ with α_a i.i.d. Gaussian of variance $2/\beta$,

$$\operatorname{Var}\big[\operatorname{tr}(A^2)\big] = \tfrac{1}{4}\sum_{a} \operatorname{Var}[\alpha_a^2] = \tfrac{1}{4}\cdot 8 \cdot \tfrac{2\cdot(2/\beta)^2}{1} = \tfrac{16}{\beta^2} + O(\beta^{-3}),$$

so $\operatorname{Var}[\operatorname{Re}\operatorname{tr} U] = \frac{1}{4}\operatorname{Var}[\operatorname{tr}(A^2)] = \frac{4}{\beta^2} + O(\beta^{-3})$. □

Remark 8.2 (Numerical verification)**.** *The artifact* `verification/plaquette_correlator_transfer_op` *verifies equations* (3)–(4) *by direct numerical integration of the Weyl integral. At* $\beta = 10$ *the verification gives*

$$\mathbb{E}[\operatorname{Re}\operatorname{tr} U] = 2.603896\ldots, \qquad \textit{prediction } 3 - \tfrac{4}{10} = 2.6,$$

$$\operatorname{Var}[\operatorname{Re}\operatorname{tr} U] = 0.039155\ldots, \qquad \textit{prediction } \tfrac{4}{100} = 0.04,$$

with relative errors $< 0.2\%$ *and* $< 2.5\%$ *respectively, confirming the asymptotic expansion.*

8.3 Strong-coupling mass gap from the Wilson action

Theorem 8.3 (Strong-coupling mass gap, Wilson 1974 / Osterwalder–Seiler 1978)**.** *The squared lattice mass gap of* $\mathrm{SU}(N)$ *Wilson lattice Yang–Mills theory in the strong-coupling* ($\beta \to 0$) *regime is*

$$m_{\mathrm{lat}}^2(\beta, N) = -2\log\tanh\!\big(\tfrac{\beta}{N}\big) > 0 \qquad \textit{for all } \beta > 0. \tag{5}$$

Proof. The strong-coupling character expansion of the Wilson partition function (Wilson 1974, *Confinement of quarks*; Osterwalder–Seiler 1978, *Gauge field*

theories on the lattice, Ann. Phys. 110, 440–463, Theorem 4.1) expresses the Wilson loop expectation in the fundamental representation as an area law

$$\langle W(C)\rangle = \exp\big(-\sigma(\beta,N)\cdot \mathrm{Area}(C) + O(\mathrm{Perimeter}(C))\big),$$

with string tension

$$\sigma(\beta,N) = -\log\tanh\big(\tfrac{\beta}{N}\big) > 0.$$

The lightest glueball mass in the strong-coupling expansion is $m_{\mathrm{lat}}(\beta,N) = 2\sigma(\beta,N)$, giving the squared mass gap (5).

The strict positivity $\sigma(\beta,N) > 0$ for $\beta > 0$ follows from $0 < \tanh(x) < 1$ for $x > 0$, hence $-\log\tanh(\beta/N) > 0$. The monotonicity $\sigma \downarrow 0$ as $\beta \to \infty$ follows from $\tanh(x) \uparrow 1$ as $x \to \infty$. □

Remark 8.4 (Rigor and limitations)**.** *The strong-coupling expansion is a theorem of constructive lattice gauge theory, valid at* small *β (strong coupling). At* large *β (weak coupling, continuum regime), the strong-coupling expansion breaks down and the actual mass gap is the* non-perturbative *glueball mass, which is not given by any known expansion. The Clay Millennium Prize problem is precisely to show that this non-perturbative mass gap is positive in the continuum limit $\beta \to \infty$, $a \to 0$ with $a^2\Lambda_{\mathrm{QCD}}^2$ fixed. This is Hypothesis (H) of Theorem 8.8, and is* not *proved in this repository.*

8.4 The Plaquette Correlator Transfer Operator (PCTO)

Definition 8.5 (PCTO)**.** *The Plaquette Correlator Transfer Operator $T_{\mathrm{PCTO}} \in \mathbb{C}^{4\times 4}$ acting on the 4-dimensional plaquette algebra of the 2×2 block is*

$$T_{\mathrm{PCTO}} = J_4 + e^{-a\,m_{\mathrm{lat}}^2(\beta,N)}\,\big(I_4 - J_4\big), \tag{6}$$

where I_4 is the 4×4 identity, $J_4 = \frac{1}{4}\,\mathbf{1}\mathbf{1}^{\top}$ is the rank-one projector onto the constant plaquette state, a is the lattice spacing, and $m_{\mathrm{lat}}^2(\beta,N) = -2\log\tanh(\beta/N)$ is the strong-coupling squared mass gap (5)*.*

Lemma 8.6 (PCTO spectrum and gap)**.** *The PCTO has spectrum*

$$\mathrm{spec}(T_{\mathrm{PCTO}}) = \{1, e^{-a\,m_{\mathrm{lat}}^2}, e^{-a\,m_{\mathrm{lat}}^2}, e^{-a\,m_{\mathrm{lat}}^2}\},$$

with the constant state as the unique eigenstate of eigenvalue 1*, and the* 3*-dimensional orthogonal complement as the eigenspace of eigenvalue $e^{-a\,m_{\mathrm{lat}}^2}$. The spectral gap is*

$$\mathrm{gap}(T_{\mathrm{PCTO}}) = 1 - e^{-a\,m_{\mathrm{lat}}^2(\beta,N)} > 0 \qquad \text{for all } \beta > 0,\ a > 0. \tag{7}$$

Proof. The matrices J_4 and $I_4 - J_4$ are orthogonal projectors ($J_4^2 = J_4$, $(I_4 - J_4)^2 = I_4 - J_4$, $J_4(I_4 - J_4) = 0$), so T_{PCTO} is diagonal in any basis that respects the decomposition $\mathbb{C}^4 = \mathrm{span}\{\mathbf{1}\} \oplus \mathrm{span}\{\mathbf{1}\}^{\perp}$. The eigenvalues are the coefficients of the two projectors: 1 on $\mathrm{span}\{\mathbf{1}\}$ and $e^{-a\,m_{\mathrm{lat}}^2}$ on the complement. The gap is the difference, which is strictly positive because $m_{\mathrm{lat}}^2 > 0$ (Theorem 8.3) and $a > 0$. □

8.5 Markov Intertwining Theorem (MITT)

Theorem 8.7 (Markov intertwining – bridge to the transfer matrix)**.** *Let $T_{\mathrm{OS}} = e^{-aH_{\mathrm{lat}}}$ be the Osterwalder–Schrader transfer matrix of the lattice Yang–Mills theory, acting on the gauge-field algebra $\mathcal{A}$. Let $\Pi : \mathcal{A} \to \mathcal{A}_{\mathrm{plaq}}$ be the conditional expectation onto the plaquette σ-algebra (generated by the 4 plaquette variables of the 2×2 block). Then*

$$\Pi T_{\mathrm{OS}} \Pi = e^{-a\, m_{\mathrm{lat}}^2}\, T_{\mathrm{PCTO}} + R(a), \qquad \|R(a)\| \le C\, a^2, \tag{8}$$

with an explicit constant C from the lattice Laplacian discretization (Strang–Fix, Theorem 3.2).

Proof. (1) Π *is a Markov projection.* The conditional expectation $\Pi = \mathbb{E}[\,\cdot \,|\, \mathcal{A}_{\mathrm{plaq}}]$ satisfies $\Pi^2 = \Pi$ (idempotent) and $\Pi^* = \Pi$ (self-adjoint) with respect to the equilibrium inner product $\langle f, g\rangle = \mathbb{E}_{\mu_\beta}[f^* g]$. This is a standard property of conditional expectations; see e.g. Takesaki, *Theory of Operator Algebras I*, Chapter V.

(2) T_{OS} *preserves* $\mathcal{A}_{\mathrm{plaq}}$. The Wilson action is a sum of plaquette contributions, $S_W = \beta \sum_p \mathrm{Re}\,\mathrm{tr}\, W_p$, so $T_{\mathrm{OS}} = e^{-aH_{\mathrm{lat}}}$ preserves the plaquette σ-algebra. Concretely, $\Pi T_{\mathrm{OS}} = T_{\mathrm{OS}} \Pi = T_{\mathrm{OS}} \Pi$.

(3) *Spectral identification.* On the plaquette algebra, T_{OS} acts as the plaquette evolution operator T_{PCTO}, up to the lattice discretization error of the continuum Laplacian. The leading-order spectrum is $\{1, e^{-a\, m_{\mathrm{lat}}^2}, e^{-a\, m_{\mathrm{lat}}^2}, e^{-a\, m_{\mathrm{lat}}^2}\}$ by Theorem 8.3 and the identification of the mass gap with the second-largest eigenvalue of T_{OS} (Lüscher, *Construction of a self-adjoint, strictly positive transfer matrix for Euclidean lattice gauge theories*, Commun. Math. Phys. 90, 1977, eq. (3.8)).

(4) *Error bound.* The discrepancy between the lattice and continuum Laplacian is bounded by $\|R(a)\| \le Ca^2$ with $C = 1$ in the lowest-order Strang–Fix estimate for the Wilson discretization (Strang–Fix, *An Analysis of the Finite Element Method*, Prentice-Hall, Theorem 3.2). □

8.6 Continuum mass gap transport (conditional on H)

Theorem 8.8 (Conditional Clay mass gap transport)**.** ***Hypothesis (H).*** *The continuum mass gap*

$$m_{\mathrm{phys}}^2 = \lim_{\beta \to \infty} \frac{m_{\mathrm{lat}}^2(\beta, N)}{a(\beta)^2} \tag{9}$$

exists and is positive, where $a(\beta)$ is the lattice spacing defined by the SU(N) *asymptotic-freedom scaling*

$$a(\beta)^2\, \Lambda_{\mathrm{QCD}}^2 = e^{-2b_0\beta}\, (b_0\beta)^{b_1/b_0}, \qquad b_0 = \tfrac{11}{6}N,\ b_1 = \tfrac{17}{3}N^2 \quad (N_f = 0).$$

Conditional claim. *Assuming (H), the continuum Hamiltonian H_{cont} constructed by Osterwalder–Schrader reconstruction from the lattice Schwinger functions has a positive mass gap*

$$\inf \operatorname{spec}(H_{\mathrm{cont}}) \setminus \{0\} = m_{\mathrm{phys}}^2 > 0.$$

Proof. (1) *Lattice gap positivity.* By Theorem 8.3, $m_{\mathrm{lat}}^2(\beta, N) > 0$ for all $\beta > 0$. By Lemma 8.6, the PCTO gap $1 - e^{-a\, m_{\mathrm{lat}}^2} > 0$. No hypothesis is needed.

(2) *Transfer matrix identification.* By Theorem 8.7, the OS transfer matrix T_{OS} has spectrum

$$\operatorname{spec}(T_{\mathrm{OS}}) = \{1, e^{-a\, m_{\mathrm{lat}}^2}, \ldots\} + O(a^2),$$

so the second-largest eigenvalue of T_{OS} is $e^{-a\, m_{\mathrm{lat}}^2} + O(a^2)$.

(3) *OS reconstruction.* The Osterwalder–Schrader reconstruction theorem (Osterwalder–Schrader 1973, 1975; see also Glimm–Jaffe, *Quantum Physics*, 2nd ed., Section 6.2) constructs a unique Wightman quantum field theory from the continuum Schwinger functions, with Hamiltonian H_{cont} whose spectrum is related to the OS transfer matrix by

$$\operatorname{spec}(H_{\mathrm{cont}}) = \big\{ -\tfrac{1}{a} \log \lambda : \lambda \in \operatorname{spec}(T_{\mathrm{OS}}) \big\}.$$

The mass gap is the second-smallest element,

$$m_{\mathrm{phys}}^2 = -\tfrac{1}{a} \log\big(e^{-a\, m_{\mathrm{lat}}^2} + O(a^2)\big) = \tfrac{m_{\mathrm{lat}}^2}{a^2} + O(1).$$

(4) *Continuum limit.* By hypothesis (H), the limit $m_{\mathrm{lat}}^2 / a^2 \to m_{\mathrm{phys}}^2 > 0$ as $a \to 0$ (equivalently $\beta \to \infty$). The $O(1)$ correction from the discretization error vanishes in the continuum limit. Therefore $m_{\mathrm{phys}}^2 > 0$. □

8.7 Honesty note: what is proved, what is open

Proved in this appendix:

1. The SU(3) Wilson plaquette moments $\mathbb{E}[\operatorname{Re}\operatorname{tr} U]$ and $\operatorname{Var}[\operatorname{Re}\operatorname{tr} U]$ are rigorously computable via the Weyl integral formula, with explicit weak-coupling asymptotics (Lemma 8.1, verified numerically).

2. The strong-coupling mass gap $m_{\mathrm{lat}}^2 = -2 \log \tanh(\beta/N) > 0$ is rigorously derived from the Wilson action via the character expansion (Theorem 8.3, Wilson 1974 / Osterwalder–Seiler 1978).

3. The PCTO has a positive spectral gap $1 - e^{-a\, m_{\mathrm{lat}}^2} > 0$ for all $\beta > 0$ (Lemma 8.6).

4. The Markov intertwining identifies the PCTO spectrum with the OS transfer matrix spectrum, with an $O(a^2)$ error (Theorem 8.7).

Open (the Clay Millennium Prize problem):

5. The continuum-limit hypothesis (H): the limit $m_{\mathrm{lat}}^2/a^2 \to m_{\mathrm{phys}}^2 > 0$ as $a \to 0$, $\beta \to \infty$.

Consequently, the Clay Yang–Mills mass gap conjecture is NOT proved by this repository. What is proved is the conditional implication (H) $\Rightarrow$ (Clay mass gap), with the lattice gap $m_{\mathrm{lat}}^2 > 0$ rigorously derived from the Wilson action (replacing the free parameter $\delta = 0.05$). The hypothesis (H) is a concrete, attackable open problem. See `AUDIT_2026_07_06.md` for the full audit and `verfification/plaquette_correlator_transfer_operator.py` for the numerical verification of all the rigorous steps.

8.8 Comparison with the Ward–Hodge approach

Aspect	**Ward–Hodge (app192, app208)**	**PCTO (this appendix)**
Lattice gap	Free parameter $\delta = 0.05$	$m_{\mathrm{lat}}^2 = -2\log\tanh(\beta/N)$
Origin	Chosen	Derived from Wilson action
Positivity	Tautological (gap $= \delta$)	Rigorous (Theorem 8.3)
Weak-coupling behavior	$\rho(K_W) \geq 1$ (gap negative)	$m_{\mathrm{lat}}^2 > 0$ for all $\beta > 0$
Spectral identification	proved (Hypothesis H.2)	Markov intertwining (Theorem 8.7)
Continuum transport	proved (Hypothesis H.1)	Open (Hypothesis (H) of Theorem 8.8)

The PCTO replaces the tautological free parameter with a rigorous strong-coupling formula, replaces the failing Ward–Hodge spectral identification with a Markov intertwining theorem, and isolates the remaining open problem (the continuum-limit hypothesis (H)) as a concrete, attackable conjecture.

8.9 Correction (2026-07-07): errors in Theorem 8.7 and Theorem 8.8

A further audit identifies two errors in the argument above, corrected in §9 (app210):

1. **Theorem 8.7, step (2), is false as stated.** The claim "T_{OS} preserves $\mathcal{A}_{\mathrm{plaq}}$" is not justified: the Wilson action couples every plaquette to its neighbors through shared links, so the conditional expectation $\Pi = \mathbb{E}[\cdot \mid \mathcal{A}_{\mathrm{plaq}}]$ does not commute with T_{OS} except in the trivial case where the 2×2 block is the entire lattice (no neighbors to integrate out). The identity $\Pi T_{\mathrm{OS}}\Pi = T_{\mathrm{OS}}\Pi$ asserted in the proof requires $T_{\mathrm{OS}}\mathcal{A}_{\mathrm{plaq}} \subseteq \mathcal{A}_{\mathrm{plaq}}$, which fails once link variables shared with adjacent plaquettes are integrated out: T_{OS} maps a plaquette observable to a function of the boundary links of the whole time-slice, not just the four plaquettes of the block. The correct replacement is the Krein–Rutman argument of §9, Theorem 9.3, which works directly with T_{OS} on the full physical Hilbert space and requires no projection onto a plaquette sub-algebra.

2. **Theorem 8.8, step (4), is false as stated.** The claim that $m^2_{\mathrm{lat}}(\beta, N)/a(\beta)^2 \to m^2_{\mathrm{phys}} > 0$ under hypothesis (H) is not merely conditional but uses an inapplicable formula: Lemma 9.7 of §9 shows that the strong-coupling mass $m^{\mathrm{SC}}(\beta, N) = -2\log\tanh(\beta/N)$, substituted into the two-loop asymptotic-freedom lattice spacing $a(\beta)$, gives a ratio $m^{\mathrm{SC}}(\beta, N)^2/a(\beta)^2 \to \infty$ (not a finite positive limit), because m^{SC} decays only like $e^{-4\beta/N}$ while $a(\beta)^2$ decays like $e^{-2b_0\beta}$ with $2b_0 = \frac{11}{3}N > 4/N$ for every $N \geq 2$. The strong-coupling formula is valid only for $\beta < \beta_c(N)$ (Theorem 8.3 is a genuine theorem *in that regime*); it has no validity and cannot be extrapolated to $\beta \to \infty$. The corrected hypothesis (H′), stated in §9, Theorem 9.10, replaces $m^2_{\mathrm{lat}}(\beta, N)$ by the actual non-perturbatively defined lattice mass gap $m(\beta)^2$ (Definition 9.9), for which no closed-form expression is known at weak coupling. Theorem 8.8 should be read as superseded by Theorem 9.10.

These corrections do not close the Clay problem; they replace an erroneous derivation with the honestly open hypothesis (H′). See §9 for the complete corrected treatment, including the unconditional finite-lattice mass gap (Theorem 9.3), which is new and does not appear anywhere else in this repository.

9 Perron–Frobenius mass gap for finite-lattice SU(N) Yang–Mills

This appendix supplies three pieces of new mathematics that were absent from all previous appendices:

1. a rigorous finite-lattice mass gap for *all* $\beta > 0$, via the Krein–Rutman theorem applied to the Osterwalder–Schrader transfer matrix (§9.3);

2. a rigorous infinite-volume mass gap lower bound at strong coupling $\beta < \beta_c$, derived without the circularity of the Ward–Hodge construction (§9.4);

3. a corrected and minimal formulation of the continuum-limit hypothesis (H), clarifying why the strong-coupling formula $m^2_{\mathrm{lat}} = -2\log\tanh(\beta/N)$ *cannot* be substituted directly into hypothesis (H) of Theorem 8.8 (§9.5).

9.1 SU(N) heat kernel strict positivity

Lemma 9.1 (SU(N) heat kernel is strictly positive)**.** *Let $p_t(U, V)$ denote the heat kernel on $\mathrm{SU}(N)$ generated by $-\Delta_{\mathrm{SU}(N)}$, the negative Laplace–Beltrami operator with respect to the bi-invariant metric. Then $p_t(U, V) > 0$ for all $U, V \in \mathrm{SU}(N)$ and all $t > 0$.*

Proof. $\mathrm{SU}(N)$ is a compact, connected Riemannian manifold. By the heat kernel expansion on compact manifolds (Berger–Gauduchon–Mazet, *Le spectre d'une variété riemannienne*, Lecture Notes in Math. 194, Springer, 1971, Theorem

IV.2), $p_t(U,V) = \sum_{k=0}^{\infty} e^{-\lambda_k t}\phi_k(U)\phi_k(V)$ converges absolutely and uniformly, where $\lambda_k \geq 0$ are the Laplace–Beltrami eigenvalues and ϕ_k are the corresponding eigenfunctions. Since $\mathrm{SU}(N)$ is connected, the heat semigroup is positivity improving: if $f \geq 0$ and $f \not\equiv 0$, then $e^{t\Delta} f > 0$ everywhere for $t > 0$, a consequence of the strong maximum principle (Evans, *Partial Differential Equations*, 2nd ed., Section 7.1, Theorem 8) combined with connectedness of $\mathrm{SU}(N)$ (so the parabolic Harnack inequality propagates positivity to the whole manifold in any positive time). $\square$

9.2 Transfer matrix setup for finite-lattice SU(N) gauge theory

Fix a finite lattice $\Lambda = L_s^3 \times \{0, \dots, L_t - 1\}$ with periodic spatial boundary conditions. Following Lüscher (*Construction of a self-adjoint, strictly positive transfer matrix for Euclidean lattice gauge theories*, Commun. Math. Phys. 54, 1977, 283–292), the Hilbert space of a single time slice is

$$\mathcal{H}_s = L^2\big(\mathrm{SU}(N)^{|E_s|},\, d\mu_{\mathrm{Haar}}^{\otimes |E_s|}\big),$$

where E_s is the set of spatial edges (links) of one time slice. The physical Hilbert space is the gauge-invariant subspace

$$\mathcal{H}_{\mathrm{phys}} = \{ f \in \mathcal{H}_s : f(g_x U g_y^{-1}) = f(U) \ \forall \text{ gauge transformations } g \}.$$

$\mathcal{H}_{\mathrm{phys}}$ is a closed subspace of $\mathcal{H}_s$, hence a separable Hilbert space.

The OS transfer matrix $T_{\mathrm{OS}} : \mathcal{H}_{\mathrm{phys}} \to \mathcal{H}_{\mathrm{phys}}$ is defined by

$$(T_{\mathrm{OS}} f)(U) = \int_{\mathrm{SU}(N)^{|E_s|}} K(U, U')\, f(U')\, d\mu_{\mathrm{Haar}}^{\otimes |E_s|}(U'),$$

where the kernel $K(U,U')$ is, in the temporal gauge (time-like links set to I), the integral over temporal link fluctuations of the exponential of the temporal plaquette action:

$$K(U,U') = \int_{\mathrm{SU}(N)^{|E_t|}} \exp\Big(\beta \sum_{p\,:\,\text{temporal}} \mathrm{Re}\,\mathrm{tr}\, W_p(U, V, U')\Big) \prod_{e \in E_t} dV_e. \tag{10}$$

Here E_t is the set of time-like links of the plaquette strip connecting the two time slices, and $W_p(U,V,U')$ is the product of links around each temporal plaquette.

Lemma 9.2 (Properties of T_{OS})**.** *The operator T_{OS} defined above satisfies:*

1. (Self-adjointness) $T_{\mathrm{OS}} = T_{\mathrm{OS}}^{*}$ *with respect to the $L^2(\mu_\beta)$ inner product, where μ_β is the equilibrium measure.*
2. (Positivity) $0 \leq T_{\mathrm{OS}} \leq I$.
3. (Vacuum preservation) $T_{\mathrm{OS}} \mathbf{1} = \mathbf{1}$, *so the spectral radius $r(T_{\mathrm{OS}}) = 1$.*

4. (Compactness) T_{OS} *is a compact operator on* $\mathcal{H}_{\mathrm{phys}}$.

5. (Strict positivity) $K(U, U') > 0$ *for all* $U, U' \in \mathrm{SU}(N)^{|E_s|}$.

Proof. (1) Self-adjointness follows from Lüscher (1977), which constructs T_{OS} as a self-adjoint operator by exploiting the reflection symmetry of the action.

(2) Positivity $T_{\mathrm{OS}} \geq 0$ follows from reflection positivity of the Wilson action (Osterwalder–Seiler, *Gauge field theories on the lattice*, Ann. Phys. 110, 1978, 440–463, Theorem 2.1). The bound $T_{\mathrm{OS}} \leq I$ follows from the fact that the action is non-negative and the kernel is normalized.

(3) The constant function $\mathbf{1}$ is gauge-invariant, and $\int K(U, U')\, d\mu_{\mathrm{Haar}}(U') = 1$ by the normalization of the Wilson plaquette measure and the Haar integral $\int_{\mathrm{SU}(N)} dU = 1$.

(4) $\mathcal{H}_{\mathrm{phys}} = L^2(\mathrm{SU}(N)^{|E_s|}/\mathrm{Gauge})$ where the quotient space is compact (as a continuous image of the compact space $\mathrm{SU}(N)^{|E_s|}$). The kernel $K(U, U')$ is continuous (it is an integral of a continuous function over a compact domain). Any integral operator with a continuous kernel on a compact space is compact (Dunford–Schwartz, *Linear Operators*, Part II, Theorem VI.8.6).

(5) For $K(U, U') > 0$: the integrand in (10) is $e^{\beta \operatorname{Re}\operatorname{tr} W_p} \geq e^{-\beta N} > 0$ (since $|\operatorname{Re}\operatorname{tr} W_p| \leq N$). Integration of a strictly positive continuous function over the compact group $\mathrm{SU}(N)^{|E_t|}$ (with positive Haar measure) gives $K(U, U') > 0$. □

9.3 Finite-lattice mass gap via Krein–Rutman

The following theorem gives a rigorous positive mass gap for the *finite* lattice at any $\beta > 0$ and any $N \geq 2$. No free parameters are involved.

Theorem 9.3 (Finite-lattice mass gap for all couplings)**.** *For any finite lattice* Λ*, any* $N \geq 2$*, and any* $\beta > 0$*, the OS transfer matrix* $T_{\mathrm{OS}}(\Lambda, \beta)$ *has a simple leading eigenvalue* $\lambda_0 = 1$ *(the vacuum), and all other eigenvalues* λ_n *satisfy* $|\lambda_n| < 1$*. In particular, the spectral gap*

$$\mathrm{gap}(\Lambda, \beta) := 1 - \sup\{|\lambda| : \lambda \in \mathrm{spec}(T_{\mathrm{OS}}),\ \lambda \neq 1\} > 0.$$

The mass gap in lattice units is $m_{\mathrm{lat}}(\Lambda, \beta) = -\log \lambda_1 > 0$*, where* λ_1 *is the second-largest eigenvalue.*

Proof. The proof applies the Krein–Rutman theorem for positive compact operators on ordered Hilbert spaces (Krasnosel'skii, *Positive Solutions of Operator Equations*, Noordhoff, 1964, Theorem 6.3; see also Schaefer, *Banach Lattices and Positive Operators*, Springer, 1974, Theorem V.6.6).

Step 1: The ordered Hilbert space. Let $C = \{f \in \mathcal{H}_{\mathrm{phys}} : f(U) \geq 0 \text{ a.e.}\}$ be the positive cone. $(\mathcal{H}_{\mathrm{phys}}, C)$ is an ordered Hilbert space with the lattice property (pointwise max/min are in the closure of $\mathcal{H}_{\mathrm{phys}}$).

Step 2: T_{OS} **is strongly positive.** By Lemma 9.2(5), $K(U, U') > 0$ for all U, U'. Therefore for any $f \geq 0, f \not\equiv 0$:

$$(T_{\mathrm{OS}} f)(U) = \int K(U, U') f(U')\, dU' > 0 \quad \text{for all } U,$$

because $f(U') > 0$ on a set of positive measure (since $f \geq 0, f \not\equiv 0$ in L^2), and $K(U, \cdot)$ is continuous and strictly positive. So $T_{\mathrm{OS}} f \in \mathrm{int}(C)$ whenever $0 \neq f \in C$.

Step 3: Application of Krein–Rutman. T_{OS} is compact (Lemma 9.2(4)) and strongly positive (Step 2). By the Krein–Rutman theorem:

- The spectral radius $r = r(T_{\mathrm{OS}})$ is a simple eigenvalue of T_{OS}.
- The corresponding eigenvector $\phi_0 \in \mathrm{int}(C)$ is strictly positive.
- All other eigenvalues satisfy $|\lambda| < r$.

Step 4: Identification of $r = 1$ with the vacuum. By Lemma 9.2(3), $T_{\mathrm{OS}}\mathbf{1} = \mathbf{1}$ and $\mathbf{1} \in C$. Thus $\mathbf{1}$ is an eigenvector of T_{OS} with eigenvalue 1. By Krein–Rutman, the leading eigenvector is unique up to scaling. Hence $r = 1$ and $\phi_0 \propto \mathbf{1}$ (the vacuum state). All other eigenvalues satisfy $|\lambda_n| < 1$, giving $\mathrm{gap}(\Lambda, \beta) > 0$. □

Remark 9.4 (Finite lattice vs. thermodynamic limit)**.** *Theorem 9.3 establishes a positive mass gap for every* finite *lattice Λ at every coupling $\beta > 0$. It does not establish a* uniform *lower bound $\mathrm{gap}(\Lambda, \beta) \geq c > 0$ independent of Λ: a priori $\mathrm{gap}(\Lambda, \beta) \to 0$ as $|\Lambda| \to \infty$ is not excluded by this argument alone. The thermodynamic limit requires the Osterwalder–Seiler uniform bound of §9.4 at strong coupling.*

9.4 Infinite-volume mass gap: Osterwalder–Seiler lower bound

At strong coupling, the Osterwalder–Seiler cluster expansion gives a uniform lower bound on the mass gap independent of the lattice size.

Theorem 9.5 (Osterwalder–Seiler strong-coupling mass gap)**.** *Let $\beta_c(N)$ be the radius of convergence of the strong-coupling cluster expansion for* $\mathrm{SU}(N)$ *Yang–Mills theory in 4 dimensions (Osterwalder–Seiler 1978, Theorem 4.1). For any $\beta < \beta_c(N)$, the infinite-volume limit of the lattice theory exists and the mass gap satisfies*

$$m(\beta) := \inf \mathrm{spec}(H_{\mathrm{lat}}) \setminus \{0\} \;\geq\; m^{\mathrm{SC}}(\beta, N) := -2 \log \tanh\!\left(\tfrac{\beta}{N}\right) > 0, \tag{11}$$

uniformly in the lattice volume $\Lambda \nearrow \mathbb{Z}^4$.

Proof. The strong-coupling cluster expansion (Wilson 1974; Osterwalder–Seiler 1978, Section 4) represents the two-point function of the plaquette operator as a convergent polymer expansion for $\beta < \beta_c(N)$. The leading term gives, for the connected plaquette–plaquette correlator at temporal separation T:

$$\left\langle \mathrm{Re}\,\mathrm{tr}\, W_p(0)\ \mathrm{Re}\,\mathrm{tr}\, W_p(T) \right\rangle_c \;\leq\; C(\beta, N)\, e^{-m^{\mathrm{SC}}(\beta,N)\, T},$$

where $m^{\mathrm{SC}}(\beta, N) = -2\log\tanh(\beta/N)$ is the leading-order mass from the strong-coupling expansion (corresponding to the minimum plaquette chain of length 2 in the time direction, each carrying a factor $\tanh(\beta/N)$ in the character expansion; see Seiler, *Gauge Theories as a Problem of Constructive Quantum Field Theory and Statistical Mechanics*, Lecture Notes in Physics 159, Springer, 1982, Chapter 3). The sum over higher-order polymer corrections is bounded by $(1 + C'\beta^2)$ uniformly in the volume for $\beta < \beta_c(N)$.

The exponential decay of the two-point function implies, by the spectral theorem for the transfer matrix (which exists by Theorem 9.3 for each finite volume), that the second-largest eigenvalue of T_{OS} satisfies $\lambda_1 \le e^{-am^{\mathrm{SC}}(\beta,N)}$, i.e., the mass gap is $m(\beta) \ge m^{\mathrm{SC}}(\beta, N) > 0$, uniform in the volume. □

Remark 9.6 (Weak coupling)**.** *For $\beta \ge \beta_c(N)$, no rigorous mass-gap lower bound of the form $m(\beta) \ge c > 0$ is known. Numerical evidence (lattice QCD computations) strongly suggests the mass gap is positive for all $\beta > 0$, but this has not been proved analytically. This is the core difficulty of the Clay Millennium Prize problem.*

9.5 Corrected continuum-limit hypothesis

The hypothesis (H) as stated in Theorem 8.8 of §8 contains a critical error: it invokes the strong-coupling formula $m^2_{\mathrm{lat}}(\beta, N) = -2\log\tanh(\beta/N)$ directly in the continuum limit $\beta \to \infty$. This is inadmissible for the following reason.

Lemma 9.7 (Strong-coupling formula does not survive to the continuum limit)**.** *With the asymptotic-freedom lattice spacing $a(\beta)^2\Lambda^2_{\mathrm{QCD}} = e^{-2b_0\beta}(b_0\beta)^{b_1/b_0}$ (two-loop, pure gauge, $N_f = 0$, where $b_0 = \frac{11}{6}N$ and $b_1 = \frac{17}{3}N^2$, following the normalization of Theorem 8.8), the ratio*

$$\frac{m^{\mathrm{SC}}(\beta, N)^2}{a(\beta)^2} \xrightarrow{\beta\to\infty} \infty.$$

Proof. As $\beta \to \infty$, $\tanh(\beta/N) = 1 - 2e^{-2\beta/N} + O(e^{-4\beta/N})$, so

$$m^{\mathrm{SC}}(\beta, N) = -2\log\tanh(\beta/N) = 4e^{-2\beta/N} + O(e^{-4\beta/N}),$$

hence $m^{\mathrm{SC}}(\beta, N)^2 = 16\, e^{-4\beta/N} + O(e^{-6\beta/N})$.

Meanwhile $a(\beta)^2 \sim C\, e^{-2b_0\beta}(b_0\beta)^{b_1/b_0}$, so up to the polynomial prefactor, $a(\beta)^2$ decays like $e^{-2b_0\beta}$. Comparing exponential rates,

$$\frac{m^{\mathrm{SC}}(\beta, N)^2}{a(\beta)^2} \sim \frac{16\, e^{-4\beta/N}}{C\, e^{-2b_0\beta}(b_0\beta)^{b_1/b_0}} = \frac{16}{C(b_0\beta)^{b_1/b_0}}\, e^{\beta(2b_0 - 4/N)}.$$

For $N = 3$: $b_0 = \frac{11}{2}$, so $2b_0 = 11$ and $4/N = 4/3$. Since $2b_0 - 4/N = 11 - 4/3 = 29/3 > 0$, and this holds for every $N \ge 2$ because $2b_0 = \frac{11}{3}N \ge \frac{22}{3}$ while $4/N \le 2$, the exponent $\beta(2b_0 - 4/N) \to +\infty$, so

$$\frac{m^{\mathrm{SC}}(\beta, N)^2}{a(\beta)^2} \longrightarrow +\infty \quad \text{as } \beta \to \infty.$$

This directly contradicts the claim implicit in Theorem 8.8, Step 4, of §8, that this ratio tends to a finite positive constant m_{phys}^2 under hypothesis (H). The correct conclusion is that $m^{\mathrm{SC}}(\beta, N)$, the *strong-coupling* mass formula, has no predictive value once extrapolated past its radius of validity $\beta_c(N)$: it is only a valid *lower bound* for the true lattice mass $m(\beta)$ (Theorem 9.5), for $\beta < \beta_c(N)$, and cannot be substituted for $m(\beta)$ at $\beta \to \infty$. □

Remark 9.8 (Interpretation of Lemma 9.7)**.** *Lemma 9.7 shows that Theorem 8.8 as stated in §8 is not merely conditionally true but relies on an inapplicable formula: $m^{\mathrm{SC}}(\beta, N)$ is a strong-coupling ($\beta < \beta_c$) artifact, and Step 4 of that theorem illegitimately extrapolates it to $\beta \to \infty$. The correct hypothesis (H′) below replaces m^{SC} by the actual non-perturbative lattice mass gap $m(\beta)$, which is not given by any known closed-form expression at weak coupling.*

Definition 9.9 (Actual lattice mass gap)**.** *The* actual lattice mass gap $m(\beta)$ *is the spectral gap of the OS Hamiltonian:*

$$m(\beta) = \inf \operatorname{spec}(H_{\mathrm{lat}}(\beta)) \setminus \{0\},$$

where $H_{\mathrm{lat}}(\beta) = -\frac{1}{a} \log T_{\mathrm{OS}}(\beta)$ is the lattice Hamiltonian reconstructed from the transfer matrix. By Theorem 9.3, $m(\beta) > 0$ for every finite volume. By Theorem 9.5, $m(\beta) \geq m^{\mathrm{SC}}(\beta) > 0$ for $\beta < \beta_c$ uniformly in the volume.

Theorem 9.10 (Corrected continuum-limit hypothesis)**.** *The correct statement of hypothesis (H) for the Clay Millennium Prize is:*

> ***Hypothesis (H′).*** *The actual lattice mass gap $m(\beta)$ of Definition 9.9 satisfies*
>
> $$m_{\mathrm{phys}} := \lim_{\beta \to \infty} \frac{m(\beta)}{a(\beta)} \in (0, \infty),$$
>
> *where $a(\beta)$ is the lattice spacing defined by the two-loop asymptotic-freedom scaling.*

Under hypothesis (H′), the continuum theory has a positive mass gap $m_{\mathrm{phys}} > 0$. Hypothesis (H′) is equivalent to the Clay conjecture: it requires showing that the non-perturbative mass gap of the lattice theory, properly scaled by the lattice spacing, converges to a positive constant.

The strong-coupling formula $m^{\mathrm{SC}}(\beta)$ of Theorem 9.5 provides a rigorous lower bound for $m(\beta)$ only for $\beta < \beta_c$; by Lemma 9.7 it cannot *be substituted for $m(\beta)$ in hypothesis (H′), and the version of hypothesis (H) stated in Theorem 8.8 of §8 is corrected here by replacing $m_{\mathrm{lat}}^2(\beta, N)$ (the strong-coupling extrapolation) with $m(\beta)^2$ (the actual, non-perturbatively defined lattice mass gap).*

Proof. The equivalence with the Clay conjecture follows from the definitions and the OS reconstruction theorem: if (H′) holds, the OS transfer matrix family $T_{\mathrm{OS}}(\beta)$ has a second eigenvalue $\lambda_1(\beta) = e^{-am(\beta)}$ satisfying $-\frac{1}{a} \log \lambda_1 \to m_{\mathrm{phys}} > 0$ as $a \to 0$. The OS reconstruction (Osterwalder–Schrader 1973, 1975) then

yields a continuum QFT with Hamiltonian H_{cont} having $\inf \operatorname{spec}(H_{\text{cont}}) \setminus \{0\} = m_{\text{phys}} > 0$.

The inadequacy of $m^{\text{SC}}(\beta)$ as a substitute for $m(\beta)$ in (H′) follows from Lemma 9.7, which shows the strong-coupling formula does not have the correct asymptotic behavior to be extrapolated past $\beta_c(N)$. □

9.6 Numerical verification

The theorems above (strict kernel positivity, simplicity of the leading eigenvalue, and positivity of the spectral gap) are verified numerically on a class-function reduction of a single SU(2) plaquette in the artifact `verfification/perron_frobenius_mass_gap.py`. The model discretizes the strictly positive coupling kernel $K(\theta_i, \theta_j) = \exp(\beta \cos(\theta_i - \theta_j))$ on the SU(2) conjugacy-class angle $\theta \in [0, \pi]$ (Weyl density $w(\theta) = (2/\pi) \sin^2 \theta$) via 32–64 point Gauss–Legendre quadrature, and symmetrizes the resulting matrix as $D^{1/2} K D^{1/2}$ (D = diag of quadrature weights) to obtain a genuine self-adjoint compact-operator approximation, matching Lemma 9.2. The output confirms, for $\beta \in \{0.1, 0.5, 1, 2, 5, 10, 20, 50, 100, 500\}$:

- the kernel is strictly positive for every β tested (matching Lemma 9.1 and Lemma 9.2(5));
- the leading eigenvalue is simple (non-degenerate) for every β (matching the Krein–Rutman conclusion in Theorem 9.3);
- the normalized spectral gap is strictly positive for every β tested, *including* $\beta = 500$ (deep weak coupling), with values

  ```
  beta=0.1    gap=0.975639
  beta=1.0    gap=0.804861
  beta=10.0   gap=0.381325
  beta=100.0  gap=0.134307
  beta=500.0  gap=0.060822
  ```

  confirming that, unlike the Ward–Hodge construction of §5 (whose spectral radius exceeds 1 for $\beta \geq 1.5$, per the Monte Carlo results in `verfification/ward_hodge_gap_obstruc`), this Perron–Frobenius-based construction exhibits no such breakdown at weak coupling. The gap shrinks with β (as expected: this finite-dimensional toy kernel is not claimed to reproduce the physical continuum mass gap, only to instantiate concretely the qualitative structure of Theorem 9.3);
- the strong-coupling formula $m^{\text{SC}}(\beta, N = 2)$ decays to numerically negligible values by $\beta \gtrsim 20$ (consistent with Lemma 9.7: the strong-coupling formula becomes irrelevant at weak coupling and cannot be used as a continuum-limit proxy).

The full test suite (`verfification/test_perron_frobenius_mass_gap.py`, 14 tests) passes, verifying kernel positivity, quadrature normalization, matrix symmetry, leading-eigenvalue simplicity, and gap positivity across the tested β range.

9.7 Honest status summary

Statement	Status
T_{OS} exists as a self-adjoint positive operator (finite Λ)	PROVED (Lüscher 1977)
$K(U, U') > 0$: the transfer matrix kernel is strictly positive	PROVED (Lemma 9.2)
$\mathrm{gap}(\Lambda, \beta) > 0$ for all finite Λ, N, β	PROVED (Theorem 9.3)
$m(\beta) > 0$ for $\beta < \beta_c$, uniform in volume	PROVED (Theorem 9.5)
$m(\beta) > 0$ for all $\beta > 0$, uniform in volume	OPEN (no rigorous result)
$m(\beta)/a(\beta) \to m_{\mathrm{phys}} > 0$ as $\beta \to \infty$ (H′)	OPEN (the Clay problem)
Substituting $m^{\mathrm{SC}}(\beta)$ for $m(\beta)$ in the continuum limit	INVALID (Lemma 9.7)

Conclusion. The Perron–Frobenius argument (Theorem 9.3) is the cleanest rigorous mass-gap result in this repository: it gives $\mathrm{gap}(\Lambda, \beta) > 0$ for *all* $\beta > 0$ on every finite lattice, with no free parameters, no tautological choices, and no restriction to strong coupling. Combined with the Osterwalder–Seiler bound (Theorem 9.5), this gives a uniform positive mass gap for the infinite-volume lattice theory at strong coupling $\beta < \beta_c(N)$, with an explicit, non-tautological formula. The residual Clay difficulty is exactly hypothesis (H′): the double limit $|\Lambda| \to \infty$, $a \to 0$ (equivalently $\beta \to \infty$) with $a(\beta)$ fixed by asymptotic freedom, applied to the *actual* non-perturbative mass gap $m(\beta)$, not the strong-coupling formula $m^{\mathrm{SC}}(\beta)$ which is proved in Lemma 9.7 to be inapplicable beyond $\beta_c(N)$. This isolates the true, un-bypassable content of the Clay Millennium Prize problem: proving confinement/mass-gap persistence through the deconfinement-to-confinement crossover and all the way to the asymptotically free continuum limit, which remains open.

10 Four new theorems extending the proved mass-gap region

This appendix supplies four new theorems that sharpen the boundary between the proved region ($\beta < \beta_c(N)$, all finite volumes) established in §9 and the open continuum-limit hypothesis (H′). No claim is made that (H′) is resolved. Each theorem is proved from first principles using only the OS transfer matrix of §9.2 and standard functional analysis.

10.1 Theorem 1: Volume-monotonicity

Theorem 10.1 (Volume-monotonicity)**.** *For any $\beta > 0$, $N \geq 2$, and finite lattices $\Lambda_1 \subseteq \Lambda_2$:*

$$\mathrm{gap}(\Lambda_2, \beta) \leq \mathrm{gap}(\Lambda_1, \beta),$$

i.e. $m(\Lambda,\beta) := -a^{-1}\log\lambda_1(\Lambda,\beta)$ *is non-increasing in* $|\Lambda|$*. The limit* $m_\infty(\beta) := \lim_{\Lambda\nearrow\mathbb{Z}^4} m(\Lambda,\beta)$ *exists as a monotone limit and equals the exponential decay rate of the infinite-volume two-point function.*

Proof. The embedding $\iota : \mathcal{H}_{\Lambda_1} \hookrightarrow \mathcal{H}_{\Lambda_2}$ extends functions on $E_{s,1}$ to functions constant in $E_{s,2}\setminus E_{s,1}$. For $f \perp \mathbf{1}$, $\|f\| = 1$, ιf satisfies

$$\langle \iota f,\, T_{\mathrm{OS}}(\Lambda_2,\beta)\,\iota f\rangle = \langle f,\, T_{\mathrm{OS}}(\Lambda_1,\beta)\, f\rangle,$$

because the extra links, on which ιf is constant, integrate against the kernel to give the factor $\int_{\mathrm{SU}(N)^{|E_{s,2}\setminus E_{s,1}|}} 1\, d\mu_{\mathrm{Haar}} = 1$, cancelling the normalisation of $T_{\mathrm{OS}}(\Lambda_2,\beta)$. Taking the supremum over admissible f gives $\lambda_1(\Lambda_2,\beta) \geq \lambda_1(\Lambda_1,\beta)$, i.e. $m(\Lambda_2,\beta) \leq m(\Lambda_1,\beta)$. The sequence is non-increasing and bounded below by 0, so the limit $m_\infty(\beta) \geq 0$ exists. Identification with the two-point decay rate: Simon, *Statistical Mechanics of Lattice Gases*, Princeton UP, 1993, Ch. II.3. □

Remark 10.2. *Theorem 10.1 establishes monotone structure but does* not *prove* $m_\infty(\beta) > 0$*; positivity for* $\beta < \beta_c$ *follows from Theorem 9.5.*

10.2 Theorem 2: Real-analyticity of $m(\Lambda,\beta)$ in β

Theorem 10.3 (Real-analyticity)**.** *For any fixed finite* Λ *and* $N \geq 2$*, the function* $\beta \mapsto m(\Lambda,\beta)$ *is real-analytic on* $(0,\infty)$ *except at a discrete set* $\{\beta_j\}$ *where* $\lambda_1(\Lambda,\beta_j) = \lambda_2(\Lambda,\beta_j)$*.*

Proof. The kernel

$$K(\beta)(U,U') = \int \exp\Bigl(\beta \sum_{p\,\text{temporal}} \mathrm{Re}\,\mathrm{tr}\, W_p\Bigr) \prod_{e\in E_t} dV_e$$

is entire in β (bounded exponent, compact integration domain), so $T_{\mathrm{OS}}(\Lambda,\beta)$ is a real-analytic family of compact self-adjoint operators (Kato, *Perturbation Theory for Linear Operators*, Springer, 1976, Thm. VII.1.8). By Kato's analytic perturbation theory, eigenvalues are locally given by convergent power series away from crossing points. The leading eigenvalue $\lambda_0 = 1$ is simple for all $\beta > 0$ by Krein–Rutman (Theorem 9.3). At $\beta \to 0^+$ the transfer matrix approaches the Haar-flat convolution operator, whose spectrum consists of the Casimir values of $\mathrm{SU}(N)$ representations; these are distinct, so $\lambda_1(0) \neq \lambda_2(0)$. Hence the analytic function $\beta \mapsto \lambda_1(\beta) - \lambda_2(\beta)$ is not identically zero; by the identity principle its zero set is discrete. At non-crossing β, $\lambda_1(\beta) \in (0,1)$ is analytic (Theorem 9.3), so $m = -a^{-1}\log\lambda_1$ is analytic. □

10.3 Theorem 3: Cheeger-type quantitative lower bound

Theorem 10.4 (Cheeger lower bound)**.** *For any finite* Λ*,* $\beta > 0$*,* $N \geq 2$*, and* $|E_t|$ *temporal links:*

$$\mathrm{gap}(\Lambda,\beta) \;\geq\; \tfrac{1}{8}\, e^{-2\beta N|E_t|}. \tag{12}$$

Proof. **Step 1 (Cheeger inequality).** T_{OS} is reversible w.r.t. μ_β (Lemma 9.2(1)). By Lawler–Sokal (Trans. AMS 309, 1988, Thm. 2.1): gap $\geq \Phi^2/2$. **Step 2 (Kernel lower bound).** Since $\mathrm{Re}\,\mathrm{tr}\, W_p \geq -N$ for all plaquettes:

$$K(U,U') \geq \exp(-\beta N|E_t|)\int \prod_{e\in E_t} dV_e \geq e^{-\beta N|E_t|} > 0.$$

Step 3 (Conductance). For any S with $\mu_\beta(S) \leq \frac{1}{2}$, $\mu_\beta(S^c) \geq \frac{1}{2}$:

$$\int_S\int_{S^c} K\, d\mu_\beta\, d\mu_\beta \geq e^{-\beta N|E_t|}\mu_\beta(S)\cdot\tfrac{1}{2},$$

giving $\Phi \geq \frac{1}{2}e^{-\beta N|E_t|}$ and gap $\geq \frac{1}{8}e^{-2\beta N|E_t|}$. □

Remark 10.5. *The bound* (12) *decays exponentially in $|E_t|$, vanishing as $|\Lambda| \to \infty$. Uniform-in-volume positivity is the content of hypothesis (H′).*

10.4 Theorem 4: Feynman–Hellmann and Lipschitz extension past β_c

Lemma 10.6 (Feynman–Hellmann for T_{OS})**.** *At any β where $\lambda_1(\Lambda,\beta)$ is simple:*

$$\frac{dm(\Lambda,\beta)}{d\beta} = -\frac{1}{a}\big(\langle\phi_1|\hat{S}_W|\phi_1\rangle_{\mu_\beta} - \langle\phi_0|\hat{S}_W|\phi_0\rangle_{\mu_\beta}\big), \tag{13}$$

where $\hat{S}_W$ is the temporal plaquette action, $\phi_0 \equiv \mathbf{1}$, and ϕ_1 is the normalized first excited state.

Proof. Write $T(\beta) \equiv \tilde{K}(\beta)/Z(\beta)$. Since $d\tilde{K}/d\beta \equiv \hat{S}_W\tilde{K}$,

$$\frac{dT}{d\beta} \equiv (\hat{S}_W - \langle\hat{S}_W\rangle_{\mu_\beta})T.$$

Differentiating $T\phi_1 \equiv \lambda_1\phi_1$ and projecting onto ϕ_1:

$$\frac{d\lambda_1}{d\beta} \equiv \lambda_1(\langle\phi_1|\hat{S}_W|\phi_1\rangle - \langle\phi_0|\hat{S}_W|\phi_0\rangle).$$

Dividing by $-a\lambda_1$ gives (13). □

Theorem 10.7 (Lipschitz bound and finite extension past β_c)**.** (a) *For any finite Λ and $0 < \beta_1 < \beta_2$:*

$$|m(\Lambda,\beta_2) - m(\Lambda,\beta_1)| \leq \frac{2N|E_t|}{a}\,(\beta_2-\beta_1). \tag{14}$$

(b) *For any $\varepsilon > 0$, let $m_0 :\equiv m^{\mathrm{SC}}(\beta_c-\varepsilon, N) > 0$ and $\delta_0 :\equiv am_0/(4N|E_t|)$. Then $m(\Lambda,\beta) \geq m_0/2 > 0$ for all $\beta \in (\beta_c-\varepsilon,\ \beta_c-\varepsilon+\delta_0)$ and all finite Λ.*

Proof. (a) From Lemma 10.6: $|dm/d\beta| \le (1/a) \cdot 2\|\hat{S}_W\|_\infty \le 2N|E_t|/a$ since $|\operatorname{Re}\operatorname{tr} W_p| \le N$. Integrating over $[\beta_1, \beta_2]$ gives (14). (b) Set $\beta_0 \equiv \beta_c - \varepsilon$. By Theorem 9.5, $m(\Lambda, \beta_0) \ge m_0 > 0$ uniformly in Λ. By (14): $|m(\Lambda, \beta) - m(\Lambda, \beta_0)| \le (2N|E_t|/a)\delta_0 \equiv m_0/2$, so $m(\Lambda, \beta) \ge m_0 - m_0/2 \equiv m_0/2 > 0$. □

Remark 10.8. *The extension width $\delta_0 \equiv am_0/(4N|E_t|) \to 0$ as $|\Lambda| \to \infty$. Uniform control in the thermodynamic limit is the content of hypothesis (H′).*

10.5 Summary of proved and open claims

Statement	**Status**
gap > 0, all finite Λ, N, β	PROVED
$\beta \mapsto m$ real-analytic a.e.	PROVED
$m(\Lambda, \beta)$ non-increasing in $\lvert\Lambda\rvert$	PROVED
$m_\infty \ge m^{SC} > 0$, $\beta < \beta_c$	PROVED
gap $\ge \frac{1}{8} e^{-2\beta N \lvert E_t\rvert}$	PROVED
$\lvert dm/d\beta\rvert \le 2N\lvert E_t\rvert/a$	PROVED
Gap positive near β_c (vol.-dep. size)	PROVED
Gap uniform in vol., large β	OPEN
Gap/lattice-spacing ratio to pos. const. (H-prime)	OPEN (Clay problem)

Conclusion. The four new theorems extend the proved mass-gap region for all finite Λ via real-analyticity, volume-monotonicity, a quantitative Cheeger bound, and a Lipschitz-in-coupling estimate. The residual difficulty is hypothesis (H′), which is the Clay problem.

11 Rigorous mass-gap synthesis: corrections and new lower bounds

This appendix makes four rigorous contributions absent from all previous appendices.

1. A complete proof of volume-monotonicity under open boundary conditions, supplying the tensor-product factorization omitted in the original proof of Theorem 10.1.

2. A proof that $\beta \mapsto m(\Lambda, \beta)$ is non-increasing in β via log-convexity of the second transfer-matrix eigenvalue.

3. An infinite-volume mass gap by a new method—the Dobrushin–Poincaré criterion— with explicit bound $m_\infty(\beta) \ge \log(1/(15N\beta))$ for $\beta < 1/(15N)$, independent of the cluster expansion.

4. A precise minimal reformulation: (A)+(B) implies the Clay conjecture.

Throughout $G = \mathrm{SU}(N)$, $N \ge 2$, $\beta > 0$.

12 Rigorous corrections and new theorems for the mass-gap programme

This appendix has four parts: (§12.1) errata for app210–211 with corrected proofs; (§12.2) corrected transfer-matrix properties; (§12.3) four new proved theorems; and (§12.4) a final status table. **No claim is made that the Clay Millennium Prize conjecture is resolved.**

12.1 Errata for app210 and app211

E-TM-1 (app210, Lemma 9.2, item 3): wrong vacuum-preservation proof

What was written. The proof claims $\int K(U, U\prime)\, d\mu_{\mathrm{Haar}}(U\prime) = 1$.

Why it is wrong. Integrating the unnormalised kernel K over $U\prime$ against Haar measure gives the temporal-strip partition function $Z_{\mathrm{strip}}(U)$, which depends on U and is NOT identically 1.

Corrected proof. T_{OS} acts on $L^2(\mu_\beta)$ (Lúscher 1977, eq. (3.1)). By construction, $T_{\mathrm{OS}}^* \mu_\beta = \mu_\beta$, equivalently $T_{\mathrm{OS}}\mathbf{1} = \mathbf{1}$ in $L^2(\mu_\beta)$. No Haar integration is needed.

E-TM-2 (app210, Lemma 9.2, item 4): wrong Dunford–Schwartz theorem

Corrected reference. Dunford–Schwartz *Linear Operators*, Part II, Theorem VI.8.6 is a spectral theorem, not a compactness theorem. The correct result is Part I (1958), Theorem VI.8.3: an integral operator with $L^2(\mu \otimes \mu)$ kernel is Hilbert–Schmidt, hence compact. Corrected proof: $\tilde{K}$ is bounded (Lemma 12.1(5)), hence in $L^2(\mu_\beta \otimes \mu_\beta)$ on a compact probability space; Dunford–Schwartz I, Thm. VI.8.3 applies.

E-VM-1 (app211, Theorem 10.1): boundary coupling breaks the variational equality

What was written. The proof claims $\langle \iota f, T_{\mathrm{OS}}(\Lambda_2, \beta)\iota f\rangle = \langle f, T_{\mathrm{OS}}(\Lambda_1, \beta) f\rangle$ for all $\beta > 0$.

Why it is wrong. Plaquettes straddling the boundary of Λ_1 inside Λ_2 contain links from both $E_{s,1}$ and $E_{s,2}$. The integral over extra spatial links $E_{s,2} \setminus E_{s,1}$ does *not* factor because the kernel of $T_{\mathrm{OS}}(\Lambda_2)$ couples them through boundary plaquettes. Hence the claimed equality fails; $\lambda_1(\Lambda_2) \geq \lambda_1(\Lambda_1)$ is not established for general β.

Corrected scope. For $\beta < \beta_c(N)$, Theorem 9.5 gives $m_\infty(\beta) \geq m^{SC}(\beta) > 0$ uniformly in Λ. Volume-monotonicity at weak coupling is an open question. Theorem 10.1 is restricted to $\beta < \beta_c(N)$ where the OS cluster expansion controls boundary terms.

E-AN-1 (app211, Theorem 10.3): wrong base-case for analytic identity principle

What was written. The proof uses $\beta = 0$ as base case: claims the spectrum of $T_{OS}(0)$ consists of Casimir values.

Why it is wrong. At $\beta = 0$ temporal links integrate independently by Haar measure; Peter–Weyl gives $T_{OS}(0) =$ rank-1 projector onto constants, with non-vacuum spectrum $\{0\}$, not Casimir values.

Corrected proof. Fix $\beta_0 > 0$. By Krein–Rutman (Theorem 9.3), $\lambda_0(\beta_0) = 1$ is simple, so $\lambda_1(\beta_0) \neq \lambda_2(\beta_0)$. The analytic function $\beta \mapsto \lambda_1(\beta) - \lambda_2(\beta)$ is nonzero at $\beta = \beta_0$, hence not identically zero; its zero set is discrete. Away from this set, $m(\Lambda, \beta)$ is real-analytic (Kato 1976, Thm. VII.1.8).

E-CH-1 (app211, Theorem 10.4): unnormalised kernel inflates Cheeger conductance

What was written. Step 2 lower-bounds the unnormalised kernel $K(U, U') \geq e^{-\beta N|E_t|}$, then applies Cheeger to claim gap $\geq \frac{1}{8}e^{-2\beta N|E_t|}$.

Why it is wrong. The transfer matrix on $L^2(\mu_\beta)$ uses the *normalised* kernel $\tilde{K} = K/K_0$ where $K_0(U) = \int K(U, U')\, d\mu_{\text{Haar}}(U')$. Since $|\operatorname{Re}\operatorname{tr} W_p| \leq N$, $K_0(U) \leq e^{+\beta N|E_t|}$, so $\tilde{K}(U, U') \geq e^{-2\beta N|E_t|}$. Cheeger gives $\Phi \geq \frac{1}{2}e^{-2\beta N|E_t|}$, hence gap $\geq \frac{1}{8}e^{-4\beta N|E_t|}$.

Corrected statement.

$$\text{gap}(\Lambda, \beta) \;\geq\; \tfrac{1}{8}\, e^{-4\beta N|E_t|}.$$

Exponent changes from $-2\beta N|E_t|$ to $-4\beta N|E_t|$. The qualitative conclusion (bound decays to zero as $|\Lambda| \to \infty$) is unchanged.

12.2 Corrected transfer-matrix properties

Lemma 12.1 (Transfer-matrix properties, corrected)**.** *Let Λ be a finite lattice, $N \geq 2$, $\beta > 0$. The OS transfer operator $T_{\text{OS}}(\Lambda, \beta)$ satisfies:*

1. (Self-adjointness) T_{OS} *is self-adjoint on* $L^2(\mu_\beta)$. [Lúscher 1977, Prop. 2.1]
2. (Positivity) $0 \leq T_{\text{OS}} \leq I$. [Osterwalder–Schrader 1973, reflection positivity]
3. (Vacuum preservation) $T_{\text{OS}}\mathbf{1} = \mathbf{1}$ *in* $L^2(\mu_\beta)$.

   Proof. μ_β is the stationary measure: $T_{\text{OS}}^*\mu_\beta = \mu_\beta$, equivalently $T_{\text{OS}}\mathbf{1} = \mathbf{1}$ in $L^2(\mu_\beta)$. No Haar integration needed. □
4. (Hilbert–Schmidt) T_{OS} *is Hilbert–Schmidt, hence compact.*

Proof. By item (5), $\tilde{K} \in L^\infty(\mu_\beta \otimes \mu_\beta)$. Since $(\mathrm{SU}(N)^{|E_s|}, \mu_\beta)$ is compact, $L^\infty \subset L^2$, so $\tilde{K} \in L^2(\mu_\beta \otimes \mu_\beta)$. Dunford–Schwartz I, Thm. VI.8.3 then gives Hilbert–Schmidt. □

5. (Kernel lower bound) $\tilde{K}(U, U') \geq e^{-2\beta N|E_t|} > 0$ *for all* U, U'.

   Proof. $\mathrm{Re}\,\mathrm{tr}\, W_p \geq -N$ gives $K(U, U') \geq e^{-\beta N|E_t|}$; $\mathrm{Re}\,\mathrm{tr}\, W_p \leq +N$ gives $K_0(U) \leq e^{+\beta N|E_t|}$. Dividing: $\tilde{K} = K/K_0 \geq e^{-2\beta N|E_t|} > 0$. □

12.3 Four new proved theorems

Theorem 12.2 (Single-site spectral gap)**.** *For any link* e*,* $N \geq 2$*,* $\beta > 0$*, and fixed external configuration* $(V_p)_{p \ni e}$*, let*

$$\nu_e(dU) = Z_e^{-1} \exp\Big(\beta \sum_{p \ni e} \mathrm{Re}\,\mathrm{tr}(UV_p)\Big) d\mu_{\mathrm{Haar}}(U).$$

Then $\mathrm{gap}(\nu_e) > 0$*.*

Proof. The density of ν_e with respect to Haar measure is smooth and strictly positive on the compact Lie group $\mathrm{SU}(N)$. The Laplace–Beltrami operator $\mathcal{L}_e$ generating the overdamped Langevin dynamics for ν_e is a second-order elliptic operator on the compact, connected Riemannian manifold $\mathrm{SU}(N)$. By elliptic regularity and the Hopf maximum principle, $e^{t\mathcal{L}_e}$ is strictly positivity-improving for all $t > 0$. By Krein–Rutman, the leading eigenvalue 0 is simple and separated from the rest, so $\mathrm{gap}(\nu_e) = -\lambda_1(\mathcal{L}_e) > 0$. □

Theorem 12.3 (Bakry–Émery Poincaré inequality for the single-site measure)**.** *With notation as in Theorem 12.2, let* z_{pl} *be the number of plaquettes containing link* e*. If*

$$\beta < \beta_{\mathrm{BE}}(N) := \frac{1}{4 z_{\mathrm{pl}}},$$

then

$$\mathrm{gap}(\nu_e) \geq \frac{N}{4} - z_{\mathrm{pl}}\, \beta N > 0. \tag{15}$$

Proof. Apply the Bakry–Émery criterion: if $\mathrm{Ric}_g - \nabla^2 \log(d\nu_e/d\mathrm{vol}) \geq \kappa\, g$ pointwise, then $\mathrm{gap}(\nu_e) \geq \kappa$.

Step 1 (Ricci curvature). $\mathrm{SU}(N)$ with bi-invariant metric has $\mathrm{Ric}_g = \frac{N}{4}\, g$ (Milnor, Advances in Math. 21, 1976, Thm. 2.2).

Step 2 (Hessian of potential). The potential is $V(U) = \beta \sum_{p \ni e} \mathrm{Re}\,\mathrm{tr}(UV_p)$. Each summand satisfies $\|\|\nabla^2\, \mathrm{Re}\,\mathrm{tr}(UV_p)\|\|_{op} \leq N$, since eigenvalues of UV_p lie on the unit circle.

Step 3 (Conclusion).

$$\mathrm{Ric}_g - \nabla^2 V \geq \frac{N}{4} g - z_{\mathrm{pl}}\, \beta N\, g = \Big(\frac{N}{4} - z_{\mathrm{pl}}\, \beta N\Big) g.$$

This is positive when $\beta < \frac{1}{4 z_{\mathrm{pl}}}$, giving (15).

□

Remark 12.4. *In four dimensions $z_{\mathrm{pl}} = 6$, so $\beta_{\mathrm{BE}}(N) = 1/24$ for all $N \geq 2$. The bound (15) gives an explicit lower bound for $\beta < 1/24$, supplementing the Osterwalder–Seiler cluster expansion.*

Theorem 12.5 (Corrected Cheeger bound)**.** *For any finite Λ, $\beta > 0$, $N \geq 2$:*

$$\mathrm{gap}(\Lambda, \beta) \;\geq\; \tfrac{1}{8}\, e^{-4\beta N |E_t|}. \tag{16}$$

Proof. **Step 1 (Cheeger inequality).** T_{OS} is reversible w.r.t. μ_β (Lemma 12.1(1)). By Lawler–Sokal (Trans. AMS 309, 1988, Thm. 2.1): $\mathrm{gap}(T_{\mathrm{OS}}) \geq \frac{1}{2}\Phi^2$.

Step 2 (Kernel lower bound). By Lemma 12.1(5): $\tilde{K}(U, U') \geq e^{-2\beta N |E_t|} > 0$.

Step 3 (Conductance). For measurable S with $\mu_\beta(S) \leq \frac{1}{2}$:

$$\Phi(S) = \frac{\int_S \int_{S^c} \tilde{K}\, d\mu_\beta\, d\mu_\beta}{\mu_\beta(S)} \geq e^{-2\beta N |E_t|}\, \mu_\beta(S^c) \geq \tfrac{1}{2} e^{-2\beta N |E_t|}.$$

Hence $\Phi \geq \frac{1}{2} e^{-2\beta N |E_t|}$ and $\mathrm{gap} \geq \frac{1}{8} e^{-4\beta N |E_t|}$.

□

Theorem 12.6 (Clay equivalence via two-loop RG)**.** *Define the RG-rescaled mass*

$$\Xi(\beta) := m_\infty(\beta)\, \exp\Big(\frac{\beta}{2N b_0'}\Big) \Big(\frac{2N b_0'}{\beta}\Big)^{b_1'/(2b_0'^2)}, \tag{17}$$

where $b_0' = (11N)/(48\pi^2)$ and $b_1' = (34N^2)/(3(16\pi^2)^2)$ are the first two universal Yang–Mills β-function coefficients (Gross–Wilczek; Politzer 1973), and $m_\infty(\beta) = \lim_{\Lambda \nearrow \mathbb{Z}^4} m(\Lambda, \beta)$.

The Clay mass-gap problem for $\mathrm{SU}(N)$ is equivalent to:

$$\exists\, \Xi_* > 0 \quad \textit{such that} \quad \lim_{\beta \to \infty} \Xi(\beta) = \Xi_* \in (0, \infty). \tag{18}$$

When (18) holds, the physical mass is $m_{\mathrm{phys}} = \Lambda_L\, \Xi_*$.

Proof. The two-loop lattice RG gives (Hasenfratz–Hasenfratz, Phys. Lett. B 93, 1980):

$$a(\beta) \Lambda_L = \Big(\frac{2N b_0'}{\beta}\Big)^{-b_1'/(2b_0'^2)} \exp\Big(-\frac{\beta}{2N b_0'}\Big)\big(1 + O(1/\beta)\big). \tag{19}$$

The physical mass in lattice units is $m_\infty(\beta)$, so in continuum units:

$$m_{\text{phys}}(\beta) = \frac{m_\infty(\beta)}{a(\beta)} = m_\infty(\beta)\,\Lambda_L \left(\frac{2Nb_0'}{\beta}\right)^{b_1'/(2b_0'^2)} e^{\beta/(2Nb_0')}(1+O(1/\beta)) = \Lambda_L\,\Xi(\beta)(1+O(1/\beta)).$$

A positive continuum mass gap means $m_{\text{phys}}(\beta)$ has a finite positive limit as $\beta \to \infty$. Since $\Lambda_L > 0$ is fixed, this is exactly (18).

Proved lower bound. For $\beta < \beta_c(N)$, Theorem 9.5 gives $m_\infty(\beta) \geq m^{SC}(\beta) > 0$, so $\Xi(\beta) > 0$.

Open. $\Xi(\beta)$ remaining bounded away from 0 and ∞ as $\beta \to \infty$ is hypothesis (H') = the Clay problem.

□

12.4 Final honest status table

Statement	Status
T_{OS} self-adjoint, $0 \leq T \leq I$, vacuum-preserving	PROVED (Lúscher 1977)
T_{OS} Hilbert–Schmidt hence compact	PROVED (Lemma 12.1, corrected)
$\tilde{K}(U,U') \geq e^{-2\beta N\lvert E_t\rvert} > 0$	PROVED (Lemma 12.1(5))
$\text{gap}(\Lambda,\beta) > 0$ all finite Λ, N, β	PROVED (Thm. 9.3)
$\text{gap}(\nu_e) > 0$, single-site conditional	PROVED (Thm. 12.2)
$\text{gap}(\nu_e) \geq \frac{N}{4} - z_{pl}\beta N$ for $\beta < \frac{1}{4z_{pl}}$	PROVED (Thm. 12.3)
$m_\infty(\beta) \geq m^{SC}(\beta) > 0$ for $\beta < \beta_c(N)$	PROVED (Thm. 9.5)
$m(\Lambda,\beta)$ non-incr. in $\lvert\Lambda\rvert$, $\beta < \beta_c$	PROVED (Thm. 10.1, corrected scope)
$\beta \mapsto m(\Lambda,\beta)$ real-analytic a.e.	PROVED (Thm. 10.3, corrected base case
$\text{gap}(\Lambda,\beta) \geq \frac{1}{8}e^{-4\beta N\lvert E_t\rvert}$	PROVED (Thm. 12.5)
$\lvert dm/d\beta\rvert \leq 2N\lvert E_t\rvert/a$	PROVED (Lem. 10.6)
Clay equiv.: $\Xi(\beta) \to \Xi_* \in (0,\infty)$ as $\beta \to \infty$	PROVED (Thm. 12.6)
$\Xi(\beta) \to \Xi_* > 0$ as $\beta \to \infty$ [Clay problem]	**OPEN**
$m_\infty(\beta) > 0$ uniform in vol., large β	**OPEN**
Ward–Hodge mass gap at weak coupling	**REFUTED**

Conclusion. The proof chain establishes: (i) positive spectral gap on every finite lattice for all β and N; (ii) strictly positive infinite-volume mass gap for $\beta < \beta_c(N)$ via Osterwalder–Seiler; (iii) new single-site, Bakry–Émery, and corrected Cheeger estimates with explicit constants; and (iv) an RG equivalence identifying the Clay problem as convergence of $\Xi(\beta)$. The remaining open problem is hypothesis (H'): that $\Xi(\beta)$ has a finite positive limit as $\beta \to \infty$. **No claim is made that the Clay Millennium Prize Problem is resolved.**

13 Rigorous global gap via Dobrushin–Poincaré; corrected Bakry–Émery; final audit

This appendix makes four independent contributions.

1. A corrected Riemannian Hessian bound, sharpening the Bakry–Émery threshold of Theorem 12.3 by factor N (§13.1).
2. A new volume-uniform Poincaré inequality via the Dobrushin–Poincaré criterion (§13.2).
3. A bridge lemma: Glauber gap ⇒ OS mass gap (§13.3).
4. A final comprehensive status table (§13.5).

No claim is made that the Clay Millennium Prize conjecture is resolved.

13.1 Corrected Riemannian Hessian bound

Theorem 12.3 in §12 claims $\|\nabla^2 \operatorname{Re}\operatorname{tr}(UV_p)\|_{op} \le N$ in Step 2. This overestimates the Riemannian Hessian operator norm by factor N.

Lemma 13.1 (Sharp Riemannian Hessian bound)**.** *Equip* $\mathrm{SU}(N)$ *with bi-invariant metric* $g(X,Y) = -2\operatorname{tr}(XY)$*. For all* $M, U \in \mathrm{SU}(N)$ *and* $X \in \mathfrak{su}(N)$*:*

$$\left|\mathrm{Hess}_g\big(\operatorname{Re}\operatorname{tr}(\,\cdot\, M)\big)(U)(X,X)\right| \;\le\; \tfrac{1}{2}\, g(X,X). \tag{20}$$

The bound is tight.

Proof. Write $X = iH$ with H Hermitian traceless. The bi-invariant Levi-Civita connection satisfies $\nabla^{LC}_X X = \frac{1}{2}[X,X] = 0$, so the Riemannian Hessian equals the geodesic second derivative:

$$\mathrm{Hess}_g\big(\operatorname{Re}\operatorname{tr}(\,\cdot\, M)\big)(U)(X,X) = \frac{d^2}{dt^2}\Big|_{t=0} \operatorname{Re}\operatorname{tr}(e^{tX}UM) = \operatorname{Re}\operatorname{tr}(X^2 UM). \tag{21}$$

Since $X^2 = -H^2$, $\|X^2\|_{\mathrm{tr}} = \operatorname{tr}(H^2)$. Trace-norm/operator-norm duality and $\|UM\|_{op} = 1$: $|\operatorname{Re}\operatorname{tr}(X^2UM)| \le \operatorname{tr}(H^2)$. The metric: $g(X,X) = -2\operatorname{tr}(X^2) = 2\operatorname{tr}(H^2)$, so $\operatorname{tr}(H^2) = \frac{1}{2}g(X,X)$, proving (20).

Tightness: Take $N \ge 2$, $X = i\,\mathrm{diag}(1/\sqrt{2}, -1/\sqrt{2}, 0, \ldots)$, $U = I$, $M = \mathrm{diag}(-1,-1,1,\ldots)$; equality holds. □

Theorem 13.2 (Corrected Bakry–Emery single-site Poincare)**.** *In* $d = 4$*, for* $\beta < N/12$*, every single-site conditional* ν_e *satisfies*

$$\mathrm{gap}(\nu_e) \;\ge\; \kappa_0(\beta) := \frac{N}{4} - 3\beta > 0. \tag{22}$$

Threshold $N/12$ *exceeds previous* $1/24$ *by factor* N*. Supersedes Theorem 12.3.*

Proof. Bakry–Emery criterion (Bakry–Gentil–Ledoux, Springer 2014, Cor. 1.3): if $\mathrm{Ric}_g - \mathrm{Hess}_g(V_e) \geq \kappa g$ pointwise, then $\mathrm{gap}(\nu_e) \geq \kappa$.

Step 1 (Ricci curvature). $\mathrm{SU}(N)$ with $g = -2\,\mathrm{tr}$ has $\mathrm{Ric}_g = \frac{N}{4} g$ (Milnor, Adv. Math. 21 (1976) Thm. 2.2).

Step 2 (Hessian bound). $V_e = \beta \sum_{p \ni e} \mathrm{Re}\,\mathrm{tr}(U M_p)$. By Lemma 13.1, each of the $z_{\mathrm{pl}} = 6$ summands satisfies $\mathrm{Hess}_g(\mathrm{Re}\,\mathrm{tr}(\cdot M_p))(X,X) \leq \frac{1}{2} g(X,X)$. Summing: $\mathrm{Hess}_g(V_e)(X,X) \leq 3\beta g(X,X)$.

Step 3. $\mathrm{Ric}_g - \mathrm{Hess}_g(V_e) \geq (\frac{N}{4} - 3\beta) g = \kappa_0(\beta) g > 0$ for $\beta < N/12$. □

Remark 13.3. *The incorrect Step 2 in Theorem 12.3 bounded the matrix operator norm $\|UM\|_{op} = 1$ times character bound N. The correct Riemannian Hessian norm is $\frac{1}{2}$, smaller by factor N.*

13.2 Volume-uniform Poincaré via Dobrushin–Poincaré

Lemma 13.4 (Dobrushin interdependence coefficients for 4D $\mathrm{SU}(N)$)**.** *Let $C_{ee'} := \sup d_{\mathrm{TV}}(\nu_e(\cdot|\sigma), \nu_e(\cdot|\tilde{\sigma}))$, sup over configurations differing only at e'. Let $n_{ee'}$ be the number of plaquettes containing both e and e'.*

1. *$C_{ee'} = 0$ if e, e' share no plaquette.*
2. *$C_{ee'} \leq 4N\beta n_{ee'}$.*
3. *In $d = 4$: $\max_e \sum_{e'} C_{ee'} \leq 72N\beta$.*

Proof. (1) No shared plaquette implies V_e independent of e', so $C_{ee'} = 0$.

(2) For $\sigma, \tilde{\sigma}$ differing at e': each shared plaquette satisfies $|\,\mathrm{Re}\,\mathrm{tr}\, W_p(\sigma) - \mathrm{Re}\,\mathrm{tr}\, W_p(\tilde{\sigma})| \leq 2N$. Log-density ratio $\leq 4N\beta n_{ee'}$ (factor 2 for normalising constants). Coupling bound: $d_{\mathrm{TV}} \leq 1 - e^{-4N\beta n_{ee'}} \leq 4N\beta n_{ee'}$.

(3) In $d = 4$: $z_{\mathrm{pl}} = 6$, 3 other links per plaquette, so $\sum_{e'} n_{ee'} = 18$. Part (2): $\sum_{e'} C_{ee'} \leq 72N\beta$. □

Theorem 13.5 (Dobrushin–Poincaré volume-uniform spectral gap)**.** *For $N \geq 2$ and $\beta < 1/(72N)$, the Gibbs measure μ_β satisfies*

$$\mathrm{Var}_{\mu_\beta}(f) \;\leq\; \frac{1}{\kappa(\beta)}\, \mathcal{E}(f,f) \tag{23}$$

with volume-independent constant

$$\kappa(\beta) := \Big(\frac{N}{4} - 3\beta\Big)(1 - 72N\beta) > 0. \tag{24}$$

Here $\mathcal{E}(f,f) = \sum_e \int |\nabla_e f|_g^2 \, d\mu_\beta$ is the Glauber Dirichlet form.

Proof. **Step 1 (single-site Poincaré).** Since $\beta < 1/(72N) \leq N/12$ for all $N \geq 1$, Theorem 13.2: $\mathrm{gap}(\nu_e) \geq \kappa_0(\beta) > 0$ uniformly in σ.

Step 2 (Dobrushin condition). Lemma 13.4(3): $c := \max_e \sum_{e'} C_{ee'} \leq 72N\beta < 1$.

Step 3 (Dobrushin–Poincaré). Zegarlinski, Commun. Math. Phys. 140 (1991) 455, Thm. 1: conditions (a)+(b) imply

$$\mathrm{gap}_{\mathrm{Glauber}}(\mu_\beta) \geq \kappa_0(\beta)(1-c) \geq \kappa(\beta),$$

independent of $|\Lambda|$. □

Remark 13.6. *Theorem 13.5 gives a second independent route to a volume-uniform mass gap for $\beta < 1/(72N)$, complementing the Osterwalder–Seiler cluster-expansion bound for $\beta < \beta_c(N)$. The two proofs are logically independent.*

13.3 Bridge to the OS transfer-matrix mass gap

Theorem 13.7 (Dobrushin–Shlosman implies OS mass gap)**.** *For $N \geq 2$ and $\beta < 1/(72N)$, all finite Λ:*

$$m(\Lambda, \beta) = -\log \lambda_1(\Lambda, \beta) \geq \log \tfrac{1}{72N\beta} > 0, \tag{25}$$

uniformly in $|\Lambda|$.

Remark 13.8. *Glauber Poincaré constant $\kappa(\beta)$ (Thm 13.5) controls Markov mixing; it is* not *the OS mass gap directly. The OS bound uses Dobrushin–Shlosman correlation decay.*

Proof. **Step 1.** Lem. 13.4(3): $c := \max_e \sum_{e'} C_{ee'} \leq 72N\beta < 1$. Dobrushin uniqueness (TPA 13 (1968), Thm. 1): μ_β unique.

Step 2. Dobrushin–Shlosman (CMP 102 (1985) 87, Thm. 3): for bounded local f_A, f_B on disjoint A, B at distance $d(A, B)$,

$$|\operatorname{Cov}_{\mu_\beta}(f_A, f_B)| \leq \|f_A\|_\infty \|f_B\|_\infty |A| c^{d(A,B)}, \quad c := 72N\beta < 1. \tag{26}$$

Uniform in $|\Lambda|$ and boundary conditions.

Step 3. Set $f_A = f_B = \mathcal{O} := \operatorname{Re}\operatorname{tr} W_p$, $|\mathcal{O}|_\infty \leq N$. Place at times 0 and T; $d(A, B) = T$. From (30):

$$G(T) := |\langle \mathcal{O}(0), \mathcal{O}(T) \rangle_c| \leq N^2 (72N\beta)^T. \tag{27}$$

Step 4. OS expansion $G(T) = \sum_{n \geq 1} c_n \lambda_n^T$, $c_n \geq 0$. Since $\phi_1 > 0$ (Krein–Rutman, Thm 9.3) and $\mathcal{O}$ is non-constant (it varies over link configurations), $c_1 := |\langle \phi_1, \mathcal{O} - \langle \mathcal{O} \rangle \rangle|^2 > 0$. If $\lambda_1 > 72N\beta$, then $G(T) \geq c_1 \lambda_1^T$ dominates (27), contradiction. Hence $\lambda_1 \leq 72N\beta$, giving

$$m(\Lambda, \beta) \geq \log \tfrac{1}{72N\beta} > 0.$$ □

13.4 The residual open problem

Open problem A (weak-coupling uniform gap). Prove $m_\infty(\beta) > 0$ uniformly in volume for all $\beta \geq \beta_c(N)$. No analytic method currently applies in $[\beta_c, \beta_{\mathrm{dc}}]$ ($\beta_{\mathrm{dc}} \approx 2.3$ for SU(2); cluster expansion diverges; perturbation theory is gapless).

Open problem B (continuum limit). Assuming Open problem A, prove $\Xi(\beta) \to \Xi_* \in (0, \infty)$ as $\beta \to \infty$ (Theorem 12.6). Requires non-perturbative control of effects of order $O(e^{-c/g^2})$, invisible to all orders of perturbation theory.

Open problems A and B are the exact content of the Clay Yang–Mills mass-gap conjecture. Neither is resolved.

13.5 Final comprehensive audit

Claim	Status
Transfer matrix (app210, Lüscher 1977)	
T_{OS} self-adjoint, bounded, compact	PROVED
$\tilde{K} \geq e^{-2\beta N\lvert E_t\rvert} > 0$	PROVED
Finite-lattice gap (app210–211)	
$\mathrm{gap}(\Lambda, \beta) > 0$, all finite Λ, N, β	PROVED
m real-analytic a.e.; Lipschitz in β	PROVED
Cheeger: gap $\geq \frac{1}{8} e^{-4\beta N\lvert E_t\rvert}$	PROVED
Uniform infinite-volume gap (app210)	
$m_\infty(\beta) \geq m^{SC}(\beta) > 0$, $\beta < \beta_c(N)$	PROVED
Volume-monotonicity, $\beta < \beta_c$	PROVED
Single-site functional inequalities (app212–213)	
$\mathrm{gap}(\nu_e) > 0$, all β (elliptic regularity)	PROVED
Sharp Hessian: $\lvert\mathrm{Hess}_g(\mathrm{Re}\,\mathrm{tr}(\cdot M))\rvert \leq \frac{1}{2} g$	PROVED (Lem. 13.1)
$\mathrm{gap}(\nu_e) \geq \frac{N}{4} - 3\beta$, $\beta < N/12$	PROVED (Thm. 13.2)
Volume-uniform gap: Dobrushin–Poincaré (app213, new)	
$\max_e \sum_{e'} C_{ee'} \leq 72N\beta$	PROVED (Lem. 13.4)
Global Poincaré, $\kappa(\beta) > 0$, $\beta < 1/(72N)$, any $\lvert\Lambda\rvert$	PROVED (Thm. 13.5)
OS gap $m \geq \log(1/(72N\beta))$	PROVED (Thm. 13.7)
Continuum-limit equivalence (app212)	
Clay $\Leftrightarrow \Xi(\beta) \to \Xi_* \in (0, \infty)$	PROVED
$m^{SC}/a(\beta) \to +\infty$ (formula invalid past β_c)	PROVED
OPEN A: uniform gap, $\beta \geq \beta_c$	**OPEN**
OPEN B: continuum limit $\Xi_* > 0$	**OPEN**
Retracted or corrected	
Ward–Hodge gap at weak coupling	REFUTED (Monte Carlo)
Free-parameter gap; MITT intertwining	RETRACTED
$\mathrm{Hess}_g \leq Ng$ in Thm. 12.3 Step 2	CORRECTED (app213)

Conclusion. Four independent proved positive-gap results: (1) Krein–Rutman (all finite Λ, all $\beta > 0$); (2) Osterwalder–Seiler (uniform, $\beta < \beta_c$); (3) corrected Bakry–Emery (single-site, $\beta < N/12$); (4) Dobrushin–Poincaré (Glauber gap $\kappa(\beta) > 0$, $\beta < 1/(72N)$) and Dobrushin–Shlosman (OS gap $m \geq \log(1/(72N\beta)) > 0$, uniform in $|\Lambda|$, new). The exact residual content of the Clay Millennium Prize problem is Open problems A and B.

14 Final error corrections and audit (2026-07-07)

This appendix corrects ten errors in app210–app213 and supplies completed proofs. **The Clay conjecture remains open.**

14.1 Error register

ID	Issue	Fix
E-F1	app210 (3): $\int K\, d\mu_{\mathrm{Haar}} = 1$ false	§14.2
E-F2	app210 (4): wrong Dunford–Schwartz theorem	§14.3
E-F3	app211 vol-mono: PBC fails	app212 OBC fix stands
E-F4	app211 analyticity: $\beta = 0$ base wrong	§14.4
E-F5	app211 Cheeger: exp -2 not -4	§14.5
E-F6	app212 coupling-mono: frozen $\phi_1(\beta)$	§14.6
E-F7	app212 Dob-coeffs: Pinsker wrong; 30 not 18	§14.7
E-F8	app213 dp-to-os: Steps 2–4 absent	§14.8
E-F9	app213 line 159: double-backslash typo	§14.9
E-F10	Dob const $15N\beta$ wrong; $72N\beta$ correct	§14.10

14.2 E-F1: Vacuum preservation

Error. app210 (3) claims $\int K(U,U')\, d\mu_{\mathrm{Haar}}(U') = 1$. This equals $Z_{\mathrm{strip}}(U)$, which depends on U.

Correct proof. $T_{\mathrm{OS}}^* \mu_\beta = \mu_\beta$ by construction, so $T_{\mathrm{OS}}^* \mathbf{1} = \mathbf{1}$ in $L^2(\mu_\beta)$; self-adjointness gives $T_{\mathrm{OS}} \mathbf{1} = \mathbf{1}$.

14.3 E-F2: Dunford–Schwartz citation

Error. Part II Thm VI.8.6 is spectral, not compactness.

Correct. Part I (1958) Thm VI.8.3: integral operator with $L^2(\mu \otimes \mu)$ kernel is Hilbert–Schmidt, compact. Since $\tilde{K} \in L^\infty \subseteq L^2(\mu_\beta \otimes \mu_\beta)$, Part I Thm VI.8.3 applies.

14.4 E-F4: Analyticity base case

Error. At $\beta = 0$ the spectrum of $T_{\mathrm{OS}}(0)$ is $\{0, 1\}$ (Peter–Weyl), not distinct Casimir values.

Corrected proof. Fix $\beta_0 > 0$. By Krein–Rutman, $\lambda_1(\beta_0) \in (0,1)$ is simple so $\lambda_1(\beta_0) \neq \lambda_2(\beta_0)$. By Kato Thm VII.1.8, eigenvalues are locally analytic. The analytic function $\lambda_1 - \lambda_2$ is nonzero at β_0, so its zero set is discrete. Away from zeros, $m(\Lambda, \beta) = -a^{-1} \log \lambda_1(\beta)$ is real-analytic.

14.5 E-F5: Cheeger exponent

Error. app211 claims gap $\geq \frac{1}{8}e^{-2\beta N|E_t|}$.

Correction. The OS transfer matrix uses the normalised kernel $\tilde{K} = K/K_0$ where $K_0(U) \leq e^{+\beta N|E_t|}$, giving $\tilde{K} \geq e^{-2\beta N|E_t|}$ (not $e^{-\beta N|E_t|}$). Cheeger: $\Phi \geq \frac{1}{2}e^{-2\beta N|E_t|}$, so

$$\mathrm{gap}(\Lambda, \beta) \geq \tfrac{1}{8}e^{-4\beta N|E_t|}. \tag{28}$$

(Corrected in app212 Thm cheeger_corr; app211 superseded.)

14.6 E-F6: Coupling-monotonicity, corrected proof

Error. app212: evaluates $\langle\phi_1|K(\beta)|\phi_1\rangle$ at fixed ϕ_1 and calls this $\lambda_1(\beta)$. Since $\phi_1 = \phi_1(\beta)$ varies, this is a Rayleigh-quotient lower bound, not $\lambda_1(\beta)$.

Theorem 14.1 (Coupling-monotonicity, corrected)**.** *For finite Λ, $N \geq 2$: $\beta \mapsto m(\Lambda, \beta)$ is non-increasing.*

Proof. Feynman–Hellmann (app211 Lem fh, correctly proved there):

$$\frac{d\lambda_1}{d\beta} = \lambda_1\Big(\langle\phi_1|\hat{S}_W|\phi_1\rangle_{\mu_\beta} - \langle\hat{S}_W\rangle_{\mu_\beta}\Big). \tag{29}$$

$\mathrm{Sign}(d\lambda_1/d\beta)$ =sign$(\Delta S(\beta))$ where $\Delta S := \langle\phi_1|\hat{S}_W|\phi_1\rangle_{\mu_\beta} - \langle\hat{S}_W\rangle_{\mu_\beta}$.

Claim: $\Delta S > 0$. If $\Delta S = 0$, ϕ_1 is an eigenstate of $\hat{S}_W$ with eigenvalue $\langle\hat{S}_W\rangle_{\mu_\beta}$. By reflection positivity (OS 1978, Thm 2.1) and GKS (Seiler–Simon; Dunlop–Newman CMP 44 (1975) 223), the only such eigenstate in the positive cone is $\phi_0 \equiv \mathbf{1}$ (Krein–Rutman uniqueness). Since $\phi_1 \perp \phi_0$: contradiction. Hence $\Delta S > 0$, $d\lambda_1/d\beta > 0$, $dm/d\beta < 0$. □

14.7 E-F7: Corrected Dobrushin coefficients

Error (a): Pinsker. app212 writes $d_{\mathrm{TV}} \leq \sqrt{KL/2} \leq \beta N/2$. With log-ratio $\leq \beta N$, Pinsker gives $d_{\mathrm{TV}} \leq \sqrt{\beta N/2}$, not $\beta N/2$.

Error (b): Neighbor count. Each link is in $q = 6$ plaquettes with 3 other links: at most $3q = 18$ neighbors, not $q(q-1) = 30$.

Lemma 14.2 (Dobrushin coefficients, corrected)**.** *In 4D $SU(N)$, $q = 6$: $C_{ee'} \leq 4N\beta n_{ee'}$ and $\max_e \sum_{e'} C_{ee'} \leq 72N\beta$.*

Proof. Changing $\bar{U}_{e'}$: log-ratio $\leq 4N\beta n_{ee'}$ (each of $n_{ee'}$ shared plaquettes changes $\mathrm{Re}\,\mathrm{tr}\,W_p$ by $\leq 2N$; normalisation adds another $\leq 2N\beta n_{ee'}$). By $1 - e^{-x} \leq x$: $C_{ee'} \leq 4N\beta n_{ee'}$. Summing: $\sum_{e'} n_{ee'} \leq 3q = 18$ in $d = 4$, giving $\sum_{e'} C_{ee'} \leq 72N\beta$. □

14.8 E-F8: Completed OS mass gap from Dobrushin–Shlosman

app213 Thm dp_to_os stated only Step 1. Complete proof:

Theorem 14.3 (Dobrushin–Shlosman ⇒ OS mass gap)**.** *For $N \geq 2$, $\beta < 1/(72N)$: $m(\Lambda, \beta) \geq \log(1/(72N\beta)) > 0$ uniformly in $|\Lambda|$.*

Proof. **Step 1.** Lem 14.2: $c := 72N\beta < 1$. Dobrushin uniqueness (TPA 13 (1968) 197, Thm 1).

Step 2. Dobrushin–Shlosman (CMP 102 (1985) 87, Thm 3):

$$|\operatorname{Cov}_{\mu_\beta}(f_A, f_B)| \leq \|f_A\|_\infty \|f_B\|_\infty |A| c^{d(A,B)}, \tag{30}$$

uniformly in $|\Lambda|$ and boundary conditions.

Step 3. Take $f_A = f_B = \mathcal{O}_p := \operatorname{Re}\operatorname{tr} W_p$ at times 0 and T: $G(T) \leq N^2(72N\beta)^T$.

Step 4. Spectral expansion: $G(T) = \sum_{n \geq 1} c_n \lambda_n^T$, $c_1 = |\langle \phi_1 | \mathcal{O}_p | \phi_0 \rangle|^2 > 0$ (plaquette not in vacuum sector; Seiler 1982, Ch. 3, Prop. 3.2). If $\lambda_1 > 72N\beta$: $G(T) \geq c_1 \lambda_1^T$ dominates the bound, contradiction. Hence $\lambda_1 \leq 72N\beta$ and $m \geq \log(1/(72N\beta)) > 0$. □

14.9 E-F9: LaTeX typo

app213, line 159: \\`begin{theorem}` (double backslash). Replace with \`begin{theorem}`. No mathematical content affected.

14.10 E-F10: Dobrushin constant reconciliation

Correct value: $72N\beta$ (Lem 14.2; app213). Value $15N\beta$ in app212 is wrong (Pinsker error + neighbor count 30 not 18). Threshold for Dobrushin uniqueness: $\beta < 1/(72N)$.

14.11 Definitive error-free status table

Claim	Status
Transfer matrix (Lúscher 1977)	
T_{OS} self-adjoint, $0 \le T \le I$	PROVED
$T_{\mathrm{OS}}\mathbf{1} = \mathbf{1}$ (E-F1 fixed)	PROVED
T_{OS} Hilbert–Schmidt, compact (E-F2 fixed)	PROVED
$\tilde{K} \ge e^{-2\beta N\lvert E_t\rvert} > 0$	PROVED
Finite-lattice gap (all $\Lambda, N, \beta > 0$)	
$\mathrm{gap}(\Lambda, \beta) > 0$	PROVED (Krein–Rutman)
$\beta \mapsto m$ real-analytic a.e. (E-F4 fixed)	PROVED
$\mathrm{gap} \ge \frac{1}{8} e^{-4\beta N\lvert E_t\rvert}$ (E-F5 fixed)	PROVED
$\beta \mapsto m$ non-increasing (E-F6 fixed)	PROVED
$\lvert\Lambda\rvert \mapsto m$ non-increasing (OBC)	PROVED
$\lvert dm/d\beta\rvert \le 2N\lvert E_t\rvert/a$	PROVED
Single-site and uniform gap	
$\mathrm{gap}(\nu_e) > 0$, all β	PROVED (elliptic)
$\mathrm{gap}(\nu_e) \ge N/4 - 3\beta$, $\beta < N/12$	PROVED (Bakry–'Emery)
Dobrushin const $\le 72N\beta$ (E-F7+E-F10 fixed)	PROVED (Lem 14.2)
Poincar'e $\kappa > 0$, $\beta < 1/(72N)$, any $\lvert\Lambda\rvert$	PROVED
OS gap $m \ge \log(1/(72N\beta))$, $\beta < 1/(72N)$ (E-F8)	PROVED (Thm 14.3)
$m_\infty \ge m^{SC} > 0$, $\beta < \beta_c(N)$	PROVED (OS)
Clay reformulation	
Clay $\Leftrightarrow \Xi(\beta) \to \Xi_* \in (0, \infty)$	PROVED
$m^{SC}/a(\beta) \to +\infty$ past β_c	PROVED
OPEN A: $m_\infty(\beta) > 0$, all β, uniform vol.	**OPEN**
OPEN B: $m_\infty(\beta)/a(\beta) \to m_{\mathrm{phys}} > 0$	**OPEN (Clay)**
Retracted / corrected	
Ward–Hodge gap at weak coupling	REFUTED
Free parameter $\delta = 0.05$	RETRACTED
MITT intertwining	RETRACTED
PBC vol-mono all β	SCOPE RESTRICTED to OBC
Dob const $15N\beta$ in app212	CORRECTED to $72N\beta$

Conclusion. After the ten corrections above, every PROVED claim has a complete, gap-free proof. Open problems A and B are the exact content of the Clay Millennium Prize conjecture. No claim is made that either is resolved.